

Diuretics

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CHAPTER OUTLINE

INDIVIDUAL CLASSES OF DIURETICS, 1437
ADAPTATION TO DIURETIC THERAPY, 1450

CLINICAL USES OF DIURETICS, 1454
ADVERSE EFFECTS OF DIURETICS, 1461

KEY POINTS

- Dietary sodium restriction to <3 g daily, selection of a diuretic with a prolonged action, or more frequent administration of the diuretic will enhance NaCl loss by limiting postdiuretic salt retention.
- During prolonged diuretic administration and in the case of diuretic resistance, subjects may be particularly responsive to another class of diuretic.
- Diuretic responsiveness in heart failure (HF) is impaired, as shown by a shift to the right in the natriuresis-excretion relationship of diuretics. Therefore higher or more frequent doses may be required or selection of an extended-release formulation.
- Compensatory distal reabsorption is usually more important than reduced delivery in driving diuretic resistance in HF. Therefore addition of a thiazide or mineralocorticoid receptor blocker or a collecting duct diuretic that act downstream can enhance natriuresis.
- In chronic kidney disease (CKD), the maximal increase in fractional excretion of Na⁺ produced by loop diuretics is maintained, but the absolute response is limited.
- The addition of an SGLT2 inhibitor may help relieve loop diuretic resistance while providing protection from progressive loss of the glomerular filtration rate in those with proteinuric CKD or from recurrent hospitalizations for congestion in those with HF.
- The risk of thiazide-induced hyponatremia increases in subjects with a variant in *SLCO2A1*, which inactivates a prostaglandin transporter and allows prostaglandin E₂ to stimulate EP₄ receptors and insert aquaporin-2 water channels, thereby enhancing electrolyte free water reabsorption and causing recurrent hyponatremia on rechallenge.

This chapter reviews the mechanisms of actions, physiologic adaptations, clinical uses, and adverse effects of diuretics. The term diuretics refers to drugs that increase urinary output. Most diuretics target sodium transport proteins in the kidney (natriuretics). Novel targets for which drugs are in preclinical development also include potassium and urea channels (urearetics; see “Novel Diuretics in Preclinical Development” later). Other targets of diuretics include enzymes (e.g., carbonic anhydrase and neprilysin) or receptors (e.g., mineralocorticoid receptors) that indirectly control renal sodium reabsorption. Vasopressin receptor antagonists (vaptans) increase diuresis by selectively inhibiting water reabsorption (aquaretics). Osmotic diuretics are filtered but not reabsorbed and create a reverse osmotic gradient that inhibits the reabsorption of water and,

indirectly, NaCl (solvent drag). Other diuretics also rely on osmotic diuresis including sodium-glucose transporter type 2 (SGLT2) inhibitors and urea channel inhibitors.

The major sodium transport targets for diuretic drugs have been defined and their genes cloned. The effects of disease on diuretic kinetics are discussed in this chapter because they predict the required dosage modifications. Loop diuretics and thiazides are the most widely used diuretics, and the physiologic adaptations to their prolonged use are described. Diuretic resistance, its management, and major adverse effects of therapy are discussed. This provides a framework for the design of strategies to maximize the desired actions while minimizing the unwanted effects. The chapter also includes a discussion of the practical use of diuretics in the treatment of specific clinical conditions.

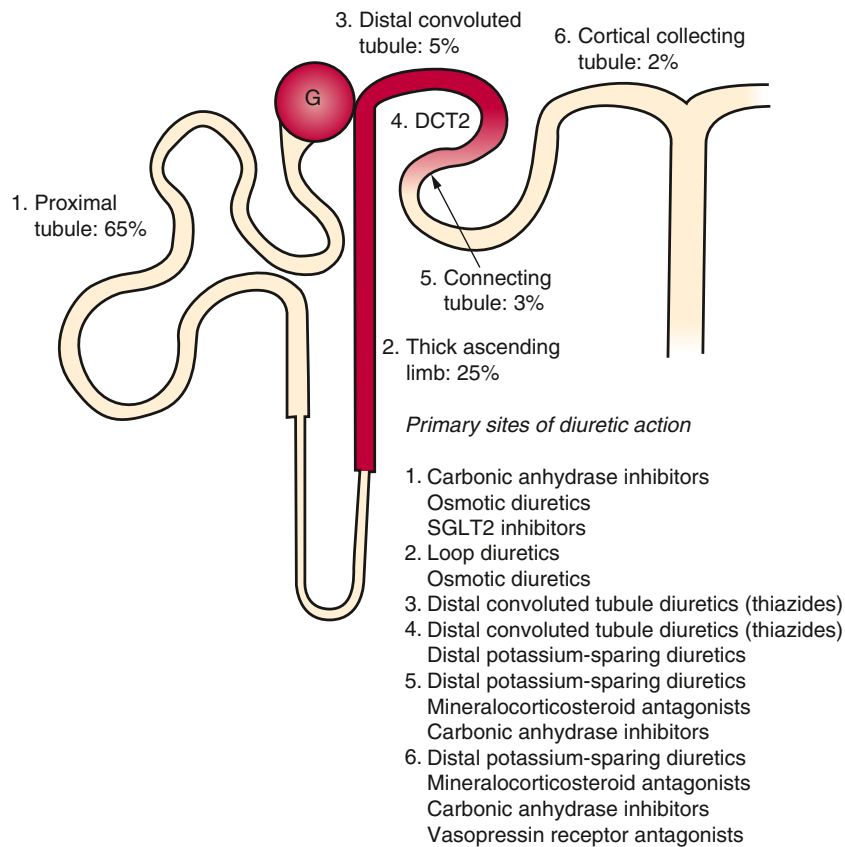


Fig. 49.1 Nephron diagram showing primary sites of diuretic action and approximate percentage of filtered sodium reabsorbed at each. DCT2, Late segment of distal convoluted tubule; G, glomerulus.

Other chapters discuss the treatment of hypertension by diuretic drugs (Chapter 48), diuretic-induced changes in potassium excretion (Chapter 16), acid-base disturbances (Chapter 15), divalent cation excretion and nephrolithiasis (Chapters 17 and 40), disorders of sodium balance (Chapter 13), the syndrome of inappropriate antidiuretic hormone (SIADH) secretion (Chapter 14), and acute kidney injury ([AKI]; Chapters 27 and 28). Diuretics have been reviewed extensively.¹⁻⁵ More extensive and historical references appeared in previous editions of this chapter; the interested reader is referred to editions 7 to 11 of *Brenner & Rector's The Kidney* for more detailed references.

INDIVIDUAL CLASSES OF DIURETICS

The major sites of action of diuretics and fractions of filtered Na^+ reabsorbed at the corresponding nephron segments are summarized in Fig. 49.1.

CARBONIC ANHYDRASE INHIBITORS

SITES AND MECHANISMS OF ACTION

Carbonic anhydrase inhibitors (CAIs) act primarily on proximal tubule cells to inhibit bicarbonate absorption (Fig. 49.2) with additional effects in the loop of Henle and along the distal nephron.^{6,7} Carbonic anhydrase (CA) is a metalloenzyme that regulates sodium-bicarbonate reabsorption and hydrogen ion secretion by renal epithelial cells.^{8,9}

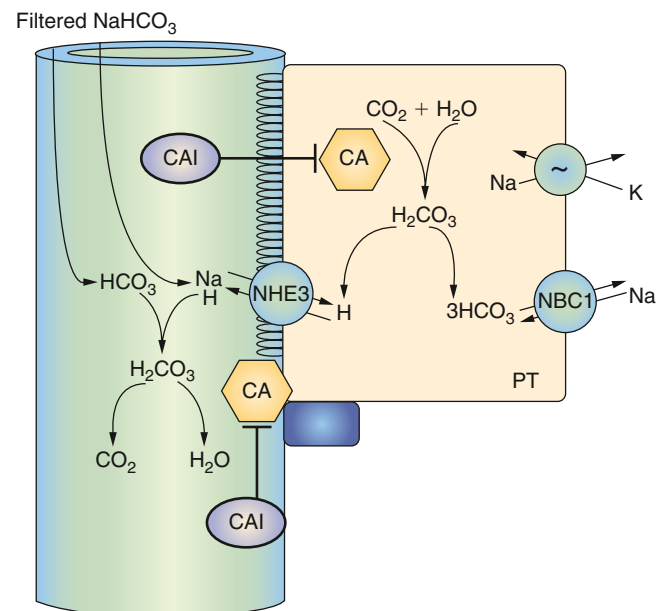


Fig. 49.2 Mechanisms of action of carbonic anhydrase inhibitors in the proximal tubule. The figure shows a functional model of proximal tubule cells; many transport proteins are omitted from the model for clarity. Inside the cell, carbonic anhydrase (CA) catalyzes the formation of HCO_3^- from OH^- and CO_2 . Bicarbonate leaves the cell via the sodium-bicarbonate transporter 1 (NBC1). A second pool of carbonic anhydrase is located in the brush border. This participates in disposing of the carbonic acid formed from filtered bicarbonate and secreted H^+ . Both pools of carbonic anhydrase are inhibited by acetazolamide and other carbonic anhydrase inhibitors (CAIs; see text for details).

CA is expressed by many tissues including erythrocytes, kidney, gut, ciliary body, choroid plexus, and glial cells. Although at least 14 isoforms of CA have been identified, two play predominant roles in renal acid-base homeostasis, CA II and CA IV. CA II is widely expressed in renal tubular cells, where it accounts for 95% of renal CA.⁹ It is present in proximal tubule cells and intercalated cells of the aldosterone-sensitive distal nephron (ASDN).⁹ CA IV is expressed at the luminal border of the cells of the proximal thick ascending limb (TAL) of the loop of Henle and α -intercalated cells of the ASDN.¹⁰

The prototypic CAI is acetazolamide. However, many diuretics have some CAI action.^{11,12} This characteristic contributes to the weak inhibition of proximal reabsorption by furosemide and chlorothiazide and to the relaxation of vascular smooth muscle cells by high-dose furosemide.¹²

CAIs block the catalytic dehydration of luminal carbonic acid at the brush border of the proximal tubule, decrease the intracellular generation of H^+ required for counter transport with Na^+ , and decrease the peritubular capillary fluid uptake.¹³ CAIs also are weak inhibitors of reabsorption in the TAL,¹⁴ but the natriuretic efficacy of CAIs and loop diuretics (see later) is additive, confirming their independent mechanisms of action.¹⁵ CAIs also inhibit bicarbonate reabsorption along the distal tubule, presumably by interfering with the action of α -intercalated cells.¹² The first administration of a CAI causes a brisk alkaline diuresis. The excretion of Na^+ , K^+ , HCO_3^- , and PO_4^{2-} increases, whereas the excretion of titratable acid and NH_4^+ decreases sharply. Excretion of Ca^{2+} remains essentially unchanged. There is substantial kaliuresis owing to the presence of nonreabsorbable HCO_3^- and high flow rates in the distal nephron. However, hypokalemia is uncommon because acidosis partitions K^+ out of cells.¹⁶ Frank hypokalemia is only reported with very-high-dose CAIs (as used in intracranial hypertension) used in combination with thiazides. Bicarbonaturia is reduced over time as the serum bicarbonate concentration declines, thereby reducing the filtered load of bicarbonate. Moreover, the tubular fluid bicarbonate concentrations are increased downstream of the proximal tubule, providing a favorable gradient for passive paracellular reabsorption.

Long-term CAI administration causes only a modest natriuresis, despite the magnitude of CA-dependent proximal Na^+ reabsorption. Several factors account for this:

1. CA is required for the reabsorption of HCO_3^- , whereas about two-thirds of the proximal Na^+ reabsorption is accompanied by Cl^- .
2. Some proximal HCO_3^- reabsorption persists, even after apparently full inhibition of CA.¹⁷
3. Some of the HCO_3^- that is delivered out of the proximal tubule can be reabsorbed in the loop of Henle and more distal sites.¹⁵
4. The metabolic acidosis that develops limits the filtered load of HCO_3^- and thereby curtails the natriuresis.
5. The increased delivery of filtered Na^+ to the macula densa elicits a tubuloglomerular feedback (TGF)-induced reduction in the glomerular filtration rate (GFR).¹⁸

Micropuncture studies of mice with deletion of the proximal Na^+-H^+ exchanger, NHE3, have shown that

inhibition of proximal Na reabsorption is largely balanced by reduced GFR,¹⁹ supporting mechanism #5. Although CAIs only cause a modest natriuresis, the combined use of a thiazide and acetazolamide results in a brisk natriuresis,²⁰ perhaps because acetazolamide inhibits the HCO_3^-/Cl^- exchanger pendrin that is upregulated secondary to upstream inhibition of the sodium chloride cotransporter (Na^+-Cl^- cotransporter [NCC]) by thiazide diuretics.^{20,21}

PHARMACOKINETICS

Acetazolamide is readily absorbed. It is eliminated with a half-life ($t_{1/2}$) of 13 hours by tubular secretion, which is diminished during hypoalbuminemia.²² Methazolamide has less plasma protein binding, a longer $t_{1/2}$, and greater lipid solubility, all of which favor penetration into aqueous humor and cerebrospinal fluid. This agent has less renal effect and therefore is used to treat glaucoma.

CLINICAL INDICATIONS

The use of CAIs as diuretics is limited by their transient action, the development of metabolic acidosis, and a spectrum of adverse effects. They can be used with $NaHCO_3$ infusion to initiate an alkaline diuresis that increases the excretion of weakly acidic drugs (e.g., salicylates and phenobarbital) or acidic metabolites (e.g., urate). Chloride depletion metabolic alkalosis is best treated by the administration of Cl^- with K^+ or Na^+ . However, if this produces unacceptable extracellular volume (ECV) expansion, acetazolamide (250–500 mg/day) and KCl can be used to increase HCO_3^- excretion.

Metabolic alkalosis due to loop diuretics or thiazides can depress respiration in patients with chronic respiratory acidosis (e.g., due to chronic obstructive pulmonary disease). This provides the rationale for the administration of acetazolamide to such subjects that can reduce the arterial partial pressure of carbon dioxide ($Paco_2$) and improve the partial pressure of oxygen (Pao_2). Because both $Paco_2$ and the plasma bicarbonate concentration ($PHCO_3$) decrease, there is little change in blood pH,²³ but a reduction in $PHCO_3$ limits the buffering capacity of blood. CAIs can increase the $Paco_2$ during metabolic acidosis or exercise, perhaps by depressing hypoxic ventilatory drive²⁴ and hypoxic pulmonary vasoconstriction,²⁵ and can cause ventilation-perfusion imbalance.²⁶ Nevertheless, acetazolamide (250 mg bid) can improve blood gas parameters in patients with chronic obstructive pulmonary disease.²⁷ Careful surveillance is required when CAIs are administered to such patients.

When used to treat glaucoma, CAIs diminish the transport of HCO_3^- and Na^+ by the ciliary process, thereby reducing intraocular pressure.²⁸ CAIs also limit the formation of cerebrospinal fluid²⁹ and endolymph.³⁰ Currently, their use to treat glaucoma is mainly as topical preparations. On the other hand, oral acetazolamide is still widely used in the management of idiopathic intracranial hypertension (IIH; also sometimes still referred to as *pseudotumor cerebri*).³¹

Acute mountain sickness is characterized by headache, nausea, drowsiness, insomnia, shortness of breath, dizziness, and malaise after an abrupt ascent. Acetazolamide is useful in dosages of 250 to 750 mg daily as prophylaxis against mountain sickness, probably through stimulating respiration and diminishing cerebral blood flow and cerebrospinal fluid formation.³²⁻³⁴

Acetazolamide, used in established mountain sickness, improves oxygenation and pulmonary gas exchange³⁵ and stimulates ventilation in patients with central sleep apnea.³⁶ In patients with IIH, acetazolamide administration with a low-sodium weight-reduction diet modestly improves visual field function.³¹

The Acetazolamide in Decompensated Heart Failure with Volume Overload (ADVOR) trial randomized patients with acute decompensated heart failure (ADHF) to add-on therapy with intravenous acetazolamide (500 mg daily for 3 days) versus placebo during intravenous loop diuretic administration.³⁷ It reported a significant ($P < 0.001$) 46% improvement in decongestion over 3 days. While encouraging, the effects of longer-term administration were not studied.

CAIs are effective in the prophylaxis of hypokalemic periodic paralysis because they diminish the influx of K^+ into cells.³⁸ Paradoxically, they are also useful in the treatment of hyperkalemic periodic paralysis.³⁹

Studies in mice showed that acetazolamide attenuates lithium-induced nephrogenic diabetes insipidus, with fewer adverse effects than thiazides and amiloride.⁴⁰ This effect was ascribed to a reduction in the GFR and activation of TGF but may also indicate a specific effect in the collecting duct. Clinical data are needed to assess the corresponding effect in humans.

Other indications for CAIs are experimental. They include possible applications in diseases as diverse as obesity, cancer, and infection.⁴¹

ADVERSE EFFECTS

Patients taking CAIs may complain of weakness, lethargy, abnormal taste, paresthesia, gastrointestinal distress, malaise, and decreased libido. These symptoms can be diminished by $NaHCO_3$, but this agent increases the risk of nephrocalcinosis and nephrolithiasis.⁴² Overall, symptomatic metabolic acidosis develops in 50% of patients with glaucoma treated with oral CAIs.⁴³

Older patients or those with diabetes mellitus or chronic kidney disease (CKD) can experience serious metabolic acidosis if given a CAI.⁴³ An alkaline urine favors partitioning of renal ammonia into blood rather than its elimination in urine. Therefore in patients with liver failure, an increase in blood ammonia may precipitate encephalopathy.⁴³

Acetazolamide increases the risk of nephrolithiasis by more than 10-fold.⁴⁴ CAIs occasionally cause allergic reactions, hepatitis, and blood dyscrasias.⁴⁵ They can cause osteomalacia when used with phenytoin or phenobarbital.⁴⁶ While kaliuresis is common, frank hypokalemia is rare.

OSMOTIC DIURETICS

SITES AND MECHANISMS OF ACTION

Osmotic diuretics are freely filtered but poorly reabsorbed.⁴⁷ Mannitol is the prototypic osmotic diuretic, although sorbitol and glycerol have similar actions. In the water-permeable nephron segments of the proximal nephron and thin limbs of the loop of Henle, fluid reabsorption concentrates filtered mannitol sufficiently to diminish tubular fluid reabsorption. Ongoing Na^+ reabsorption lowers the tubular fluid (Na^+) and creates a gradient for back-flux of reabsorbed Na^+ into the tubule. Increased distal flow stimulates K^+ secretion.

Mannitol is a hypertonic solute that abstracts water from cells. The increase in total renal blood flow (RBF) relates in part to hemodilution and a decrease in hematocrit and viscosity. Mannitol increases the medullary blood flow and decreases the medullary solute gradient, thereby preventing urinary concentration. The rise in renal plasma flow and drop in plasma colloid osmotic pressure can increase the GFR.⁴⁸

PHARMACOKINETICS AND DOSAGE

Mannitol is distributed exclusively to the extracellular fluid. It is filtered freely at the glomerulus. Consequently, the $t_{1/2}$ for plasma clearance of mannitol depends on the GFR and is prolonged from 1 to 36 hours in those with advanced CKD.⁴⁹ It can be infused intravenously in daily doses of 50 to 200 g as a 15% or 20% solution or 1.5 to 2.0 g/kg of 20% mannitol over 30 to 60 minutes to treat elevated intraocular or intracranial pressure.⁴⁷

CLINICAL INDICATIONS

Mannitol has been evaluated for the prophylaxis of AKI, but controlled trials of its use in patients at risk for AKI have not had positive results.⁴⁷⁻⁵¹ However, mannitol can protect against AKI in cadaveric kidney transplant recipients.⁴⁷ The use of diuretics to convert oliguric to nonoliguric AKI is discussed later (see “Clinical Uses of Diuretics”).

A trial of mannitol therapy for cerebral edema complicating hepatic failure has demonstrated a markedly better survival of 47%, compared with only 6% in the control group.⁵² Mannitol is recommended for the management of severe head injury with cerebral edema.^{53,54} It is more effective than loop diuretics or hypertonic saline in reducing brain water content.⁵⁵ Mannitol can reverse the dialysis disequilibrium syndrome.⁵⁶

ADVERSE EFFECTS

The effects of mannitol on plasma electrolyte concentrations are complex. The osmotic abstraction of cell water initially causes hypertonic hyponatremia and hypochloremia. Later, after the excess extracellular fluid (ECF) is excreted, the decrease in cell water concentrates K^+ and H^+ within cells, thereby increasing the gradient for their diffusion into the ECF and leading to hyperkalemic acidosis. Normally, these electrolyte changes are corrected rapidly by the kidney if renal function is adequate. Later, hypernatremic dehydration may develop if free water is not provided because urinary concentrating ability is inhibited.⁵⁷

Expansion of ECV, hemodilution, and hyperkalemic metabolic acidosis occur in patients with renal failure who cannot eliminate the drug. Circulatory overload, pulmonary edema, central nervous system depression, and severe hyponatremia require urgent hemodialysis.⁵⁸ Doses of more than 200 g/day can cause renal vasoconstriction and AKI.⁴⁷

LOOP DIURETICS

SITES AND MECHANISMS OF ACTION

The primary action of loop diuretics occurs from the luminal aspect of the TAL (Fig. 49.3). An electroneutral Na - K - $2Cl$ cotransporter, termed “NKCC2,” is located at the luminal membrane.^{59,60} This cotransporter, a member of the solute carrier family 12 (gene symbol *SLC12A1*),

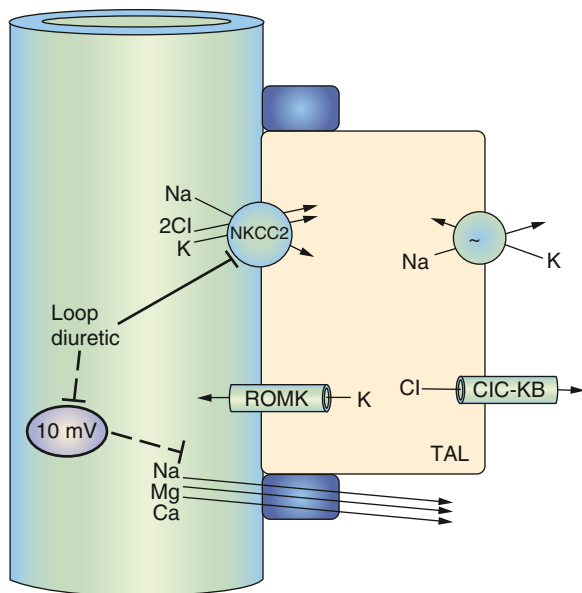


Fig. 49.3 Mechanisms of diuretic action along the loop of Henle. A model of thick ascending limb cells. Na^+ and Cl^- are reabsorbed across the apical membrane via the loop diuretic-sensitive Na-K-2Cl cotransporter 2 (NKCC2). Loop diuretics bind to and block this pathway directly. Note that the transepithelial voltage along the thick ascending limb is oriented with the lumen positive relative to blood (circled value, given in millivolts [mV]). This transepithelial voltage drives a component of Na^+ (and calcium and magnesium; see Fig. 49.4) reabsorption via the paracellular pathway. This component of Na^+ absorption is also reduced by loop diuretics because they reduce the transepithelial voltage. CIC-KB, Chloride channel protein; ROMK, renal outer medullary K^+ channel; TAL, thick ascending limb.

mediates Na^+ and Cl^- movement across the cell. A high luminal K^+ conductance, via the renal outer medullary K^+ (ROMK) channel, allows most of the K^+ to recycle across the luminal membrane.⁶¹ This effectively converts NKCC2 into an electrogenic NCC transporter that is coupled with the electrogenic exit of Cl^- across the basolateral membrane. Thus NKCC2 generates a transepithelial voltage, oriented positively with the lumen relative to interstitial fluid. The primary energy for transport across TAL cells is provided via the basolateral $\text{Na}^+\text{-K}^+\text{-adenosine triphosphatase}$ (ATPase), which maintains a low intracellular (Na^+). Additional details concerning mechanisms of solute reabsorption by TAL cells can be found in Chapter 6. Loop diuretics are organic anions that bind to the NKCC2 from the luminal surface.

NKCC2 is expressed on the apical membranes of medullary and cortical TALs and macula densa segments.^{62,63} Its abundance is increased by prolonged infusion of saline or furosemide.⁶² A closely related gene, *NKCC1*, encodes a protein that is widely expressed in transporting epithelia.⁵⁹ In contrast to NKCC2, *NKCC1* is implicated in the uptake and secretion of Cl^- and NH_4^+ at the basolateral membrane of the medullary CDs.⁶⁴

Hormones that stimulate cyclic adenosine monophosphate (cAMP), such as arginine vasopressin (AVP), enhance TAL reabsorption and should enhance the response to loop diuretics.⁶⁵ In contrast, those that stimulate cyclic guanosine monophosphate (cGMP), such as nitric oxide and atrial natriuretic peptide (ANP), those that increase intracellular (Ca^{2+}), such as 20-hydroxyeicosatetraenoic acid (20-HETE),

or those that activate the Ca^{2+} (polyvalent cation)-sensing protein⁶⁶ inhibit TAL reabsorption and reduce the response to loop diuretics.⁶⁷

The rat TAL also transports NH_4^+ ,⁶⁸ which can substitute for K^+ on NKCC2. There is a luminal $\text{Na}^+\text{-H}^+$ countertransporter in the rat nephron that contributes to tubular fluid acidification. Loop diuretics block the luminal entry of Na^+ via NKCC2, but not the peritubular exit via the $\text{Na}^+\text{-K}^+\text{-ATPase}$, and thereby reduce the intracellular (Na^+) sufficiently to promote luminal Na^+ uptake via the $\text{Na}^+\text{-H}^+$ countertransport process. This is one reason that furosemide stimulates acid excretion in the rat.^{69,70} However, furosemide did not affect net acid excretion or urine pH in normal human subjects.⁷¹

Loop diuretics reduce proximal fluid reabsorption modestly. This effect has been ascribed to a weak CAI action. However, furosemide depresses proximal reabsorption in tubules perfused with HCO_3^- -free solutions.⁷² Moreover, bumetanide, which is a much less potent inhibitor of CA, also impairs proximal fluid reabsorption.⁷³

Furosemide exerts two contrasting effects on reabsorption in the superficial distal tubule. Increased delivery to the unsaturated distal tubule reabsorption process increases Na^+ reabsorption.⁶⁹ Loop diuretics also inhibit NaCl transport in short descending limbs of the loop of Henle,⁷⁴ and although the TAL is clearly the major site of action of loop diuretics, some studies suggest other nephron segments may make minor contributions.^{75,76} Reabsorption of solute from the water-impermeable TAL segments dilutes the tubular fluid. Its inhibition by loop diuretics impairs free water excretion during water loading and free water reabsorption during dehydration.⁷⁷

Loop diuretics increase the fractional excretion of Ca^{2+} by up to 30%.⁷⁸ The predominant mechanism is a decrease in the magnitude of the lumen-positive transepithelial potential (see Figs. 49.3 and 49.4). A large fraction of transepithelial Ca^{2+} transport along the TAL traverses a paracellular pathway involving claudins 16 and 19 and is driven by the lumen-positive transepithelial potential.⁷⁹ By reducing the magnitude of this potential, loop diuretics lower passive calcium absorption along this segment.

The loop of Henle is a major nephron segment for the reabsorption of Mg^{2+} .⁷⁸ Mg^{2+} also traverses a paracellular pathway that involves claudins and is driven by the transepithelial potential difference. Loop diuretics can increase fractional Mg^{2+} excretion by more than 60%⁷⁸ (see Figs. 49.3 and 49.4 and see later discussion of adverse effects). Loop diuretics reduce urate clearance that is largely secondary to volume depletion.⁸¹ There is a reduction in RBF and GFR of rats infused acutely with furosemide that depends on a steep rise in intrarenal pressure that can collapse capillaries and also on a reflex effect from renal afferent nerve activation.⁸² Likewise, human subjects given loop diuretics experience a 20% fall in GFR in the first day.⁸³ However, during more prolonged administration there is normally a well-maintained RBF and GFR⁸⁴ but a marked redistribution of blood flow from the medulla and inner cortex to the outer cortex.⁸⁵ The fall in papillary plasma flow depends on angiotensin II.⁸⁶ Furosemide increases the renal generation of prostaglandins.⁸⁷ Blockade of cyclooxygenase prevents acute furosemide-induced renal vasodilation.⁸⁸

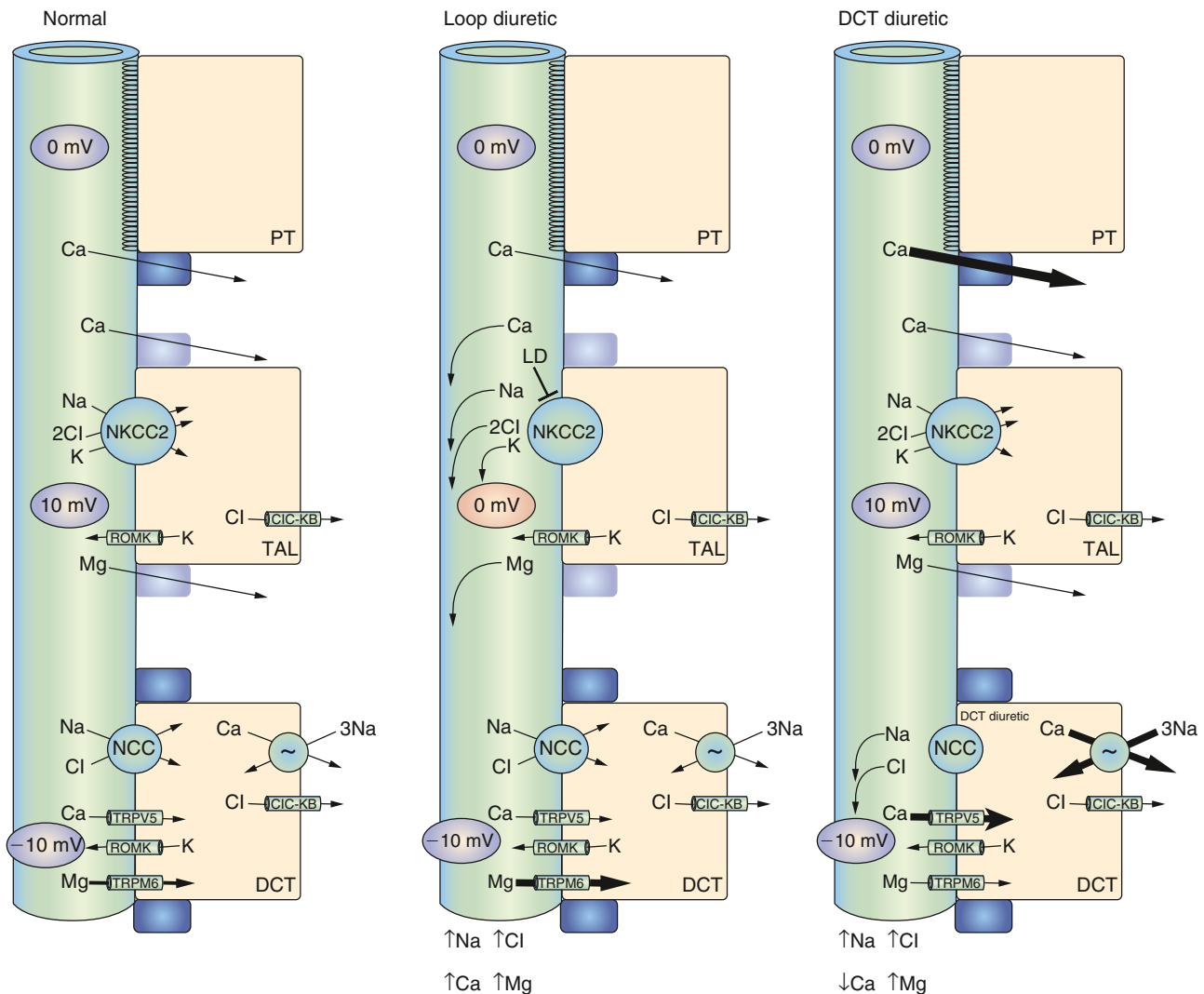


Fig. 49.4 Possible mechanisms of diuretic effects on calcium and magnesium excretion. Typical cells from the proximal tubule (PT), thick ascending limb (TAL), and distal convoluted tubule (DCT) are shown. Calcium reabsorption occurs along the DCT largely via a transient receptor potential channel (TRPV5) and magnesium reabsorption via a transient receptor potential channel (TRPM6). Transepithelial voltages (representative but arbitrary values, given in millivolts [mV]) are shown. Net effects on electrolyte excretion are shown at the bottom. Normal conditions are at the left. Treatment with loop diuretics (LDs) is shown in the middle; treatment with DCT diuretics is shown on the right. Loop diuretics reduce the magnitude of the lumen-positive transepithelial voltage, thereby retarding passive calcium and magnesium reabsorption via the paracellular pathway. Long-term treatment, especially with DCT diuretics, increases proximal Na^+ and Ca^{2+} reabsorption; thus less calcium is delivered distally. Enhanced distal calcium absorption, driven by DCT diuretics, also occurs. Effects of DCT diuretics to increase magnesium excretion remain incompletely understood. CIC-KB, Chloride channel protein; NCC, Na-Cl transporter; NKCC2, Na-K-2Cl cotransporter 2; ROMK, renal outer medullary K^+ channel; $\uparrow$, increase(d); $\downarrow$, decrease(d).

Reabsorption via NKCC2 on the luminal surface of the macula densa cells activates the TGF but inhibits renin secretion. An increase in NaCl delivery to the macula densa as during a reduction in proximal tubular reabsorption increases NaCl entry into macula densa cells. This increases the production of adenosine that interacts with adenosine 1 receptors on vascular smooth muscle and/or extraglomerular mesangial cells to activating phospholipase C. The ensuing cellular depolarization activates voltage-dependent Ca^{2+} channels, which contract afferent arterioles and reduce the GFR (and thereby activate the TGF response).⁸⁹ Loop diuretics block NaCl entry into macula densa cells and thereby block TGF completely.⁹⁰

This can explain the initial fall in GFR with drugs such as CAIs or SGLT2 inhibitors that reduce proximal reabsorption and thereby enhance delivery to the macula densa⁹¹ or the initial rise in RBF observed in some acute studies of loop diuretics in human subjects.⁹² However, furosemide exacerbates impaired renal function and oxygenation in a rat model of AKI.⁹³ An analysis of data from the Renal Optimization Strategies Evaluation Acute Heart Failure (ROSE) trial of subjects with ADHF receiving a mean daily intravenous dose of furosemide of 560 mg revealed that renal function deteriorated significantly in 21% yet there was no increase in markers of tubular damage.⁹⁴ This was attributed to a “beneficial” or “physiologic” reduction in

GFR in response to successful therapy with ACEIs or SGLT2 is rather than an adverse effect of furosemide. Moreover, furosemide may reduce RBF and thereby O_2 delivery, yet this can be offset by reduced NaCl reabsorption and thereby O_2 usage. Indeed, furosemide increases renal medullary oxygenation in normal human subjects.⁹⁵ Clearly, the causes for an increase in serum creatine during diuresis in clinical practice can be complicated but do not necessarily portend bad outcomes, renal tubular damage, or adverse effects on renal hemodynamics.

Loop diuretics also stimulate renin secretion. Although due in part to ECV depletion, a major component is from direct effects of loop diuretics to inhibit NaCl reabsorption in the macula densa. Macula densa reabsorption inhibits renin secretion acutely and inhibits renin synthesis chronically.⁹⁶ Macula densa cells express cyclooxygenase-2 (COX-2) constitutively. Addition of loop diuretics to fluid bathing a macula densa cell line acutely increases the release of prostaglandin E2 (PGE2), followed by a delayed induction of COX-2 expression⁹⁶ and renin secretion mediated by a rapid phosphorylation of p44/42 and p38 MAP kinases following activation of the prostaglandin E2 type 4 (EP4) receptor. Consequently, blockade of COX2 >I with celecoxib or nonsteroid antiinflammatory drugs (NSAIDs)⁹⁷ inhibits renin secretion strongly.

PHARMACOKINETICS AND DIFFERENCES BETWEEN DRUGS

Loop diuretics are absorbed promptly after ingestion, but their bioavailabilities vary. Because bumetanide and torsemide are more completely absorbed than furosemide, changing from intravenous to oral dosing may require a doubling of the dose of furosemide dose but does not require changing the dose of bumetanide or torsemide. Moreover, there is considerable variation in furosemide absorption, both between patients and over time,⁹⁸ that is accentuated by food intake.⁹⁸⁻¹⁰⁰

The highly variable bioavailability of furosemide compared with torsemide and the absence of hypokalemia with usual doses, combined¹⁰¹ with the superior outcomes of torsemide in two clinical trials of patients with heart failure (HF),^{102,103} have prompted consideration that torsemide become the loop diuretic of choice, at least for HF.¹⁰⁴⁻¹⁰⁶ However, the Torsemide vs. Furosemide After Discharge in all-Cause Mortality in Patients Hospitalized with Heart Failure (TRANSFORM-HF) pragmatic trial reported no difference in one-year mortality or all-cause hospitalizations.¹⁰⁷ The reasons for these discrepant findings are unexplained. Nevertheless, a class problem shared with all generic loop diuretics is their short duration of action of 2 to 4 hours.¹⁰¹ The resulting torrential diuresis (termed the “Niagara effect”) can be troubling, especially for patients with prostatism or stress incontinence. Moreover, the limited time of action of a daily dose of diuretic on the tubules provides some 20 hours for the kidneys to regain salt and water losses (see Fig. 49.12). An extended-release formulation of torsemide that delivers torsemide to the circulation over 8 to 12 hours doubled salt and water losses in normal volunteers in the day after a single dose, without increasing potassium excretion or reducing the creatinine clearance.⁸³ Remarkably, one dose of the extended-release torsemide given to subjects

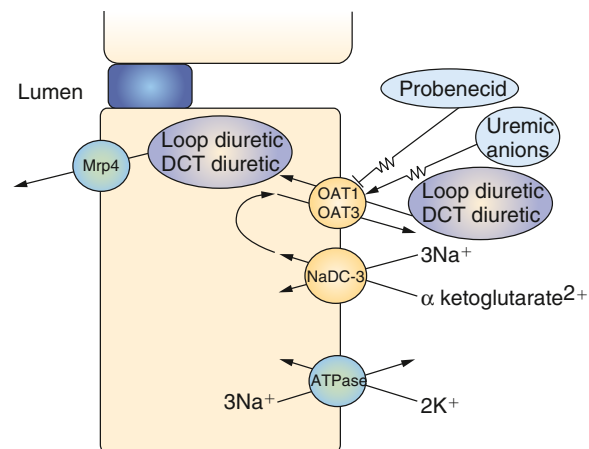


Fig. 49.5 Mechanisms of diuretic secretion by proximal tubule cells. Cell diagram of the S2 segment of the proximal tubule shows secretion of anionic diuretics including loop diuretics and distal convoluted tubule (DCT) diuretics. Peritubular uptake by an organic anion transporter (primarily OAT1, although OAT3 may play a smaller role) occurs in exchange for α -ketoglutarate, which is brought into the cell by the Na^+ -dependent cation transporter NaDC-3. Luminal secretion can occur via a voltage-dependent pathway or in exchange for luminal hydroxyl or urate. A portion of the luminal transport traverses multidrug resistance-associated protein 4 (Mrp4). ATPase, Adenosine triphosphatase.

ingesting 300 mmol of Na^+ daily caused a negative daily Na^+ balance, whereas prior studies with furosemide at this high level of salt intake reported no net loss of Na^+ because of postdiuretic salt retention.¹⁰⁸ Extended-release torsemide is likely to be especially beneficial for the >50% of patients with CHF who have symptoms of overactive bladder or urgency incontinence.

Once absorbed, loop diuretics circulate largely bound to albumin (91%–99%). This limits their clearance by glomerular filtration. The diuretic volume of distribution varies inversely with the serum albumin concentration,¹⁰⁹ but this is not usually a major determinant of diuretic responsiveness (see later).¹¹⁰ The metabolism of loop diuretics is composed of both hepatic and renal mechanisms; the relative fractions that are cleared by each mechanism differ among agents. Loop diuretics, thiazides, and CAIs are all secreted avidly by a probenecid-sensitive organic anion transporter (OAT) in proximal tubule cells (Fig. 49.5).^{111,112} Diuretics gain access to tubular fluid almost exclusively by proximal secretion. Studies have characterized this weak organic anion (OA^-) transport process. Four isoforms of an OAT have been cloned and are expressed in the kidney.^{112,113} Peritubular uptake by an OAT is a tertiary active process (see Fig. 49.5). Energy derives from the basolateral Na^+ - K^+ -ATPase providing a low intracellular $[Na^+]$ that drives an uptake of Na^+ coupled to α -ketoglutarate (α -KG) to maintain a high intracellular level of α -KG. This, in turn, drives a basolateral OA^- - α -KG countertransporter. OAT1 is expressed on the basolateral membrane of the S2 segment of the proximal tubule.¹¹⁴ OAT1 knockout dramatically impairs renal OA^- secretion and is accompanied by furosemide resistance.¹¹⁵ A similar effect was observed in OAT3-deficient mice, suggesting that both OAT1 and OAT3 mediate the secretion of loop diuretic by proximal cells.¹¹⁶

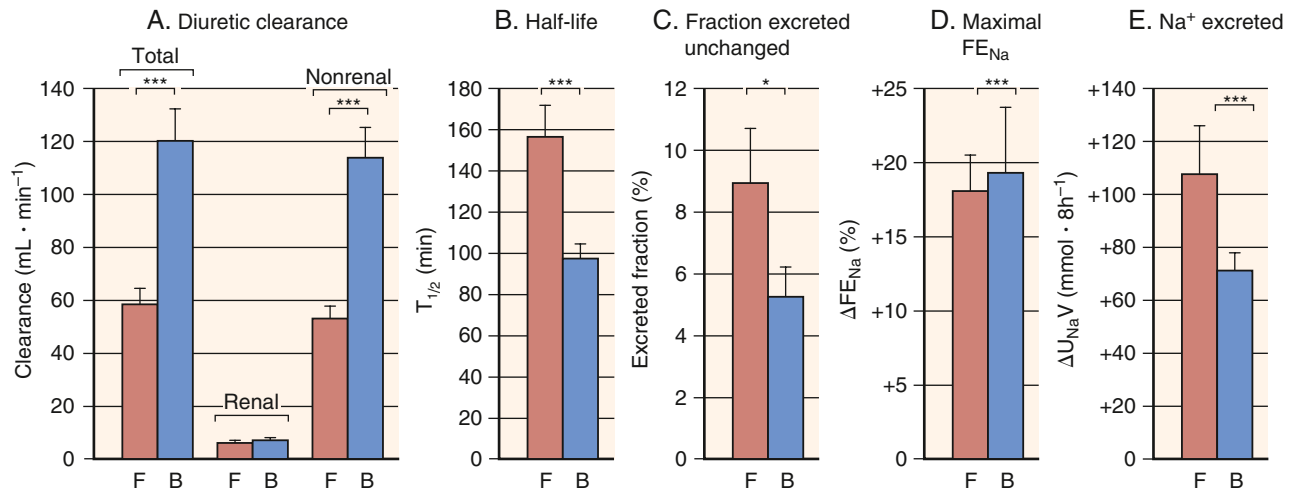


Fig. 49.6 (A–E) Comparison of the pharmacokinetics and dynamics of furosemide (F, 160 mg; metabolically inactivated in the kidney) and bumetanide (B, 4 mg; metabolically inactivated in the liver) in 10 subjects with chronic kidney disease (mean creatinine clearance, 12 ± 2 mL/min). Significance of difference: *, $P < .05$; ***, $P < .005$. FE_{Na} , Fractional excretion of sodium; $T_{1/2}$, half-life; U_{Na} , urinary sodium. (Redrawn from data in Voelker JR, Cartwright-Brown D, Anderson S, et al. Comparison of loop diuretics in patients with chronic renal insufficiency. *Kidney Int.* 1987;32:572–578.)

OATs translocate diuretics into the proximal tubule cell, where they can be sequestered in intracellular vesicles. They are secreted across the luminal membrane by a voltage-driven OA transporter¹¹⁷ and by a countertransporter in exchange for urate or OH^- .¹¹³ The orphan transporter hNPT4 (human sodium phosphate transporter 4; SLC17A3) has been identified as an organic anion efflux transporter that likely also secretes furosemide and bumetanide.¹¹⁸ In addition, the multidrug resistance–associated protein 4 (MRP4) is a third transporter involved in the urinary excretion of diuretics.¹¹⁹ Approximately 50% of furosemide is eliminated by metabolism to the inactive glucuronide. Only the unmetabolized and secreted fraction is available to inhibit NaCl reabsorption. In contrast, bumetanide and torsemide are metabolized in the liver.^{120,121} Unlike bumetanide or torsemide,¹²² the elimination of furosemide is reduced greatly in patients with CKD because its metabolism to the inactive glucuronide occurs predominantly in the kidney; in contrast, metabolic inactivation of bumetanide and torsemide in the liver are unaffected by uremia.^{102,123} This difference prolongs the $t_{1/2}$ of furosemide in CKD, leading to drug accumulation. However, CKD increases the fraction of a dose of furosemide that is excreted unchanged, leading to an enhanced natriuretic response (Fig. 49.6). Therefore there is a tradeoff in the selection of a loop diuretic in CKD: Furosemide can accumulate and cause ototoxicity at high doses, whereas bumetanide and torsemide retain their metabolic inactivation but are somewhat less potent.

Renal clearance of the active form of furosemide is reduced in CKD in proportion to the creatinine clearance.¹²⁴ There is competition for both peritubular uptake¹¹² and luminal secretion¹¹⁷ with other OAs including urate, hippurate, and other organic anions that accumulate in uremia. Metabolic acidosis depolarizes the membrane potential of proximal tubule cells,¹²⁵ which decreases OA secretion,¹¹⁷ an effect that may explain why diuretic secretion is enhanced by alkalosis.¹²⁶ Therefore the increased plasma levels of OAs and urate

and the metabolic acidosis of CKD impair proximal tubule secretion of diuretics and hence impair their delivery to active sites in the nephron.

Despite its high plasma protein binding, the proximal secretion of active furosemide is potentiated by albumin.¹²⁷ In the rabbit, an equal fraction of administered furosemide is taken up by probenecid-sensitive mechanisms in the S2 segment of the proximal tubule, where it is secreted into tubular fluid or by the S1 segment of the proximal tubule, where it is conjugated and excreted as the inactive glucuronide (Fig. 49.7).¹²⁸ Hypoalbuminemia inhibits the uptake and secretion of active furosemide by the S2 segment yet increases the uptake and metabolism by the S1 segment. Therefore it enhances furosemide metabolism¹²⁸ and decreases the tubular secretion of the active diuretic.¹²⁷ The consequences of this process are described later (see “Nephrotic Syndrome”).

There is a sigmoidal relation between fractional sodium excretion and the log of the urinary diuretic concentration that quantitates the amount of diuretic delivered to its active site on the tubular lumen (Fig. 49.8). Inhibition of proximal secretion with probenecid shifts the curve of the plasma dose-response to the right.¹³⁰ Thus natriuresis is related to the urinary, but not the plasma concentration, of diuretic. The administration of indomethacin or other nonsteroidal antiinflammatory drugs (NSAIDs) reduces the responsiveness of the tubule to furosemide.¹³¹ This reduction is due predominantly to reduced generation of PGE2 because a natriuretic response to furosemide can be restored in indomethacin-treated rats by the infusion of PGE2.¹³² A reduced dietary salt intake diminishes the renal tubular response to furosemide¹⁰⁸ (see Fig. 49.8).

CLINICAL INDICATIONS AND ADVERSE EFFECTS OF LOOP DIURETICS

These are discussed later (see “Clinical Uses of Diuretics” and “Adverse Effects of Diuretics”).

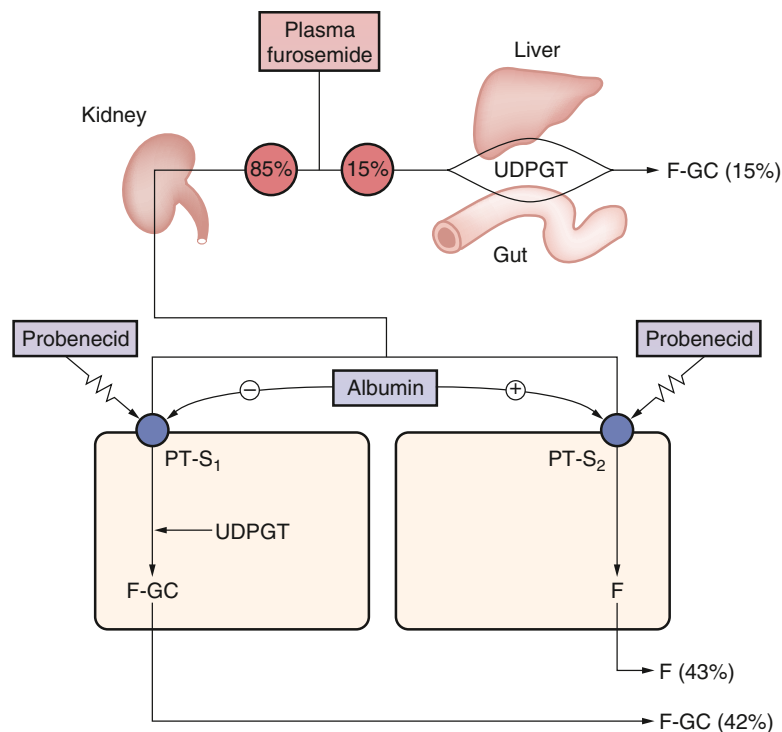


Fig. 49.7 Diagrammatic representation of the disposition of intravenous furosemide and the effects of hypoalbuminemia or probenecid in normal or hypoalbuminemic rabbits. After intravenous administration of furosemide, 15% is metabolized by uridine diphosphate glucuronyl transferase (*UDPGT*) in the liver and gut to the inactive furosemide glucuronide (*F-GC*). Of the remainder, 85% is transported by the kidney. Some 42% is taken up in the S1 segment (*PT-S₁*) and metabolized to the inactive glucuronide, and the remainder is taken up by the S2 segment (*PT-S₂*) and secreted in active form into the lumen. Both uptake processes are inhibited by probenecid. The plasma albumin concentration facilitates uptake and secretion by *PT-S₂* but inhibits uptake and metabolism by *PT-S₁*. (Modified from Pichette V, Geadah D, du Souich P. The influence of moderate hypoalbuminemia on the renal metabolism and dynamics of furosemide in the rabbit. *Br J Pharmacol*. 1996;119:885–890.)

RENAL RESPONSE TO FUROSEMIDE IN HUMANS: EFFECTS OF INDOMETHACIN, SALT RESTRICTION, AND REPEATED FUROSEMIDE ADMINISTRATION

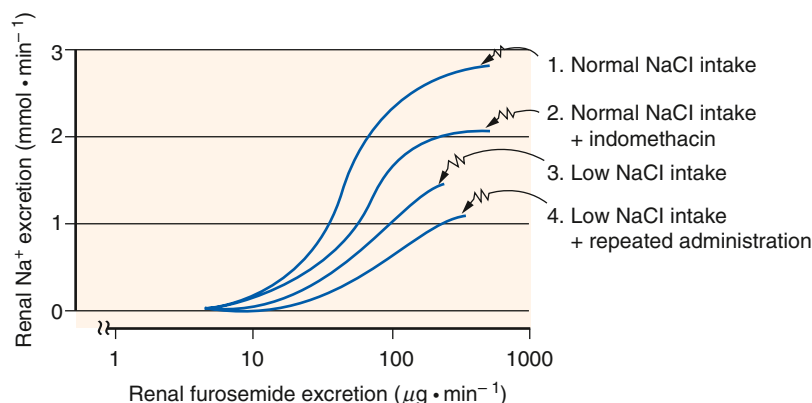


Fig. 49.8 Relationship between the excretion of Na⁺ and furosemide (log scale) following a bolus intravenous injection of 40 mg furosemide in normal subjects with a normal NaCl intake (1), with a normal NaCl intake after indomethacin (2), with a low Na⁺ intake (20 mmol/24 hours) (3), and for the third day of furosemide administration with a low Na⁺ intake (4). (Modified from Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med*. 1983;102:450–458; and Chennavasin P, Seiwel R, Brater DC. Pharmacokinetic-dynamic analysis of the indomethacin-furosemide interaction in man. *J Pharmacol Exp Ther*. 1980;215:77–81.)

THIAZIDES AND THIAZIDE-LIKE DIURETICS (DISTAL CONVOLUTED TUBULE DIURETICS)

SITES AND MECHANISMS OF ACTION

Thiazides and thiazide-like diuretics are moderately active natriuretic drugs that increase the excretion of sodium, chloride, and potassium while reducing the

excretion of calcium. Their major site of action is the distal convoluted tubule (DCT), where they block coupled reabsorption of Na⁺ and Cl⁻ (Fig. 49.9).^{75,100,133} The true thiazides (benzothiadiazines) include chlorothiazide, hydrochlorothiazide, and bendroflumethiazide, whereas metolazone and chlorthalidone have similar actions but distinct molecular structures. Chlorthalidone is

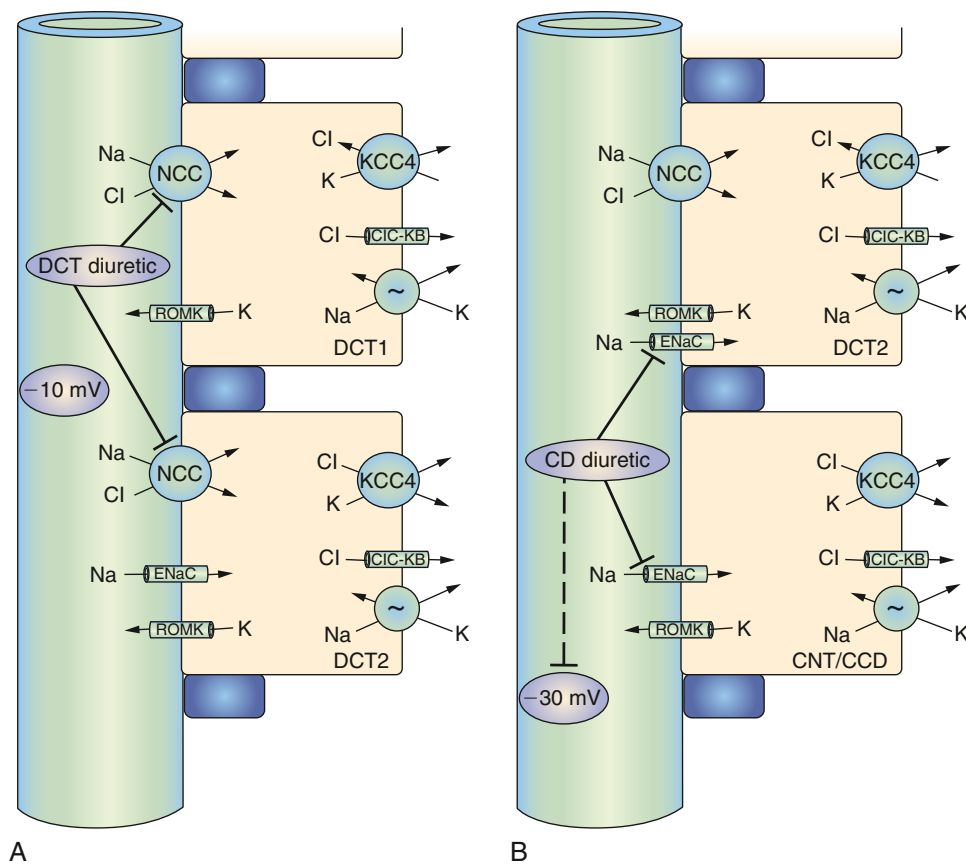


Fig. 49.9 Mechanisms of distal convoluted tubule (DCT) and collecting duct (CD) diuretics. (A) Mechanism of action of DCT diuretics. In rat, mouse, and humans, two types of DCT cells have been identified, referred to here as DCT1 and DCT2. Na⁺ and Cl⁻ are reabsorbed across the apical membrane of DCT1 cells only via the thiazide-sensitive Na⁺-Cl⁻ cotransporter (NCC). This transport protein is also expressed by DCT2 cells where Na⁺ can also cross through the epithelial Na⁺ channel (ENaC; see text for details). Thus the transepithelial voltage along the DCT1 is near to 0 mV, whereas it is finite and lumen negative along the DCT2. (B) Mechanism of action of CD diuretics. The late distal convoluted tubule cells (DCT2 cells) and connecting tubule (CNT) or cortical collecting duct (CCD) cells are shown. Na⁺ is reabsorbed via ENaC, which lies in parallel with a renal outer medullary K⁺ channel (ROMK). The transepithelial voltage is oriented with the lumen negative, relative to the interstitium (shown in the circled value), generating a favorable gradient for transepithelial K⁺ secretion. Drugs that block the epithelial Na⁺ channel reduce the voltage toward 0 mV (effect indicated by dashed line), thereby inhibiting K⁺ and H⁺ secretion. CIC-KB, Chloride channel protein; KCC4, Potassium-chloride cotransporter 4.

also an inhibitor of CA activity and has a prolonged half-life. Due to these properties, it is used widely as an antihypertensive.

The predominant effect of the thiazide diuretics is to inhibit NCC (gene symbol *SLC12A3*) in the DCT^{80,134,135} with additional natriuretic effects from a variable degree of inhibition of CA.¹³⁶ Patients with Gitelman syndrome, who have a loss-of-function mutation in the NCC, and mice lacking NCC demonstrate a dramatically impaired natriuretic response to thiazides,¹³⁷ confirming that the principal effect of these drugs is to inhibit NCC. Na⁺ reabsorption by the proximal tubule is enhanced during long-term treatment with thiazides.¹³⁸

Thiazides are organic anions that bind to the NCC transport protein from the luminal surface. Inhibition has been studied using two approaches. The avid binding of [³H]-metolazone and [³H]-bumetanide to kidney membrane proteins is inhibited competitively by Cl⁻, suggesting that Cl⁻ and diuretics compete for the same binding site.¹³⁹⁻¹⁴¹ Findings have delineated the crystal structure of NCC, allowing for assessment of the 3D binding

sites for ions and thiazides. These studies demonstrate that the thiazide binding site overlaps the proposed chloride binding site.¹⁴²

Thiazides increase potassium excretion indirectly^{100,143} from hyperaldosteronism, increased distal flow, and increased CNT/CCD sodium delivery.¹⁴⁴ They reduce the clearance of urate secondary to enhanced proximal reabsorption from ECV depletion⁸¹ and competition for uptake of urate from blood to PT cells.¹¹¹ Thiazides reduce Ca²⁺ excretion by three nonredundant mechanisms (see Fig. 49.4).^{145,146} First, blockade of luminal NaCl entry reduces the tubule cell intracellular [Na⁺] sufficiently to enhance basolateral Na⁺-Ca²⁺ exchange.¹⁴⁷ Second, thiazide-induced blockade of luminal NaCl entry reduces cell Cl⁻ concentration, thereby hyperpolarizing the membrane voltage that increases calcium entry via the transient receptor potential channel subfamily V, member 5 (TRPV5) channel.^{148,149} Third, thiazides stimulate proximal reabsorption of Ca²⁺ secondary to ECV depletion,¹⁵⁰ which accounts for a continuing effect of thiazides to reduce Ca²⁺ excretion in mice with TRPV5 deleted.¹⁵¹ However, mice deficient in the DCT-specific

protein parvalbumin fail to increase sodium excretion with hydrochlorothiazide but still develop hypocalcemia.¹⁵² Similarly, hypocalcemia occurs independent of hypovolemia in patients with Gitelman syndrome, who have a genetic inactivation of NCC.¹⁵³ Thiazides cause a sustained reduction in renal Ca^{2+} excretion that is accompanied by a small rise in serum Ca^{2+} concentration.

Mg^{2+} excretion is enhanced by prolonged thiazide therapy¹⁵⁴ (see Fig. 49.4). Long-term thiazide treatment of mice diminishes expression of the transient receptor potential channel melastatin 6 (TRPM6) substantially.¹⁵¹ This would be expected to reduce Mg^{2+} reabsorption.^{155,156} The Mg^{2+} depletion that can occur during chronic thiazide administration may augment K^{+} depletion.¹⁵⁴

The aquaporin 2 (AQP2) water channel is at the terminal diluting segment of the kidney in the connecting tubule. Thiazides impair solute excretion and thereby maximal urinary dilution but not maximal urinary concentration.¹⁵⁷ However, they also enhance water absorption from inner medullary collecting ducts in an AVP-independent manner¹⁵⁸ that is accompanied by an increase in AQP2 expression.¹⁵⁹ These effects summate with central effects on thirst and enhanced production of PGE2 to induce hyponatremia (see later discussion of adverse effects).

Clinical Relevance

Implications of Differential Effect of Diuretics on Electrolyte and Acid-Base Handling

The classes of diuretics have different effects on acid-base and electrolyte balance. Hyponatremia is a specific adverse effect of thiazide diuretics. Loop and thiazide diuretics increase urinary magnesium and potassium excretion, whereas this is reduced by distal K^{+} -sparing diuretics. Urinary calcium excretion is increased by loop diuretics and decreased by thiazide diuretics. All diuretics tend to cause metabolic alkalosis, except for carbonic anhydrase inhibitors, mineralocorticoid receptor antagonists (MRAs), and collecting duct diuretics that cause metabolic acidosis. Diuretic combinations may be used to prevent clinically significant acid-base and electrolyte disturbances.

PHARMACOKINETICS, PHARMACOGENETICS, AND DIFFERENCES AMONG THIAZIDES

Thiazides are readily absorbed and extensively bound to plasma proteins. They are eliminated largely through secretion by the S2 segment of the proximal tubule, mostly via OAT1 and OAT3.^{112,116} The $t_{1/2}$ is prolonged in renal failure and in older adults, accompanied by a reduced natriuretic efficacy.¹⁶⁰ However, chlorthalidone remains effective in CKD IV.¹⁶¹ The more lipid-soluble drugs (e.g., bendroflumethiazide and polythiazide) are more potent, have a more prolonged action, and are more extensively metabolized.¹⁶² Chlorthalidone has a particularly prolonged action.¹⁶² Indapamide is sufficiently metabolized to limit accumulation in renal failure.¹⁶² Extrarenal effects of thiazide diuretics including effects on platelet aggregation and vascular permeability vary among the types of thiazide diuretics. While thiazide-like diuretics (e.g., chlorthalidone and indapamide) were believed to have superior

cardioprotective effects compared with thiazide-type diuretics (e.g., chlorothiazide and hydrochlorothiazide),^{163–165} the large VA diuretic comparison project that contrasted outcomes for cardiovascular disease (CVD) in hypertensives randomized to hydrochlorothiazide versus chlorthalidone reported no differences and a higher rate of hypokalemia with chlorthalidone.¹⁶⁶

While there is little known about the pharmacogenetics of diuretics, available studies have suggested that this may account for some differences in individual responses. Diuretic therapy in carriers of a variant of α -adducin, a cytoskeleton protein important for the function of renal $\text{Na}^{+}\text{-K}^{+}\text{-ATPase}$, was associated with a lower risk of combined myocardial infarction and stroke than other antihypertensive therapies.¹⁶⁷ Polymorphisms in with-no-lysine kinase 1 (WNK1), which is involved in the regulation of NCC, also affected the response to a thiazide diuretic.¹⁶⁸ Finally, polymorphisms in Nedd4-2, a ubiquitin ligase that regulates NCC,¹⁶⁹ predict the blood pressure (BP) response to hydrochlorothiazide in White subjects and cardiovascular outcome with hydrochlorothiazide in Black subjects.¹⁷⁰

CLINICAL INDICATIONS AND ADVERSE EFFECTS OF THIAZIDE DIURETICS

The clinical indications for using these drugs are discussed later (see “Clinical Uses of Diuretics”), as are their adverse effects (see “Adverse Effects of Diuretics”).

DISTAL POTASSIUM-SPARING DIURETIC AGENTS

Distal K^{+} -sparing diuretics comprise those that block the ENaC directly (amiloride and triamterene) and those that inhibit the mineralocorticoid receptor (finerenone, spironolactone and eplerenone). Finerenone is the first of a class of nonsteroidal mineralocorticoid receptor antagonists (MRAs) to be released for the treatment of HF and prevention of CKD decline in patients with diabetic kidney disease based on results from the FIDELIO-DKD and FIGARO-DKD trials.^{171–174}

SITES AND MECHANISMS OF ACTION

Distal K^{+} -sparing diuretics act on the ASDN comprising the late DCT, connecting tubule, and cortical CD, where they inhibit luminal Na^{+} entry via ENaC (see Fig. 49.9).^{133,175} They depolarize the lumen-negative transepithelial voltage, thereby diminishing the electrochemical gradient for K^{+} and H^{+} secretion and increasing the serum potassium concentration (S_{K}) and reducing the serum bicarbonate concentration ($S_{\text{HCO}_3^{-}}$).^{133a}

Both amiloride and triamterene are organic cations that block ENaC directly from the luminal surface. Amiloride also inhibits NHE3, but its affinity for NHE3 is sufficiently low that the distal effects predominate. Amiloride appears to bind ENaC,¹⁷⁶ where it competes with Na^{+} ¹⁷⁷ predominantly at a short amino acid stretch within the extracellular loop.¹⁷⁸ Amiloride and triamterene also reduce the excretion of Ca^{2+} and Mg^{2+} .^{154,179}

Spironolactone and eplerenone are antagonists of the mineralocorticoid receptor (MRAs) that were developed on the basis of the steroidal structure of aldosterone and shown to block the MR competitively. Eplerenone was developed as an MRA with fewer side effects from blockade of androgen

receptors. Finerenone was developed as a nonsteroidal, dihydropyridine-based MR antagonist.¹⁸⁰

These drugs were used for many years primarily in combination with other diuretics to reduce the excretion of K⁺ and net acid. Later, it was found that long-term administration of spironolactone was more effective than furosemide in reducing cirrhotic ascites.¹⁸¹ The PATHWAY-2 trial randomly crossed over patients with uncontrolled hypertension despite taking three antihypertensives including a diuretic through four 12-week periods of add-on therapy with spironolactone (25–50 mg daily), the β -adrenoreceptor antagonist bisoprolol (5–10 mg), the α -adrenoreceptor antagonist doxazosin in a modified release formulation (4–8 mg) or placebo.¹⁸² The placebo-corrected fall in systolic blood pressure (SBP) with spironolactone that averaged 8.7 mm Hg was twice as great as for the other drugs. Spironolactone is now recommended as the first choice of add-on therapy for resistant hypertension.¹⁸³ The Randomized Aldactone Evaluation Study (RALES) reported that add-on therapy with spironolactone for 2 years for patients with HF with reduced ejection fraction (HFrEF) lessened the risk of death or hospitalization for HF significantly by >30%.¹⁸⁴ Generally, similar conclusions were drawn from trials of spironolactone in patients with HF with preserved ejection fraction (HFpEF)¹⁸⁵ or of eplerenone in HFrEF.¹⁸⁶ Thus MRAs have become drugs of first choice for the treatment of resistant hypertension and HF, but the extent to which these effects are due to their diuretic action remains unclear.

PHARMACOKINETICS

Triamterene is well absorbed. It is hydroxylated rapidly to active metabolites.¹⁸⁷ The drug and its metabolites are secreted by the organic cation pathway in the proximal tubule³ and eliminated with half-lives of 3 to 5 hours. Triamterene and its active metabolites accumulate in patients with cirrhosis because of decreased biliary secretion¹⁸⁸ and in older adults^{160,187} and patients with CKD¹⁶⁰ because of decreased renal excretion.

Amiloride is incompletely absorbed. Its duration of action is approximately 18 hours. It is secreted into the tubular fluid by the organic cation transport pathway.¹⁸⁹ Other organic cations, such as cimetidine, inhibit its secretion and prolong its half-life.¹⁸⁹ It accumulates in renal failure¹⁹⁰ and may worsen renal function.¹⁹¹ Spironolactone is readily absorbed and circulates bound to plasma proteins. Its intrinsic half-life is short, but it is metabolized to active compounds with considerably prolonged actions. Spironolactone is metabolized to canrenones ($t_{1/2}$ = 16 hours) and to sulfur-containing metabolites, predominantly 7- α -thiomethylspironolactone ($t_{1/2}$ = 13 hours).¹⁹² Canrenones are metabolized by the cytochrome P-4503A system.¹⁹³ This accounts for an effective $t_{1/2}$ for spironolactone of approximately 20 hours and the need for only once-daily dosing. It takes 10 to 48 hours to become maximally effective.¹⁹⁴ It is lipid soluble and enters distal renal tubules from the plasma.¹⁹⁵ Eplerenone has fewer antiandrogenic and proestrogenic effects.^{196,197} However, it is metabolized with a half-life of 3 hours and therefore should be given twice daily.¹⁹⁸ Finerenone is highly selective for MR and has a similarly short half-life as eplerenone.¹⁸⁰

CLINICAL INDICATIONS

Distal K⁺-sparing agents are used to prevent or treat the hypokalemic alkalosis that can arise from thiazide or loop diuretic therapy.¹⁹⁹ Spironolactone is indicated as a first-line agent for ECV expansion in the setting of cirrhotic ascites,^{181,200,201} where it is used in combination with furosemide at a dose ratio of spironolactone 100 mg/furosemide 40 mg.²⁰² It is also indicated for HF¹⁸⁴ (in doses of 25 and 50 mg/day), to treat hypertension associated with hyperaldosteronism and for resistant hypertension.^{203–205} It reduces proteinuria and progressive loss of kidney function in CKD.^{206,207} Amiloride reduces proteinuria to an equivalent degree (however, see the following discussion of adverse effects).²⁰⁸ Eplerenone is indicated to prevent cardiac remodeling and systolic dysfunction in the setting of recent myocardial infarction.^{209,210} Finerenone is indicated for treatment of diabetic kidney disease. It slows the progression of CKD and improves cardiovascular outcomes in this patient group.^{173,174}

ADVERSE EFFECTS AND DRUG INTERACTIONS

Hyperkalemia is the most common complication of the distal K⁺-sparing diuretics and MRAs. The risk is dose dependent and increases considerably in patients with CKD or in those receiving K⁺ supplements, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), NSAIDs, β -blockers, heparin, or ketoconazole. The incidence of hyperkalemia-associated morbidity and mortality in Canada rose sharply after the publication of the RALES trial.²¹¹ While finerenone does increase the risk of hyperkalemia, it may do so less frequently than spironolactone.¹⁷³

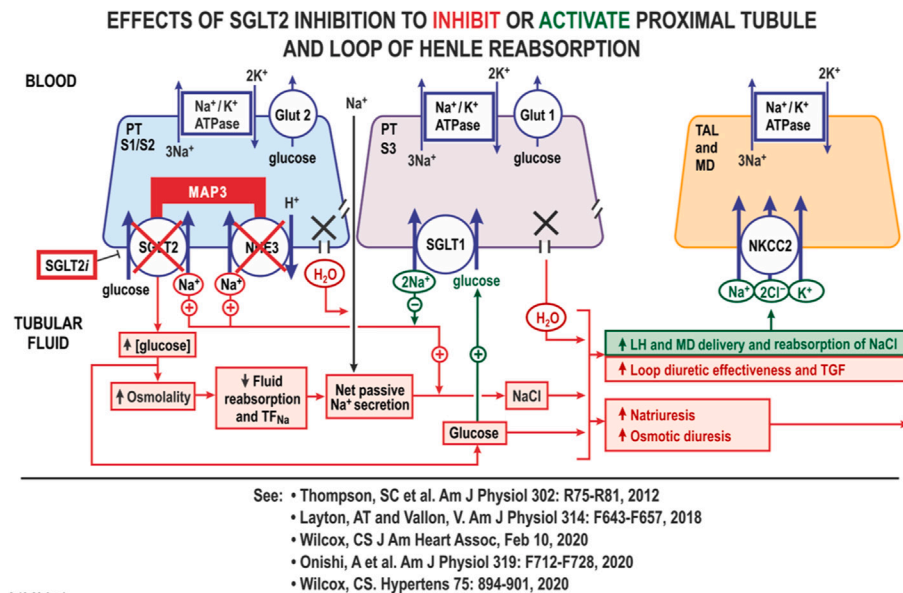
Gynecomastia and reduced libido may occur in men treated with spironolactone, especially as the dose is increased.^{212,213} Women may experience menstrual irregularities, hirsutism, or swelling and tenderness of the breast. Impaired net acid excretion can cause metabolic acidosis,²¹⁴ which worsens hyperkalemia.

Amiloride and triamterene accumulate in renal failure,^{187,190} and triamterene accumulates in cirrhosis.¹⁸⁸ Therefore these drugs should be used with caution in patients with these conditions. Triamterene occasionally precipitates in the urinary collecting system and causes obstruction.²¹⁵ It can cause AKI when given with indomethacin.²¹⁶

MISCELLANEOUS AGENTS

DOPAMINERGIC AGENTS

Dopamine infused into normal subjects in low doses (1–3 μ g/kg per minute) causes a modest increase in the GFR, reduces proximal reabsorption via a cAMP-induced inhibition of the Na⁺-H⁺ antiporter, and increases Na⁺ excretion.²¹⁷ Fenoldopam is a selective dopamine type 1 receptor agonist, with little cardiac stimulation.²¹⁷ Unfortunately, these beneficial effects are lost in patients who are critically ill and/or receiving vasopressors.²¹⁸ A comprehensive review of controlled trials concluded that low-dose dopamine universally fails to improve renal outcomes in patients at high risk for AKI and that it had no effect on renal function, need for dialysis, or mortality in critically ill patients with early renal dysfunction.²¹⁸ Thus there is currently no justification for the use of low-dose dopamine for renal protection.²¹⁹



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Fig. 49.10 Effects of sodium-glucose-linked transport 2 inhibition on proximal tubule and loop of Henle reabsorption. A simplified functional model of the early (S1 and S2) and late (S3) segments of the proximal tubule and the thick ascending limb of loop of Henle. SGLT2 is linked functionally to the sodium hydrogen exchanger 3 and to the urate transporter (URAT 1) by the microtubule-associated protein 3 (MAP3). Inhibition of SGLT2 blocks NHE3, thereby delivering more sodium from the proximal tubule while also reducing glucose reabsorption, thereby increasing tubular fluid osmolality to create an osmotic diuresis. More glucose is delivered to SGLT1 in the S3 segment that is activated to reabsorb sodium coupled in a 2:1 ratio with glucose that thereby offsets the natriuresis. The fall in tubular fluid sodium concentration promotes passive sodium back leak from the interstitium to the tubular fluid. Increased delivery of sodium chloride to the thick ascending limb (TAL) of the loop of Henle enhances responsiveness to loop diuretics while increased delivery to the macula densa activates the tubuloglomerular feedback mechanism (TGF).

Dopamine infusion at higher rates has a role as a pressor agent in septic shock or refractory HF, but the benefits can be offset by arrhythmias.²¹⁸

VASOPRESSIN RECEPTOR ANTAGONISTS

Vasopressin receptor (VR) antagonists are nonpeptide molecules that competitively inhibit one or more of the human VRs, V1aR, V1bR, and V2R.²²⁰ Conivaptan is a combined V1aR/V2R antagonist for intravenous use, whereas tolvaptan, mozavaptan, and lixivaptan are orally active V2R-selective antagonists. All these agents cause a free water diuresis without appreciable natriuresis or kaliuresis (“aquaretics.”)²²¹ This effect is mainly attributed to the inhibition of V2R in the collecting duct, which prevents vasopressin from recruiting AQP2 water channels to increase water reabsorption. Therefore vasopressin receptor antagonists can be used to treat hypervolemic or euvoletic hyponatremia. Coinhibition of V1aR, which is located in vascular smooth muscle, could be beneficial to reduce coronary vasoconstriction, myocyte hypertrophy, and vascular resistance in patients with HF,²²² but definitive studies are lacking. The results of many clinical trials in patients with liver cirrhosis, HF, or hyponatremia secondary to SIADH²²³ report that VR antagonists effectively raised serum sodium levels (S_{Na}) and help to correct hyponatremia with reduction in body weight, dyspnea, and ascites in patients with HF.²²⁴⁻²²⁶ However, the Efficacy of Vasopressin Antagonist in Heart Failure Outcome Study with Tolvaptan (EVEREST) trial failed to show a beneficial effect of tolvaptan on the primary outcome of death and rehospitalization.²²⁶ Thus V2R antagonists can effectively

correct hyponatremia but have not yet shown a benefit in primary outcomes for HF.

Nesiritide

Nesiritide is the recombinant form of B-type natriuretic peptide, which can be administered intravenously in acute decompensated congestive HF (see later discussion of clinical uses of diuretics in congestive HF). This agent causes both a natriuresis and a relaxation of smooth muscle.

NEPRILYSIN INHIBITORS

Neprilysin inhibitors reduce the metabolism of natriuretic peptides and therefore contribute to natriuresis. The combined use of an ARB and neprilysin inhibitor reduced cardiovascular death and hospitalization for HF in patients with class II, III, or IV HF.²²⁷

SODIUM-GLUCOSE-LINKED TRANSPORTER 2 INHIBITORS

Empagliflozin, canagliflozin, and dapagliflozin are three SGLT2 inhibitors that were traditionally used for the treatment of type 2 diabetes mellitus; however, they have now been identified as mainstays in the treatment of HF and proteinuric CKD.^{232,238} SGLT2 inhibitors block the coupled reabsorption of sodium and glucose in the S2 segment of the PT, thereby delivering about half of the filtered glucose to the urine and inducing an osmotic diuresis and natriuresis (Fig. 49.10). Despite blockade of almost 100% of glucose reabsorption by SGLT2, these drugs increase the excretion of glucose by only about 50% because of increased glucose

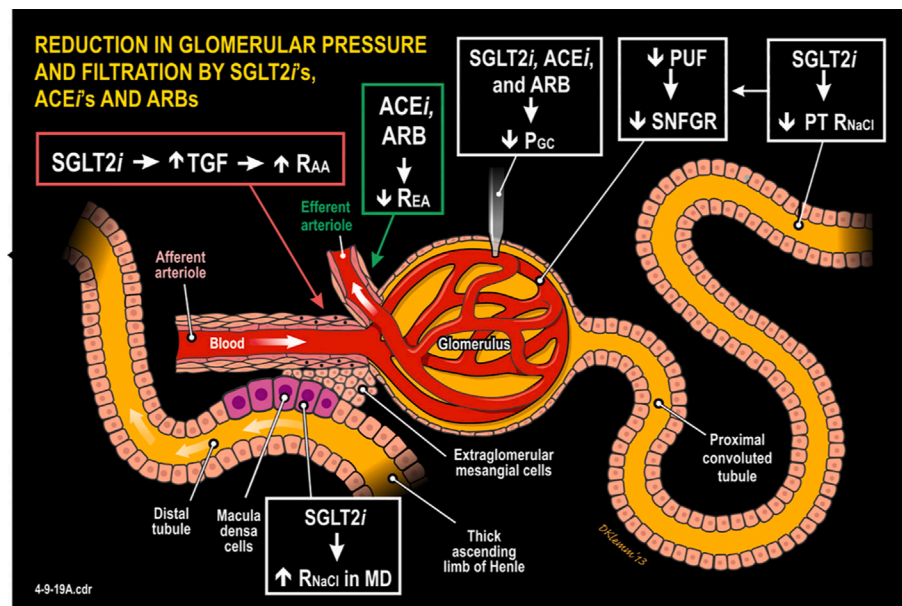


Fig 49.11 A diagrammatic depiction of mechanisms that may reduce glomerular capillary pressure and reversal of glomerular hyperfiltration by a sodium-glucose-linked transport type 2 inhibitors (SGLT2is), angiotensin-converting enzyme inhibitors (ACEis) and angiotensin receptor blockers (ARBs). SGLT2i reduces reabsorption of sodium and chloride in the proximal tubule and increases their delivery and reabsorption at the macula densa segment. This activates the tubuloglomerular feedback (TGF) response to increase the resistance of the afferent arteriole and reduce the hydrostatic pressure in the glomerular capillaries. ACEis and ARBs lead to a greater reduction in resistance of the efferent than the afferent arteriole and reduce the hydrostatic pressure in glomerular capillaries by a different mechanism.

reabsorption by the downstream SGLT1 in the S3 segment. Since SGLT2 transports Na:glucose in a 1:1 stoichiometric ratio, yet SGLT1 in a 2:1 ratio, the net effect of SGLT2 inhibitors should be to reduce PT reabsorption of glucose but to have little effect in the reabsorption of Na⁺. However, an SGLT2 inhibitor given to diabetic rats caused a sustained >25% reduction in Na⁺ and fluid reabsorption in the PT when measured by micropuncture.²³¹ Moreover, these drugs can increase Na⁺ excretion and reduce plasma volume in patients with HF.^{232,233} An explanation for this paradox is the finding that SGLT2 inhibitors coinhibit sodium hydrogen exchanger 3 (NHE3), which is the principal site for Na⁺ reabsorption in the PT.²³⁴ While binding between NHE3 and SGLT2 has not been established, they may be linked via the multifunctional peptides MAP17 and PDZK1. MAP17 binds to SGLT2, as well as the fourth PDZ domain of PDZK1, thereby potentially associating it with NHE3 to form a signaling complex. The result is a greater inhibition of proximal tubule sodium reabsorption by SGLT2 inhibitors than anticipated from inhibition of sodium-glucose cotransport alone.^{91,234-236} The increased renal Na⁺ excretion activates TGF, increases preglomerular vascular tone and reduces the GFR (Fig. 49.11). The resulting reduction in glomerular capillary pressure and glomerular blood flow reduces hyperfiltration and, in combination with a fall in BP, may account for the slowing of the progressive loss of the GFR in patients with diabetic kidney disease or CKD treated with SGLT2is.^{237,238} This effect is likely what also leads to the significant mortality benefits of these medications in this patient population. However, there are a number of other theories as to the mechanism of the significant clinical benefits of these medications in CKD and HF.^{229,239-241} Moreover, the prolonged increase in Na⁺

delivery from the proximal tubule leads to an adaptive increase in NaCl reabsorption in the loop of Henle²³¹ that could account for the observation of an adaptive synergistic natriuresis when a loop diuretic is given to subjects adapted to an SGLT2 inhibitor.²⁴² Thus the addition of an SGLT2 inhibitor may relieve loop diuretic resistance while providing some protection from progressive loss of the GFR.^{243,244} Additionally, an important effect of SGLT2 inhibitors is the sustained increase in urate clearance, likely due to inhibition of URAT1 by luminal glucose.

NOVEL DIURETICS IN PRECLINICAL DEVELOPMENT

Several novel diuretics are being developed targeting urea channels, ROMK, or pendrin. A urea channel inhibitor UT-B (SLC14A1) induced osmotic diuresis in rats by inhibiting urea reabsorption in the renal medulla²⁴⁵ without commensurate urinary sodium and potassium loss.²⁴⁶ Dimethylthiourea inhibits the urea channels UT-B and UT-A1. It produced a greater diuresis in rats than furosemide while maintaining S_{Na}. It prevented hyponatremia in a model of SIADH.²⁴⁷ ROMK inhibitors induce a robust natriuresis in rats that exceeds that of a maximal dose of furosemide but does not cause kaliuresis. A ROMK inhibitor caused greater diuresis and natriuresis when combined with hydrochlorothiazide or amiloride but not bumetanide²⁴⁸ and blocks the TGF response,²⁴⁹ confirming that the diuretic target is predominantly the TAL and MD. Pendrin is a Cl⁻/HCO₃⁻ exchanger expressed in the luminal membrane of intercalated cells involved in regulation of Na⁺ reabsorption and acid-base balance.²⁵⁰ Small-molecule inhibitors of pendrin are not effective diuretics when given alone²⁵¹ but enhance furosemide diuresis by 30% to 60%.²⁵¹ Thus these drugs could become valuable to address loop diuretic resistance.²⁵²

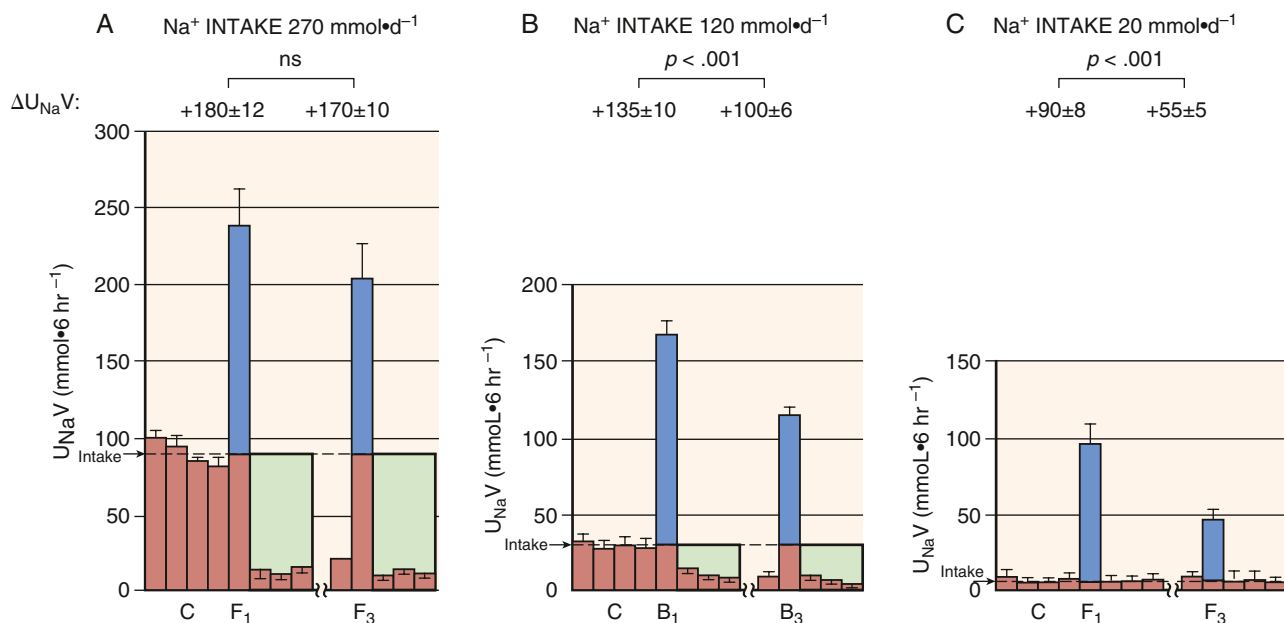


Fig. 49.12 Effects of dietary salt intake on the diuretic braking phenomenon. Renal Na⁺ excretion (mmol/6 hours) for 24 hours before and after the first (F₁) and third (F₃) daily doses of furosemide (A and C; 40 mg intravenously) and (B) bumetanide (B₁, B₃, 1 mg intravenously) in groups of 8 to 10 normal subjects equilibrated to fixed daily Na⁺ intakes. The average level of Na⁺ intake (mmol/6 hours) is shown by *broken horizontal lines*. Negative Na⁺ balance is indicated by *blue bars* and positive Na⁺ balance by *light green areas*. The mean $\pm$ standard error of the mean (SEM) values for diuretic-induced increases in Na⁺ excretion above baseline values ($\Delta U_{Na}V$) for 6 hours after the administration of the diuretic are shown at the *top*. (Modified from Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med*. 1983;102:450–458.)

ADAPTATION TO DIURETIC THERAPY

Diuretics entrain a set of homeostatic mechanisms that limit their fluid-depleting actions and contribute both to resistance to their adverse effects.

DIURETIC BRAKING PHENOMENON

The first dose of a diuretic normally produces a reassuring diuresis. However, a new equilibrium is rapidly attained when body weight stabilizes and daily fluid and electrolyte excretion no longer exceeds intake¹⁰⁸ termed the “diuretic braking phenomenon.” The effects of dietary salt intake on the diuretic braking phenomenon during 3 days of loop diuretic administration to normal human subjects are shown in Fig. 49.12.^{108,253–255} During high Na⁺ intake (270 mmol/24 hours), the first dose of furosemide (F₁) causes a large negative Na⁺ balance over the ensuing 6 hours (*blue bars* in Fig. 49.12A), followed by 18 hours during which Na⁺ excretion is reduced well below intake (postdiuresis salt retention), which results in a positive Na⁺ balance (*light green areas* in Fig. 49.12A) that offsets the preceding negative Na⁺ balance. The natriuresis caused by the third daily dose of furosemide (F₃) is comparable with that caused by the first dose and is also followed by a restoration of Na⁺ balance. Consequently, at high levels of Na⁺ intake, subjects regain neutral Na⁺ balance within 24 hours of each dose of furosemide and maintain their original body weight. A similar diuretic braking phenomenon occurs in normal subjects with torsemide⁸³ and during established furosemide

therapy.⁸⁴ During severe dietary Na⁺ restriction (20 mmol/24 hours; Fig. 49.12C), the first dose of furosemide produces a blunted natriuresis but Na⁺ balance cannot be restored because of the low level of dietary Na⁺ intake. Consequently, virtually all the Na⁺ lost during the diuretic phase is represented as a negative Na⁺ balance for the day. A low-salt diet induces tolerance manifest as a 40% reduction in the natriuretic response to the drug over 3 days. However, despite a blunted initial response and the development of tolerance, all subjects lose Na⁺ and body weight. A loop diuretic given during an Na⁺ intake of 120 mmol/24 hours (equivalent to a salt-restricted diet) causes Na⁺ loss, but the loss is curtailed by a combination of postdiuretic renal salt retention and diuretic tolerance (see Fig. 49.12B).²⁵⁵ Furosemide kinetics and the GFR are unchanged over 3 days of furosemide administration, but during a low NaCl intake, the natriuresis related to furosemide excretion is reduced by the third day of diuretic administration (see Fig. 49.8), indicating a blunting of diuretic responsiveness.

Clinical Relevance

Diuretic Braking Phenomenon

The diuretic braking phenomenon refers to the observation that repeated doses of diuretics fail to produce a negative sodium balance. It is explained by decreased diuretic responsiveness combined with postdiuretic renal salt retention and enhanced distal reabsorption. Consequently, dietary sodium restriction to about 120 mmol or 3 g of Na⁺ daily or the addition of a second diuretic can overcome the diuretic braking phenomenon.

One month of furosemide therapy for hypertension reduces the natriuretic response to a test dose of furosemide by 18%.⁸⁴ This tolerance cannot be ascribed to aldosterone nor to a fall in plasma or ECV because tolerance to furosemide is not prevented by spironolactone and does not develop during thiazide therapy, which causes similar reductions in body fluids. In fact, the natriuretic response to a test dose of a thiazide is augmented during furosemide therapy. Thus tolerance to furosemide is class specific and depends on increased NaCl reabsorption at a downstream, thiazide-sensitive nephron site.

Furosemide activates the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system (SNS). However, postdiuretic Na⁺ retention in normal subjects is not blunted by preventing RH increases in plasma angiotensin II or aldosterone by an ACE inhibitor^{253,256,257} or by blocking adrenergic receptors with prazosin even when an ACE inhibitor and prazosin are given in combination.²⁵⁴

Micropuncture studies have shown that the blunted natriuretic response to furosemide during repeated administration on a restricted salt diet can be attributed to three factors: 1. reduced NaCl delivery to the site of furosemide action; 2. limited inhibition of NaCl reabsorption by furosemide in the loop of Henle; and 3. enhanced ability of the distal tubule to reabsorb the extra NaCl load delivered during furosemide's upstream action.²⁵⁸ Renal tubular resistance is the primary driver of loop diuretic resistance in ADHF.²⁵⁹

Rats receiving prolonged infusions of loop diuretics have considerable structural hypertrophy of the DCT, connecting tubule, and α intercalated cells of the CD²⁶⁰ that is partially dependent on angiotensin II.²⁶¹ This is accompanied by increases in the mRNA expression for insulin-like growth factor-binding protein-1²⁶², Na⁺-K⁺-ATPase²⁶³, H⁺-ATPase,²⁶⁴ and NCC.²⁶¹ The Na⁺-K⁺-ATPase activity of rat cortical CD segments increases abruptly following an increase in cellular (Na⁺), owing to the mobilization of a latent pool of enzymes.²⁶⁵ Micropuncture studies of rats adapted to prolonged diuretic infusion have shown enhanced, aldosterone-independent distal Na⁺ and Cl⁻ absorption and K⁺ secretion.²⁶⁶ Therefore loop diuretics induce considerable structural and functional adaptations of downstream nephron segments, apparently in response to increased rates of NaCl delivery and, to some extent, to RAAS activation. Nephronal adaptation could underlie the inappropriate renal Na⁺ retention that can persist for up to 2 weeks after abrupt cessation of diuretic therapy.²⁶⁷

Normal subjects fully eliminate a modest NaCl load (100 mmol) over 2 days.²⁶⁸ However, when these subjects are challenged with the same NaCl load delivered after the administration of 1 mg of intravenous bumetanide given with infusion of sufficient fluid, Na⁺, and Cl⁻ to prevent any losses, elimination of the load is prevented.²⁶⁸ Thus diuretics can entrain an ECV-independent NaCl retention. Even a single dose of loop diuretic can cause a Cl⁻ depletion "contraction" alkalosis, which may contribute to diuretic tolerance and the braking phenomenon.⁷¹ In a study of normal subjects in whom mild metabolic alkalosis was produced by the equimolar substitution of NaHCO₃ for NaCl, bumetanide-induced natriuresis was reduced despite enhanced delivery of bumetanide to the urine.¹²⁶ This finding implies a profound defect in tubular responsiveness

Box 49.1 Strategies to Overcome Diuretic Braking

1. Restrict dietary salt to approximately 120 to 140 mmol daily to prevent postdiuretic salt retention.
2. Consider adding another class of diuretic.
3. Consider multiple daily dosing or a diuretic with prolonged action.
4. Do not stop diuretic therapy abruptly.
5. Prevent or reverse diuretic-induced metabolic alkalosis and hypochloremia.

to the diuretic. Two of the mechanisms that may contribute are:

1. The Na-K-2Cl cotransporter has affinities for Na⁺, K⁺, and Cl⁻ of 7.0, 1.3, and 67 mM, respectively.²⁶⁹ The (Cl⁻) of tubular fluid may be sufficiently low during Cl⁻ depletion alkalosis to limit reabsorption by this transporter and thereby limit the responsiveness to loop diuretics.
2. Alkalosis increases the expression of the thiazide-sensitive NCC in the rat DCT by 40%.²⁷⁰

Findings from these studies have several clinical implications (Box 49.1):

1. Excessive dietary salt intake must be restricted, even in subjects receiving powerful loop diuretics, to obviate postdiuretic salt retention and ensure the development of a negative NaCl balance.
2. Subjects may be particularly responsive to another class of diuretic during prolonged diuretic administration.
3. Diuretic therapy should not be stopped abruptly unless dietary salt intake is curtailed because the adaptive mechanisms limiting salt excretion persist for days after diuretic discontinuation.
4. Selection of a diuretic with a prolonged action, or more frequent administration of the diuretic, will enhance NaCl loss by limiting the time available for postdiuretic salt retention. Indeed, a continuous infusion of a loop diuretic is somewhat more effective than the same dose given as a bolus injection in volunteers²⁷¹ and in patients with CKD,²⁷² despite a similar delivery of diuretic to the urine. However, there are no improvements in outcomes.
5. Prevention or reversal of diuretic-induced hypochloremic metabolic alkalosis may enhance diuretic efficacy.

There are similar patterns of furosemide-induced K⁺ loss followed by renal K⁺ retention.^{273,274} In contrast, loop diuretics induce day long renal K⁺ losses from hyperaldosteronism during severe salt restriction that can be countered by distal, K⁺-sparing diuretics.²⁷³

HUMORAL AND NEURONAL MODULATORS OF THE RESPONSE TO DIURETICS

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Diuretic therapy increases plasma renin activity (PRA) and serum aldosterone concentration (SAC). The initial rise in PRA with loop diuretics is independent of volume depletion or the SNS and is related to the inhibition of NaCl

reabsorption at the macula densa,²⁷⁵ perhaps following prostacyclin and prostaglandin E₂ (PGE₂) release.²⁷⁶ During longer-term use of loop diuretics, the increases in PRA are augmented by ECV depletion, a fall in blood pressure and activation of the SNS.

Activation of the RAAS in patients treated with diuretics and salt restriction for edema can limit the natriuresis.²⁷⁷ ACE inhibition potentiates the diuretic and natriuretic responses to furosemide in HF, despite a drop in BP.²⁷⁸ However, severe volume depletion and azotemia can complicate overzealous combined therapy with ACE inhibitors and high doses of diuretics or in those with stenosis of both renal arteries or the artery to a single or dominant kidney.²⁷⁹ Thus the combination of diuretics and ACE inhibitors can be effective but requires careful surveillance.

Diuretic-induced increases in SAC promote renal K⁺ losses²⁸⁰ that can be countered by ACE inhibitors and MRAs.²⁸¹

EICOSANOIDS

PGE₂ acting on EP₄Rs inhibits NaCl reabsorption via NKCC2²⁸² and free water and Na⁺ reabsorption in the CDs via changes in cAMP.^{283,284} Loop diuretics, thiazides, triamterene, and spironolactone increase renal prostaglandin production substantially.²⁸⁵ Inhibition of PG synthesis by NSAIDs can diminish the natriuresis and diuresis induced by furosemide,²⁸⁶ hydrochlorothiazide,²⁸⁷ spironolactone,²⁸⁵ or triamterene (see Fig. 49.12).²⁸⁸ Microperfusion of the loop segment with PGE₂¹³² restores the response to furosemide in indomethacin-treated rats. Indomethacin also blunts furosemide-induced renal²⁸⁹ and capacitance vessel vasodilation²⁹⁰ and renin release.²⁹¹ NSAIDs blunt furosemide-induced natriuresis while a COX-2 inhibitor blocks furosemide-induced renin secretion.²⁹² Natriuresis is facilitated by COX1 in the distal nephron.²⁹³ Thus COX-1 products mediate a part of furosemide-induced natriuresis, whereas COX-2 products mediate much of the renin secretion.

ARGININE VASOPRESSIN

Arginine vasopressin (AVP) increases after the administration of furosemide in response to volume depletion.²⁹⁴ Plasma AVP levels are increased in many edematous states, such as HF and liver cirrhosis,²⁹⁵ but are decreased in patients who develop hyponatremia during thiazide treatment.²⁹⁶ AVP stimulates K⁺ secretion in the rat distal tubule.²⁹⁷ Diuretic-induced AVP release contributes to hypokalemia because the kaliuretic response to furosemide is reduced by 40% in subjects whose AVP release is suppressed by a water load.²⁸⁰ AVP can increase the activity of both NCC and ENaC, thereby possibly aggravating sodium retention in edematous states.^{298,299}

CATECHOLAMINES AND SYMPATHETIC NERVOUS SYSTEM

While a dose of furosemide raises the heart rate and plasma catecholamine concentrations,^{126,254} blockade of α_1 -adrenergic receptors with prazosin does not modify the ensuing renal salt retention.²⁵⁴ However, blockade of B1 adrenoreceptors does blunt renin release.²⁷⁵ Short-term, furosemide-induced ECV depletion in the conscious rat activates the SNS that stabilizes the BP.³⁰⁰

Box 49.2 Common Causes of Diuretic Resistance

- Incorrect diagnosis (e.g., venous or lymphatic edema)
Inappropriate NaCl or fluid intake
Inadequate drug reaching tubule lumen in active form because of:
- Noncompliance
 - Dose inadequate or too infrequent
 - Poor absorption (e.g., due to uncompensated HF)
 - Decreased renal blood flow (e.g., due to HF or cirrhosis of liver, or in older patients)
 - Decreased functional renal mass (e.g., due to AKI or CKD, or in older patients)
 - Proteinuria (e.g., due to nephrotic syndrome)
- Inadequate renal response because of:
- Low glomerular filtration rate (e.g., due to AKI, CKD)
 - Decreased effective extracellular fluid volume (e.g., due to edematous conditions)
 - Activation of renin-angiotensin-aldosterone system (e.g., due to edematous conditions)
 - Nephron adaptation (e.g., due to prolonged diuretic therapy)
 - Nonsteroidal antiinflammatory drugs (e.g., indomethacin and aspirin)

AKI, Acute renal failure; CKD, chronic kidney disease; HF, heart failure.

Atrial natriuretic peptide

Administration of furosemide to dogs with HF reduces plasma levels of ANP, and their restoration³⁰¹ with an infusion of ANP restores furosemide-induced natriuresis and blunts the increase in PRA and fall in GFR. Thus a drop in ANP levels contributes to postdiuretic renal NaCl retention and adverse renal hemodynamic changes.³⁰¹

DIURETIC RESISTANCE

Diuretic resistance implies an inadequate clearance of edema or pulmonary congestion because of an inadequate natriuresis, despite a full dose of diuretic.³⁰² The principal causes are summarized in Box 49.2. The first step is to select the appropriate target response (e.g., a specific body weight, volume of diuresis, or degree of natriuresis) and ensure the edema is not due to lymphatic or venous obstruction or redistribution (Fig. 49.13) or to the use of dihydropyridine calcium channel blockers.³⁰³ The next step is to exclude poor adherence, severe blood volume depletion, or concurrent nonsteroidal antiinflammatory drug use. Thereafter, dietary NaCl intake can be quantitated from measurements of 24-hour Na⁺ excretion. For most patients, a daily Na⁺ intake of 80 to 120 mmol is a reasonable goal. Urine volume and body weight can be measured directly in patients who are hospitalized. If a diuresis does not occur, the next step is to double the dose until an effective dose or the maximum safe dose is reached.

The next step is to give two daily doses of the diuretic. Furosemide, bumetanide, and torsemide act for only 3 to 6 hours. Two daily half-doses, by interrupting postdiuretic salt retention, produce a greater response than the same total dose given once daily. Severe HF may impair absorption of the diuretic. Thus a more bioavailable diuretic, such as torsemide, may be preferable to furosemide.¹⁰² Diuretic

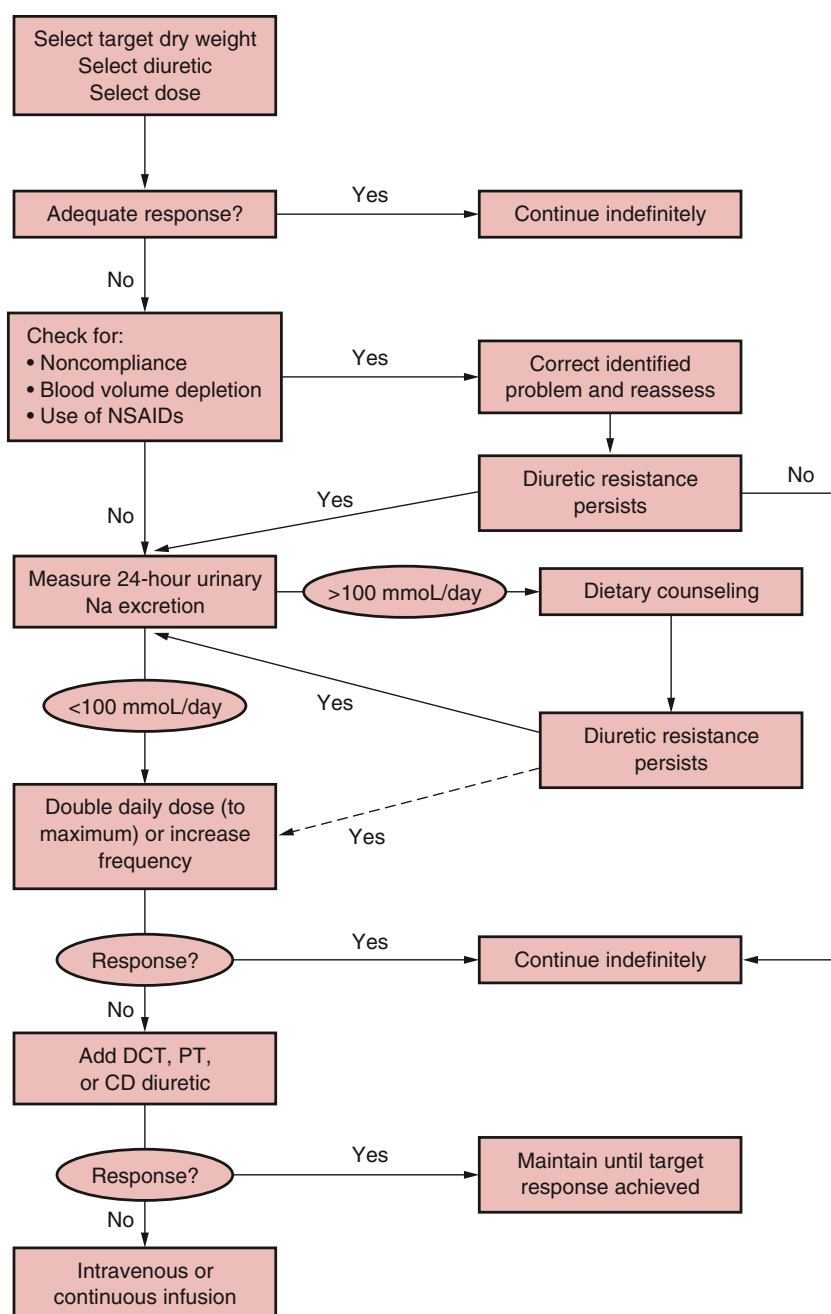


Fig. 49.13 Diagrammatic representation of an approach to the management of a patient with resistance to a loop diuretic. CD, Collecting duct diuretic (e.g., amiloride, triamterene, or spironolactone); DCT, distal convoluted tubule diuretic (e.g., thiazide); NSAIDs, nonsteroidal antiinflammatory drugs; PT, proximal tubule diuretic (e.g., acetazolamide). (From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. ed 2. London: Elsevier Science; 2003.)

resistance is often accompanied by a pronounced hypochloremic, hypokalemic metabolic alkalosis,¹²⁶ which may be reversed by the administration of KCl or addition of a distal K⁺-sparing diuretic or MRA.

An alternative approach in hospitalized patients is to assess the natriuretic response directly. Thus the natriuretic response (in mmol per 6 hours after loop diuretic administration) of >50 mmol has been classified as poor or severe diuretic resistance; 50 to 100 mmol as suboptimal or partial resistance; and >150 mmol as excellent or no resistance.³⁰⁴ This response can be predicted accurately without the need for a 6-hour urine collection from the

natriuretic response prediction equation.³⁰⁵ This equation uses a plasma sample and a single spot urine 1 to 2 hours after an intravenous dose of bumetanide for measurements of urinary and serum creatinine (U_{Cr} and S_{Cr}) and urine sodium (U_{Na}):

$$\text{Na} + \text{output (mmol)} = \text{eGFR} \times \left(\frac{\text{BSA}}{1.73} \right) \times \left(\frac{S_{Cr}}{U_{Cr}} \right) \times 60 \text{ min} \times 2.5 \text{ hours} \times (U_{Na}/1000 \text{ mL})$$

Alternatively, a urinary (Na^+) >83 mmol/L 2 hours after a test dose of furosemide (1 mg/kg for diuretic-naïve patients

and 1.5 mg/kg for diuretic-treated patients) can be used to predict a clinical diuretic response in hospital diuretic therapy.³⁰⁶

A progressive increase in diuretic dosage may produce an inadequate reduction in body fluids because of activation of NaCl-retaining mechanisms. RAAS blockade can sometimes restore a diuresis in resistant patients with HF,²⁷⁷ but a fall in BP often limits the response. Adaptive changes in downstream nephron segments during prolonged diuretic therapy^{84,260} provide a rational basis for combining diuretics (see the following section). Highly resistant patients can be admitted for a trial of intravenous infusion of a loop diuretic or ultrafiltration.³⁰⁷

DIURETIC COMBINATIONS

The use of several diuretics acting on separate mechanisms may be synergistic.^{5,308}

LOOP DIURETICS AND THIAZIDES

A loop diuretic and a thiazide or thiazide-like drug (e.g., hydrochlorothiazide and metolazone) are synergistic in normal subjects and subjects with edema or renal insufficiency.^{308–312} Metolazone is equivalent to bendroflumazide in enhancing NaCl and fluid losses in furosemide-resistant subjects with HF or nephrotic syndrome but is available for intravenous use.³¹³ Chlorothiazide is another intravenous option in decompensated HF. Prolonged furosemide therapy augments the response to a thiazide.^{84,314} Even patients with advanced CKD (eGFR <30 mL/min/1.73 m²) who are unresponsive to a loop diuretic can have a marked natriuresis when a thiazide is added to loop diuretic therapy,³¹¹ but close surveillance is required because of a high associated incidence of hypokalemia, excessive ECV depletion, and azotemia.³¹⁵

LOOP DIURETICS OR THIAZIDES AND DISTAL POTASSIUM-SPARING DIURETICS

Amiloride or triamterene increase furosemide natriuresis only modestly but curtail the excretion of K⁺ and net acid⁶⁹ and preserve total body K⁺.³¹⁶ Distal K⁺-sparing agents are generally contraindicated in advanced renal failure because they may cause severe hyperkalemia and acidosis.

CLINICAL USES OF DIURETICS

EDEMATOUS CONDITIONS

The first aim in treatment of edema is to reverse the primary cause by restoring hemodynamics and cardiac output in patients with HF (e.g., use of vasodilators or elimination of cardiac depressant drugs), improving hepatic function in patients with cirrhosis and ascites (e.g., discontinuing alcohol intake), or diminishing proteinuria in patients with nephrotic syndrome (e.g., administration of ACE inhibitors or ARBs). Although the GFR is reduced over the first day of low-dose diuretics in normal subjects, the GFR and RBF return to baseline over 1 month.¹²⁶ However, renal hemodynamics may be impaired by loop diuretics in those with CKD or those who experience an abrupt fall in BP, especially if this is complicated by orthostatic hypotension.

Moreover, overzealous diuresis decreases the cardiac output, BP, and renal function and stimulates the RAAS, SNS, prostaglandins, and AVP, all of which may compromise the desired hemodynamic and renal responses.³¹⁷ Therefore diuretic therapy for edema should be initiated with the lowest effective dose.³¹⁸ Additional drugs can be used to counteract unwanted actions. For example, ACE inhibitors, ARBs, or MRAs can prevent RAAS activation and enhance fluid losses yet diminish K⁺ depletion (see Fig. 49.19). The use of a second diuretic can have a synergistic action, whereas the use of a distal K⁺-sparing agent may counteract unwanted hypokalemia, alkalosis, or Mg²⁺ depletion (see “Adaptation to Diuretic Therapy” earlier).

Dietary Na⁺ intake should be restricted to <3 g daily (corresponding to 80–120 mmol/24 hours). A meta-analysis of patients with HF concluded that more severe dietary salt restriction for patients with HF may even increase the rate of readmission in the long term. This may be a tradeoff between the benefit of reducing dietary salt to restrict postdiuretic sodium retention and the adverse effect of more extreme neurohumoral activation. Some resistance to diuretic therapy should be anticipated in all patients with CKD and those with more than mild edema (Fig. 49.14).

HEART FAILURE

HF can occur with reduced or preserved ejection fraction (HFrEF or HFpEF), yet both forms require diuretics as cornerstones of therapy. The approach to cardiac failure depends on the cause and whether there is acute decompensation or a compensated chronic state.^{319,320} This section first reviews the role of diuretics in acute decompensated HF (ADHF) and HF secondary to acute coronary syndrome and thereafter discusses the role of diuretics as maintenance therapy in stable chronic HF. The reader is also referred to a review by Ellison and Felker³²¹ and the joint guidelines published by the American College of Cardiology and American Heart Association for more detailed recommendations.³²²

Acute decompensated heart failure

In the absence of obvious causes such as acute coronary syndrome and valve abnormalities, ADHF often results from an imbalance in the neurohumoral systems that regulate cardiac and renal function.³²³ For example, decompensated HF stimulates the RAAS and AVP,³¹⁷ predisposing to many deleterious effects including hypokalemia, hypomagnesemia, hyponatremia, and arrhythmias. Therefore it is rational to target these systems with selective therapy. After initial stabilization, the mainstays of treatment are vasodilator and diuretic therapy. Intravenous vasodilators such as nitroglycerine or nitroprusside counteract the effects of baroreceptor-dependent increases in sympathetic tone, angiotensin II and aldosterone, endothelin, and AVP. Nesiritide is a recombinant B-type natriuretic peptide that failed to impact death or HF hospitalizations in the Acute Study of Clinical Effectiveness of Nesiritide in Decompensated Heart Failure (ASCEND-HF)³²⁴ trial and is now largely discontinued.³²⁵ Vasodilators are usually combined with intravenous loop diuretics (e.g., furosemide, 40 mg; bumetanide, 1 mg; torsemide, 10–20 mg).³²⁶ Although “aggressive decongestion” during ADHF can worsen renal function, this may improve survival.³²⁷

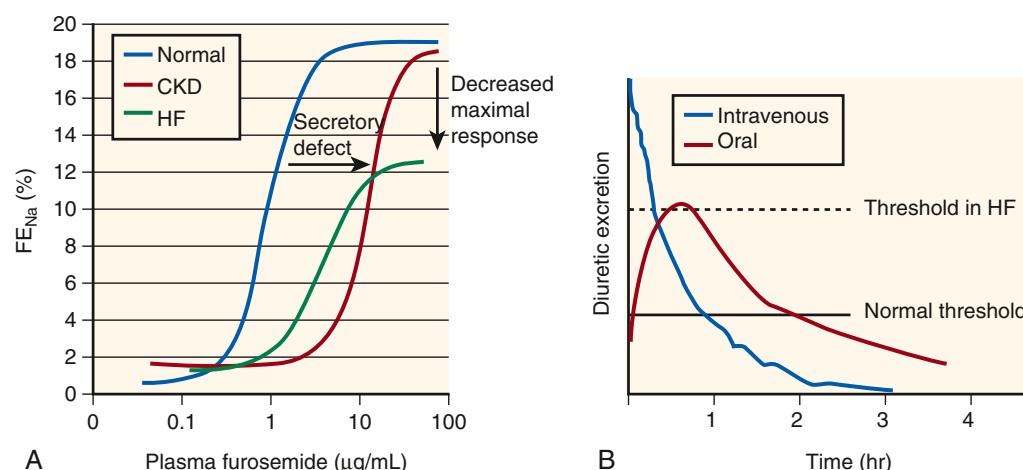


Fig. 49.14 Dose-response curves for loop diuretics. (A) The fractional Na⁺ excretion (FE_{Na}) as a function of plasma loop diuretic concentration. Compared with normal subjects, patients with chronic kidney disease (CKD) show a rightward shift in the curve owing to impaired diuretic secretion. The maximal response is preserved when expressed as FE_{Na}, but not when expressed as absolute Na⁺ excretion. Patients with congestive heart failure (HF) demonstrate a rightward and downward shift, even when expressed as FE_{Na}, and thus are relatively diuretic-resistant. (B) Comparison of the response to intravenous and oral doses of loop diuretics. In a normal individual, an oral dose may be as effective as an intravenous dose because the time during which this individual is above the natriuretic threshold (indicated by the *Normal threshold* line) is approximately equal. If the natriuretic threshold increases (as indicated by the *Threshold in HF* line), then the oral dose may not provide a high enough plasma level to elicit natriuresis.

In a carefully performed randomized trial, no differences between continuous dosing and bolus dosing of furosemide were observed in patients with ADHF.³²⁸ Furthermore, no differences in global symptom improvement were observed between a low dose (equivalent to an oral dose) and a high dose (2.5 times the oral dose).³²⁸ It should be noted, however, that this trial did not study patients shown to be diuretic-resistant and the low dose of furosemide averaged 117 mg daily. Longer-acting loop diuretics, such as torsemide³²⁹ and azosemide,³³⁰ and especially extended-release torsemide³³¹ may be preferable.

Diuretic kinetics are impaired in ADHF.^{332,333} The bioavailability of furosemide, unlike that of bumetanide or torsemide, is erratic in HF.³³⁴ However, The Effect of Torsemide vs. Furosemide After Discharge in All-Cause Mortality in Patients Hospitalized with Heart Failure (TRANSFORM-HF) trial detected no difference in 1-year mortality in patients with ADHF randomized to furosemide or torsemide.¹⁰⁷ There is decreased plasma diuretic clearance in decompensated HF because of a decreased RBF³³² that can diminish the response.

The value of venovenous ultrafiltration as treatment for ADHF remains unresolved.³³⁵ A randomized trial comparing diuretic therapy with ultrafiltration reported that plasma creatinine increases more in the ultrafiltration group despite similar volume losses.³³⁶ However, a meta-analysis concluded that ultrafiltration results in better clinical decongestion than intravenous loop diuretics but does not improve rates of rehospitalization or mortality³³⁷ and may cause serious product-related adverse events.³³⁸

In addition, some trials have demonstrated the efficacy of adding acetazolamide or thiazide diuretics to intravenous furosemide in the treatment of ADHF. Addition of either medication resulted in increased diuresis and

natriuresis.^{37,339} Finally, a trial has shown no added benefit of low-dose dopamine or nesiritide on decongestion or kidney function.³⁴⁰

Acute decompensated heart failure in acute coronary syndrome

Patients with acute myocardial infarction (AMI) require the rapid establishment of coronary reperfusion (e.g., by thrombolysis and percutaneous coronary intervention) and treatment of arrhythmias. The aim of concomitant treatment is to counter the increase in left ventricular end-diastolic pressure that enhances wall tension and O₂ usage, as well as the accumulation of pulmonary edema without further curtailing the cardiac output. Judicious use of diuretics may meet these requirements. In one study, intravenous furosemide for left ventricular failure (LVF) complicating acute AMI reduced the left ventricular filling pressure from 20 to 15 mm Hg within 5 to 15 minutes and increased the venous capacitance by 50%.³⁴¹ This rapid venodilation is blocked by NSAIDs²⁹⁰ and ACE inhibitors.³⁴² The ensuing diuresis reduces left ventricular end-diastolic pressure further.

A study of first-line therapy for 48 patients with acute LVF following AMI compared the responses to intravenous furosemide with a venodilator (isosorbide dinitrate), an arteriolar dilator (hydralazine), or a positive inotrope (prenalterol).³⁴³ The venodilator and furosemide both reduced left ventricular filling pressure while maintaining the cardiac index and heart rate. The investigators concluded that these were the best first-line agents but that they should be combined with an arteriolar vasodilator. In contrast, a study randomly assigned 110 patients with acute LVF to receive either high-dose isosorbide dinitrate and low-dose furosemide (3 mg isosorbide intravenously every 5 minutes) or high-dose furosemide and low-dose isosorbide

dinatriate (80 mg furosemide intravenously every 15 minutes).³⁴⁴ An adverse end point occurred more frequently in those receiving high-dose furosemide (46%) than in those receiving high-dose nitrate (25%). The investigators cautioned against the use of high-dose furosemide in acute LVF. Although intravenous furosemide decreases left ventricular filling pressure in patients with LVF, the shape of the Frank-Starling ventricular function curve predicts little change in cardiac output at elevated filling pressures. Moreover, the accompanying neurohumoral activation can enhance afterload and cardiac oxygen requirement. Nevertheless, most investigators recommend a trial of loop diuretics after ruling out ECV depletion by targeted volume boluses.³⁴⁵

Furosemide can be given as an intravenous bolus of up to 100 mg or as a short-term infusion to limit the risk of ototoxicity. Although controversial, a continuous infusion produces a slightly greater natriuresis in most comparative studies.³⁴⁶ Ideally, lower doses are used initially and titrated up to achieve the desired effect (e.g., a pulmonary capillary wedge pressure of 16 mm Hg).³⁴⁶

Chronic heart failure

Diuretics are used widely in the long-term management of chronic HFpEF. Avid renal NaCl and fluid retention or redistribution of fluid from the capacitance vessels expands the blood volume that leads to pulmonary edema limiting ventilation and cardiac dilation limiting cardiac function and causing increased wall tension and O₂ usage. This combination can create a spiral of decreasing oxygenation and cardiac output. A study that combined a diuretic and vasodilator reported reduced left and right atrial volumes, corrected atrioventricular valvular regurgitation, and improved stroke volume by 64%.³⁴³ A meta-analysis of trials for HF has concluded that odds ratios were reduced to 0.25 for mortality and 0.31 for hospitalization for subjects randomly selected to receive diuretics.³⁴⁷ These data strongly support the use of diuretics in HF.

On the other hand, the failing heart has a decreased capacity to regulate its contractility in response to changes in venous return. Consequently, severe diuretic therapy can decrease effective blood volume leading to orthostatic hypotension, weakness, fatigue, decreased exercise ability, and prerenal azotemia. This is especially true for patients with HFpEF and diastolic dysfunction.³⁴⁸ Therefore salt-depleting therapy requires continual reassessment and judicious use of other measures (e.g., vasodilators, ACE inhibitors, ARBs, MRAs, and SGLT2is). Mild HF often responds to dietary Na⁺ restriction (140 mmol/day) and low doses of a thiazide diuretic, but as cardiac failure progresses, larger, more frequent doses of loop diuretics are required. It is important to emphasize that diuretic responsiveness is impaired in patients with advanced HF, as shown by a shift to the right in the natriuresis-excretion relationship of diuretics (see Fig. 49.14).³³³ For the refractory patient, the addition of a second diuretic acting at the proximal tubule (e.g., acetazolamide or an SGLT2i) or a downstream site (e.g., a thiazide) can produce a dramatic diuresis, even in individuals with impaired renal function.^{349,350} By comparing the fractional lithium and sodium excretion after intravenous loop diuretics, one study confirmed that

distal tubular compensatory sodium reabsorption is the primary driver of diuretic resistance in HF.³⁵¹

Drug therapy for chronic HF should be individualized.³⁵² MRAs should be used within guidelines²¹¹ to prevent hyperkalemia, but if encountered it can often be managed with increased doses of loop diuretics or a K⁺ binder such as patiromer.³⁵³ MRAs can improve ventricular remodeling,^{354,355} and in patients with HFpEF, spironolactone may improve left ventricular diastolic dysfunction.^{356,357}

Despite these options, HF often progresses and introduces a vicious cycle involving other organ systems such as the liver and kidney. Therefore new therapies are required because a decrement in renal function predicts a bad outcome in patients treated for HF.³⁵⁸ The addition of an SGLT2i can reduce recurrent episodes of ADHF, enhance diuresis significantly with loop diuretics, and greatly improve cardiovascular outcomes.³⁵⁹ Renal dysfunction can be ameliorated by an ARB, provided that BP is maintained. The combination of an ARB with a neprilysin inhibitor, which prevents the breakdown of natriuretic peptides, improves cardiovascular and renal outcomes.^{227,360} In the scenario in which deteriorating HF worsens kidney function, often referred to as the “cardiorenal syndrome,” therapy must be aimed at improving cardiac function (if possible) and treating congestion with loop diuretics, MRAs, or SGLT2is.³⁶¹

Right ventricular failure

The requirement for diuretic therapy in patients with pure right HF or cor pulmonale is not compelling. A decrease in venous return induced by vigorous diuresis may worsen right heart function. Furosemide administration increases angiotensin II-induced hypoxic pulmonary vascular resistance.³⁶² Therefore the emphasis should be on the reversal of chronic hypoxemia.

CIRRHOSIS OF THE LIVER

Most patients with cirrhotic ascites and peripheral edema have expansion of the ECV owing to arteriolar underfilling and diminished effective arterial blood volume from peripheral vasodilation.^{363–366} This can increase proximal reabsorption,^{363–366} which limits the response to loop diuretics and thiazides.³⁶⁷ However, the RAAS is activated in cirrhosis and contributes to salt retention, portal hypertension, and likely to progressive liver fibrosis.³⁶⁶ Thus the use of MRAs, often in high concentration, are rational for cirrhosis and are usually well tolerated.¹⁸¹ The American Association for the Study of Liver Disease practice guidelines have suggested that first-line treatment of patients with cirrhosis and ascites should consist of sodium restriction (2 g or 88 mmol daily) and diuretics (oral spironolactone, with or without oral furosemide).^{368–370} This can be initiated with 100 mg spironolactone and 40 mg furosemide, with titration upward maintaining the same diuretic ratio^{368,369} to maximally recommended doses of 400 mg of spironolactone and 160 mg of furosemide.

Patients with cirrhosis and ascites usually cannot tolerate ACE inhibitors or ARBs because of a fall in BP.³⁷¹ While sodium restriction is indicated in cirrhotic ascites, fluid restriction is only required for developing hyponatremia.^{368,372} Ascitic fluid is largely cleared by the lymphatics. Diuretics reduce thoracic duct lymph flow,³⁷³ perhaps

because of reduced lymph production. Furosemide, but not ethacrynic acid, inhibits lymphatic contractions directly by blockade of NKCC1.³⁷⁴ Thus diuretics decrease ascites formation by decreasing venous and portal hydraulic pressures and concentrating the plasma proteins.^{373,376,385}

The maximal daily ascites drainage into the systemic circulation is limited to 300 to 900 mL.³⁷⁷ Therefore the maximum daily weight loss in nonedematous patients should not exceed 0.3 to 0.5 kg. However, in patients with ascites and edema, daily diuretic-induced weight losses of 1 to 3 kg do not perturb the plasma volume or renal function.³⁷⁸ In contrast, the same diuretic regimen maintained after the peripheral edema has cleared or been given to nonedematous patients reduces plasma volume by as much as 24% and raises the risks of hyponatremia, alkalosis, and azotemia. Furthermore, the reduced serum albumin levels and RH increased portal venous pressure, coupled with preexisting diuretic use, can lead to true “underfill edema.” Thus a diuretic prescription that is initially safe and effective must be reviewed continuously. Patients with ascites but without peripheral edema are more prone to the development of adverse effects of diuretics.³⁷⁸

The most common problems with furosemide in cirrhosis are electrolyte disturbances and volume depletion. Hypokalemia, which is related to preexisting K⁺ depletion and hyperaldosteronism, can be countered with spironolactone, eplerenone, or a distal K⁺-sparing agent. More severe diuretic resistance requires paracentesis. Controlled trials in patients with refractory ascites have shown that large-volume paracentesis is more effective than diuretic therapy in reducing hospital stay and electrolyte complications but does not influence mortality.³⁷⁹ Even repeated, large-volume paracenteses (4–6 L/day) are safe if intravenous albumin (40 g with each procedure) is administered.³⁷⁹ Most investigators, however, recommend paracentesis only for cases that are relatively resistant to diuretics and dietary Na⁺ restriction (Fig. 49.15).³⁸⁰

Patients with mild cirrhosis of the liver have a somewhat reduced natriuretic response to furosemide, with little change in diuretic kinetics.² However, in those with advanced disease, furosemide absorption is slowed³⁸¹ and its volume of distribution is increased because of hypoalbuminemia and an expanded ECV. Additionally, its elimination is delayed because of hypoalbuminemia, which limits proximal tubule diuretic secretion, and a low RBF, which limits renal clearance.² However, resistance to loop diuretics in early cirrhosis is largely due to decreased responsiveness to the drug, which correlates with elevated serum aldosterone levels.¹⁸¹ With the development of ascites, a further decrease in natriuretic response correlates with decreased delivery of furosemide to the urine³⁸² and with further stimulation of the RAAS.¹⁸¹

Diuretic resistance is common in advanced cirrhosis. In addition to the usual causes (see Box 49.2), it may herald the development of infection, bleeding, or a critical drop in cardiac output. Patients whose disease is refractory and who are disabled by recurrent paracentesis may show response to body compression³⁸³ or a transjugular intrahepatic portosystemic shunt.³⁸⁴ Intravenous loop diuretics are generally discouraged because they may precipitate hepatorenal syndrome.

NEPHROTIC SYNDROME

Urinary albumin losses and reduced hepatic albumin synthesis in the nephrotic syndrome eventually lead to hypoalbuminemia. The ensuing fall in plasma oncotic pressure increases the flux of fluid into the interstitial spaces, leading to underfill edema.^{385,386} Additionally, primary renal salt retention can lead to overfill edema (Fig. 49.16). Patients with minimal change disease often have contracted plasma volume and stimulated RAAS, whereas those with other causes of nephrotic syndrome usually have an expanded plasma volume and suppressed RAAS.³⁸⁷ Micropuncture studies of sodium-retaining animal models of the nephrotic syndrome have demonstrated pronounced NaCl reabsorption in the distal nephron and TAL.^{388,389} The proteinuric kidney of a rat model of unilateral nephrotic syndrome has an enhanced Na⁺ reabsorption in the CDs³⁹⁰ and a diminished response to ANP³⁹¹ reinforced by hyperaldosteronism. Renin and aldosterone levels are highly variable in patients with nephrotic syndrome.³⁹²

One study has found that a fractional sodium excretion of 0.2% differentiates volume-contracted from volume-expanded nephrotic syndrome in children; treatment with diuretics alone has proven effective and safe for the volume-expanded group.³⁹³ Hypoalbuminemia reduces the binding of furosemide to plasma proteins and thereby enlarges its volume of distribution.³⁹⁴ Whereas one study has reported that premixing furosemide with albumin in the syringe before intravenous injection enhances the diuresis of patients with the nephrotic syndrome,¹⁰⁹ this finding has not been confirmed.^{110,395,396} Indeed, two studies have shown that patients with a serum albumin level of 2 g/dL can deliver normal quantities of furosemide into the urine.^{394,397} Iso-oncotic plasma volume expansion with albumin in patients with nephrotic syndrome fails to induce negative NaCl balance³⁹⁸ or enhance the response to furosemide³⁹⁵ and is not generally recommended.^{395,399}

A more logical approach to diuretic resistance is to limit albuminuria with an ACE inhibitor, ARB, or SGLT2 inhibitor⁴⁰⁰ that also may combat the associated edema and progressive loss of renal function. The addition of a loop diuretic to an ACE inhibitor or an ARB reduces proteinuria further but increases the serum creatinine concentration.⁴⁰¹

The tubular secretion of CAIs²² and loop diuretics¹²⁸ by the S2 segment of the proximal tubule depends on albumin. However, in the rabbit, the uptake of loop diuretics into the S1 portion of the proximal tubule, where furosemide is inactivated by glucuronidation, is inhibited by albumin (see Fig. 49.7).^{128,402} Albumin infusion into nephrotic patients does indeed increase renal furosemide excretion, whereas hypoalbuminemia enhances its metabolic clearance.⁴⁰³

The interaction of furosemide with its target in the lumen of the TAL is restricted by binding to filtered albumin.⁴⁰⁴ Addition of albumin to the tubular perfusate of the loop of Henle attenuates the response to perfused furosemide and is reversed by coprefusion with warfarin, which displaces it from its albumin-binding site.⁴⁰⁵ However, Agarwal and colleagues⁴⁰⁶ reported that displacing furosemide from albumin via the coadministration of sulfisoxazole did not affect natriuresis in patients with nephrotic syndrome. However, this study's results are not definitive because subjects did not have diuretic resistance.

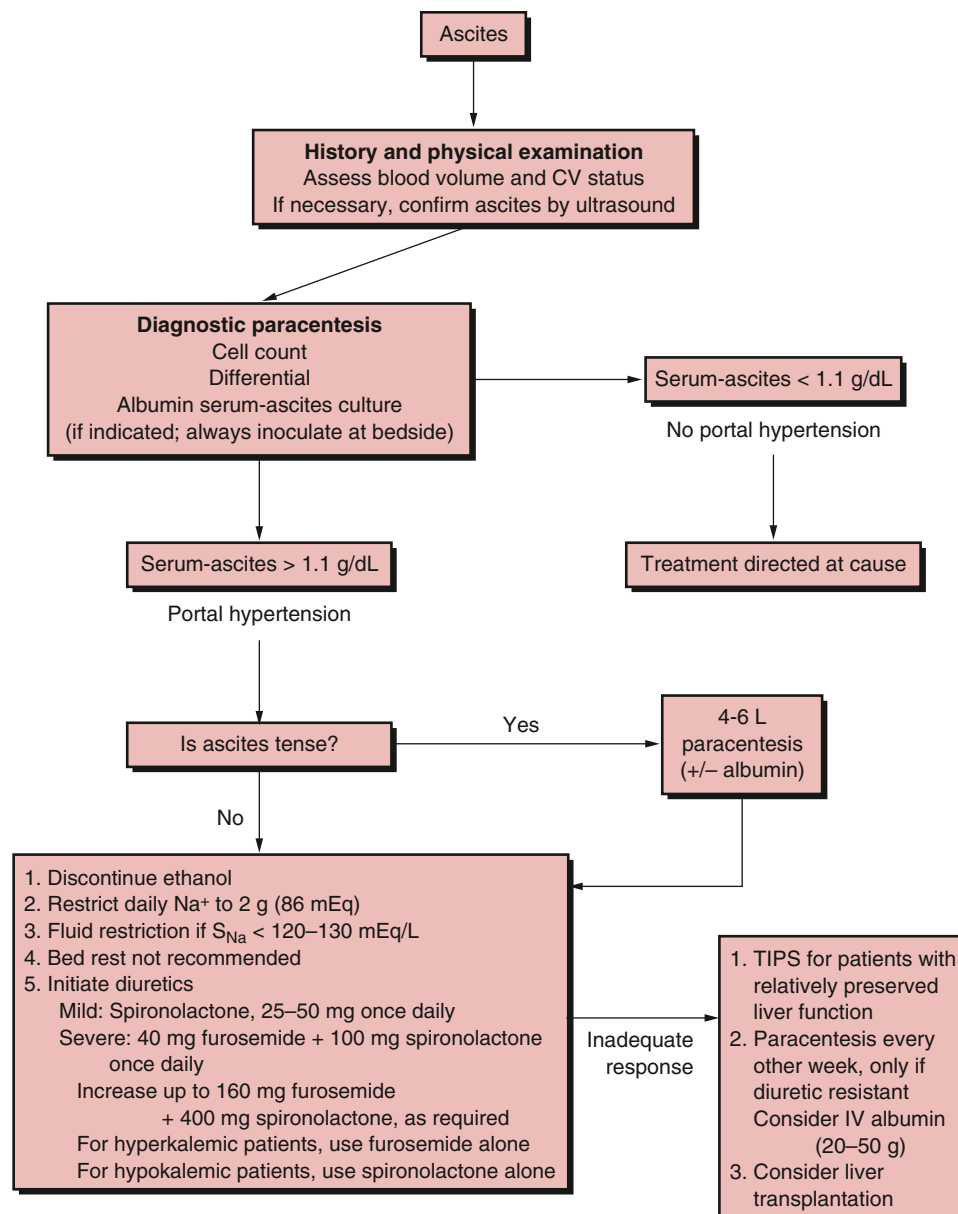


Fig. 49.15 Treatment algorithm for the management of fluid retention in patients with hepatic cirrhosis and ascites. CV, Cardiovascular; IV, intravenous; S_{Na^+} , serum Na^+ concentration; TIPS, transjugular intrahepatic portosystemic shunt. (From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. ed 2. London: Elsevier Science; 2003.)

Animal studies have demonstrated five mechanisms that could impair the responsiveness to loop diuretics in patients with nephrotic syndrome: 1 and 2. decreased delivery and/or decreased tubular secretion of the diuretic; 3. increased renal furosemide metabolism; 4. decreased blockade of tubular transport by the diuretic; and 5. increased $NaCl$ reabsorption by other nephron segments.

Nephrotic edema is best managed with dietary salt and fluid restriction. Most patients show an initial response to a loop diuretic when it is required. Spironolactone or eplerenone is effective in some patients.³⁸⁶ Decreasing renal function²⁹⁴ or the administration of indomethacin²⁸⁶ causes marked resistance to loop diuretics in these patients. The combination of a thiazide diuretic with furosemide dissipates edema but at the expense of marked kaliuresis.⁴⁰⁷ Studies have suggested that primary sodium retention in

nephrotic syndrome is related to the presence of plasma proteinases in nephrotic urine, such as plasmin, which can cleave the extracellular loop of ENaC that projects into the tubular lumen, leading to ENaC activation.^{408,409} Some studies have shown that this mechanism is also present in patients with proteinuric CKD.^{410,370} Importantly, the ENaC blockers amiloride and triamterene have now been reported to reduce albuminuria in patients with proteinuric kidney disease.⁵⁸

IDIOPATHIC EDEMA

Idiopathic edema predominantly affects women. It causes fluctuating salt retention and edema, exacerbated by orthostasis.⁴¹¹ The effect of diuretic withdrawal during controlled salt intake were studied in 10 such patients.⁴¹² Although their body weight increased by 0.5

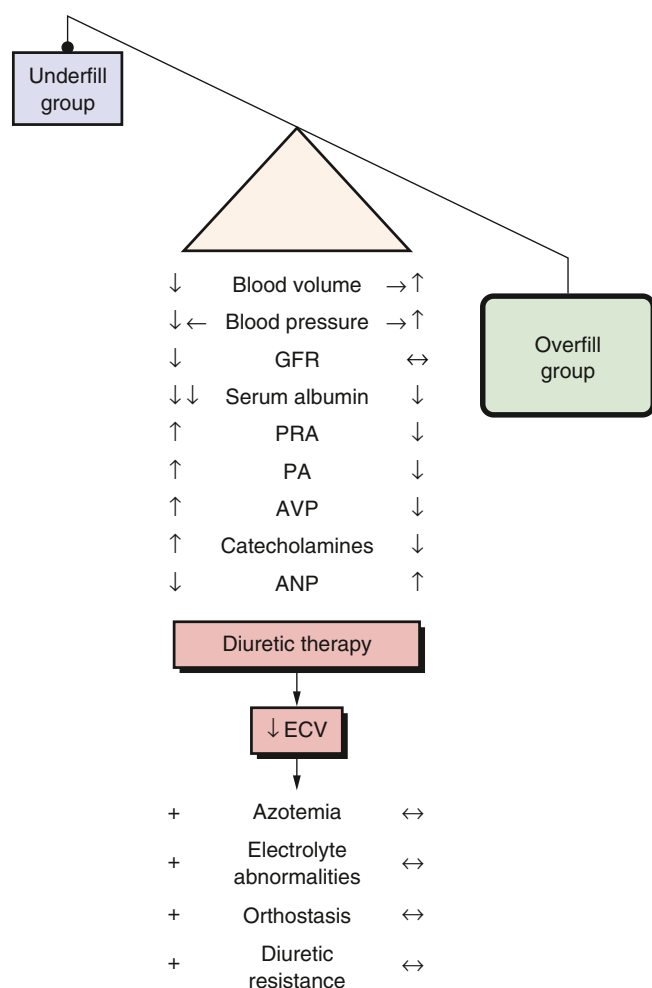


Fig. 49.16 Comparison of clinical and biochemical characteristics and responses in patients with nephrotic syndrome and underfill edema versus overfill edema. ANP, Atrial natriuretic peptide; AVP, arginine vasopressin; ↓ECV, decrease in extracellular fluid volume; GFR, glomerular filtration rate; PA, plasma aldosterone; PRA, plasma renin activity; ↑, increase; ↓, decrease; ↔, no effect; +, potential side effect. (Modified from Schrier RW, Fassett RG. A critique of the overfill hypothesis of sodium and water retention in the nephrotic syndrome. *Kidney Int.* 1998;53:1111–1117.)

to 5.0 kg within 2 to 8 days, 7 returned to their original weight by 3 weeks, without the reinstitution of diuretic therapy. The investigators concluded that diuretic abuse could cause idiopathic edema. However, this conclusion has been challenged.⁴¹³ Nevertheless, normal subjects given a single dose of bumetanide with simultaneous saline infusion to prevent volume depletion are unable to excrete a salt load for at least 2 days.²⁶⁸ Remarkably, 83% of habitual furosemide abusers who consume high doses over prolonged periods demonstrate medullary nephrocalcinosis and tubulointerstitial fibrosis.⁴¹⁴ Patients with idiopathic edema are best treated with salt restriction.

NONEDEMATOUS CONDITIONS

HYPERTENSION

Hypertension is discussed in [Chapters 46 to 48](#).

ACUTE KIDNEY INJURY

A review of 11 randomized trials of loop diuretics or mannitol for prophylaxis or treatment of established AKI found no benefit.⁴¹⁵ However, diuretics can be used to convert oliguric to nonoliguric AKI that may facilitate management. One study of 58 patients reported that sustained diuresis can be provoked in most patients given 1 g furosemide orally three times daily, but this large dose produced deafness in two patients, which was permanent in one¹²⁸ and therefore cannot be recommended. Older observational and randomized studies have indicated that furosemide does not improve the prognosis of AKI,^{416–418} and some have suggested that it may even be worsened.⁴¹⁹ Evidence of adverse effects of ECV overload in critically ill patients has been growing. The Fluid and Catheter Treatment Trial randomized patients with respiratory distress syndrome to different treatment strategies. Those with AKI exhibited reduced mortality with treatments that reduced central venous pressure (CVP) with higher diuretic doses. Thus loop diuretics can be used safely in this population.⁴²⁰ Furosemide can reduce the need for dialysis by diminishing hyperkalemia, acidosis, or fluid overload.⁴¹⁵ One protocol is to give 40 mg of furosemide, 1 mg of bumetanide, or 25 mg of torsemide intravenously and to double the dose every 60 minutes if there is no response, up to a total daily dose of 1 g of furosemide or the equivalent.⁴¹⁵ Bumetanide and torsemide are metabolized by the liver and so may need to be dosed at a lower level in patients with cirrhosis, whereas furosemide is metabolized by the kidney and therefore accumulates to a greater degree in patients with renal insufficiency (see [Fig. 49.6](#)). The management of AKI is discussed further in [Chapter 28](#).

CHRONIC KIDNEY DISEASE

The fractional reabsorption of NaCl and fluid by the renal tubules of subjects who are in balance is reduced in proportion to the fall in the GFR. The renal clearance of loop diuretics falls in parallel with the GFR because of a decreased renal mass and the accumulation of organic acids that compete for proximal secretion.⁴²¹ Thus although the maximal increase in fractional excretion of Na⁺ produced by furosemide is maintained in CKD,^{6,422,423} the absolute response to diuretics is limited by a reduction in the absolute rate of NaCl reabsorption and a reduction in the delivery of the diuretic to its target (see [Fig. 49.14](#)). Although CKD decreases proximal reabsorption, there is enhanced fractional reabsorption in the loop segment, distal tubule, and CDs,⁴²⁴ with a relative increase of threefold to fourfold per residual nephron in the expression of the NKCC2 in the TAL and the NCC in the DCT.⁴²⁵

Torsemide has a greater oral bioavailability than furosemide in CKD.¹²¹ For refractory cases, a loop diuretic infusion (e.g., bumetanide, 1 mg/hour for 12 hours) produces a greater natriuresis and less myalgia than two bolus injections.⁴²⁶ While thiazides have been used effectively in mild CKD for many years, one randomized, placebo-controlled trial has demonstrated that thiazides are also effective in CKD stage 4.^{161,427} Moreover, when used in combination with a loop diuretic that increases NaCl delivery and reabsorption at the distal tubule, larger doses of thiazides are effective in patients with moderate azotemia, although at the cost of a sharp further rise in

Table 49.1 Ceiling Doses of Loop Diuretics (in mg)

Condition	Furosemide		Bumetanide, IV or PO	Torsemide, IV or PO
	IV	PO		
Chronic renal insufficiency:				
• Moderate (GFR = 20–50 mL/min/1.73 m ²)	80–160	160	4–8	20–50
• Severe (GFR <20 mL/min/1.73 m ²)	160–200	240	8–10	50–100
Nephrotic syndrome with normal GFR	80–120	240	2–3	20–50
Cirrhosis with normal GFR	40–80	80–160	1	10–20
Heart failure with normal GFR	40–80	80–160	1–2	10–20

GFR, Glomerular filtration rate; IV, intravenous; PO, oral.

Data from Brater DC. Diuretic therapy. *N Engl J Med.* 1998;339:387–395.

the serum creatinine and blood urea concentrations and a high incidence of hypokalemia and electrolyte disorders.³¹⁵ Moreover, high plasma levels of furosemide can cause ototoxicity.⁴²⁸ Therefore care should be taken not to exceed the ceiling dose (Table 49.1).^{422,423} Epidemiologic studies have correlated diuretic use with kidney failure,⁴²⁹ but this is likely a noncausal association.⁴³⁰ In fact, continuing diuretics in patients undergoing dialysis who have residual renal function was associated with lower interdialytic weight gain, less hyperkalemia, and lower cardiac-specific mortality.⁴³¹ The presence of hyponatremia in CKD patients using diuretics indicates volume overload, not depletion, and is associated with an earlier initiation of renal replacement therapy.⁴³² (See Wilcox⁴ and Sica and Gehr⁴³³ for further reading on diuretics in CKD.)

RENAL TUBULAR ACIDOSIS

Furosemide increases the distal delivery of NaCl and fluid and stimulates aldosterone secretion and phosphate elimination, all of which enhance acid elimination.⁴³⁴ In addition, there is a direct effect of both furosemide and thiazide on distal acidification via an increased abundance of the H⁺-ATPase B1 subunit.⁴³⁵ Hence furosemide can be used in patients with hyperkalemic (type IV) renal tubular acidosis to increase renal acid excretion.^{436,437} Because this is usually due to hypoaldosteronism, mineralocorticoid therapy may also be indicated.⁴³⁸

HYPERCALCEMIA

Ca²⁺ excretion is increased by osmotic or loop diuretics but is decreased by thiazides and distal agents. Hypercalcemia activates the Ca²⁺-sensing receptor^{66,439} that inhibits fluid and NaCl reabsorption in the TAL and impairs renal concentrating ability. The ensuing diuresis and frequent ECV depletion further limit Ca²⁺ excretion by reducing the GFR and enhancing proximal fluid and Ca²⁺ reabsorption. Therefore the initial therapy for hypercalcemia is volume expansion with saline, with or without bisphosphonates or steroids, depending on the cause. Loop diuretics may help prevent or treat fluid overload, but there is little evidence to support a role in the treatment of hypercalcemia.⁴⁴⁰

NEPHROLITHIASIS

The thiazide-like diuretic indapamide reduced stone formation in hypercalciuric and even normocalciuric patients by reducing the excretion of Ca²⁺ and oxalate.⁴⁴¹ However, a randomized controlled trial did not identify significant differences in the symptomatic or radiologic recurrence of calcium-containing kidney stones with placebo, 12.5, 25, or 50 mg of hydrochlorothiazide, although there was a reduction in radiologic recurrence alone with the two highest doses.⁴⁴² Of note, once-daily dosing, high sodium intake, and the fact that not all patients were stone free at baseline may have offset some of the positive effects of hydrochlorothiazide. Combined therapy with thiazides and citrate may be effective in patients who don't respond to thiazides alone.⁴⁴³ Ca²⁺ reabsorption can be enhanced further by the addition of amiloride⁴⁴⁴ or a low-salt diet. KHCO₃ produces a greater reduction in Ca²⁺ excretion than KCl when given with hydrochlorothiazide.⁴⁴⁵

OSTEOPOROSIS

Bone cells express an Na⁺-Cl⁻ cotransporter⁴⁴⁶ that, when blocked by a thiazide, enhances Ca²⁺ uptake into bone.⁴⁴⁷ Thiazides inhibit osteocalcin, an osteoblast-specific protein that retards bone formation,⁴⁴⁸ and directly stimulate the production of the osteoblast differentiation markers.^{449,450} They inhibit bone reabsorption⁴⁵¹ and augment bone mineralization, independently of parathyroid hormone.⁴⁵² Thus thiazides may promote bone mineralization both by reducing renal Ca²⁺ excretion and by direct effects on bone. These biologic effects are supported by data from epidemiologic studies and clinical trials. Thiazide therapy in older persons is associated with an increase in bone mineral density and a reduction in hip and pelvic fractures.^{453–455} In a placebo-controlled trial in postmenopausal women,⁴⁵⁶ hydrochlorothiazide (50 mg/day) significantly slowed cortical bone loss. Surprisingly, despite having opposite effects on Ca²⁺ excretion, a thiazide and a loop diuretic both enhance bone formation in postmenopausal women, at least in the short term.⁴⁵⁷ However, loop diuretics alone have been associated with hip bone loss in older men, increased risk of fractures in postmenopausal women, and increased risk of

revision following primary total hip arthroplasty in men and women and are best avoided for this indication.^{458–460}

GITELMAN SYNDROME

Potassium-sparing diuretics, often given with potassium supplements, may be used in Gitelman syndrome to treat hypokalemia. Spironolactone (200–300 mg/day) was shown to be more effective than amiloride (10–30 mg/day).⁴⁶¹ A later crossover trial compared the efficacies of indomethacin (75 mg/day), amiloride (30 mg/day), and eplerenone (150 mg/day) in correcting hypokalemia.⁴⁶² Although all three drugs increased plasma potassium levels significantly, indomethacin was most effective but also associated with the most adverse effects.

DIABETES INSIPIDUS

Thiazides can reduce urine flow by up to 50% in patients with central or nephrogenic diabetes insipidus.^{463,464} This paradoxical effect is related to decreased GFR, enhanced water reabsorption in the proximal and distal nephron,^{465,466} and an increase in papillary osmolarity leading to distal water reabsorption. A small placebo-controlled crossover trial and an animal study have shown that amiloride prevents lithium-induced polyuria.^{467,468} This effect is attributed to amiloride blocking the cellular entry of lithium into the principal cell via ENaC, where it can downregulate the water channel AQP2 via glycogen synthase kinase 3.⁴⁶⁹

ADVERSE EFFECTS OF DIURETICS

FLUID AND ELECTROLYTE ABNORMALITIES

EXTRACELLULAR VOLUME DEPLETION AND AZOTEMIA

Furosemide or torsemide given to normal subjects or infused into rats reduces the GFR and RBF over the first hours or first day independent of volume losses or angiotensin II. However, when given over weeks or months, diuretics normally do not decrease the GFR,^{84,470} although a decrease in GFR can be precipitated by vigorous diuresis in patients with CKD, severe edema, or cirrhosis and ascites. A rise in the ratio of blood urea nitrogen to creatinine suggests ECV depletion. This change can be ascribed to a combination of a decreased renal urea clearance because of greater urea reabsorption in the distal nephron⁴⁷¹ and increased urea generation due to greater arginine uptake by the liver, which is metabolized by arginase.^{472–474} Combining diuretics with ACE inhibitors or NSAIDs raises the risk of AKI.^{475,476}

HYPONATREMIA

This effect is relatively specific for thiazides that inhibit urinary dilution, whereas loop diuretics inhibit urinary concentration and dilution.⁴⁷⁷ Indeed, thiazides are 12-fold more likely than loop diuretics to cause hyponatremia.⁴⁷⁸ The mechanism of thiazide-induced hyponatremia has long remained elusive. Although one could postulate that the natriuretic effect causes plasma volume depletion with AVP release, patients with thiazide-induced hyponatremia actually gain weight, suggesting that they enhance renal water reabsorption.⁴⁷⁹ Clark and colleagues⁴⁸⁰ have shown that older age and thiazide diuretics are additive in impairing maximal free water excretion following a water

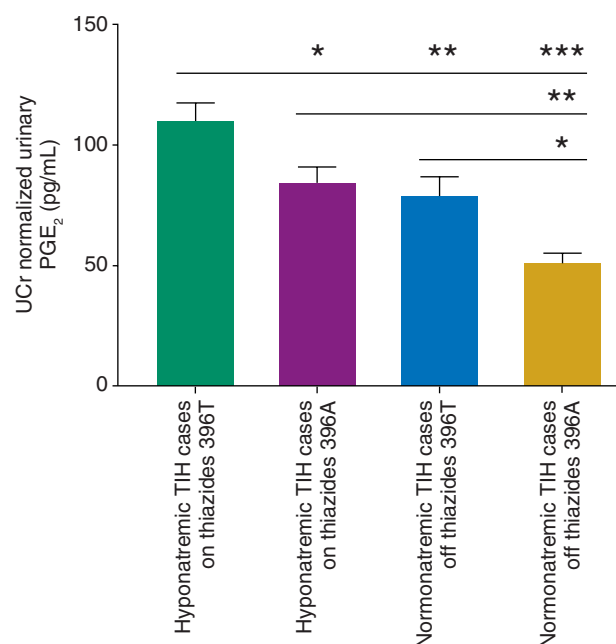


Fig. 49.17 Urinary prostaglandin E₂ (PGE₂) concentration in thiazide-induced hyponatremia (TIH) cases, compared with normonatremic thiazide- and non-thiazide-treated controls, stratified according to their genotype at position 396 of the prostaglandin transporter gene, *SLCO2A1*. TIH cases who carry at least one copy of the risk allele (396T) have significantly elevated urinary PGE₂ levels relative to those homozygous for 396A, whereas no such effect was observed in normonatremic controls. Data are represented as mean $\pm$ SEM. * $P < .05$; ** $P < .01$; *** $P < .001$. UCr, Urinary creatinine. (From Ware JS, Wain LV, Channavajhala SK, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127(9):3367–3374.)

load. Some 80% of thiazide-induced hyponatremia occurs in females,⁴⁸¹ most of whom are older and with low body mass.^{482,483} Hyponatremia can develop during rechallenge with a thiazide,^{479,484,485} often within the first 2 weeks of therapy.^{478,484} One study has shown that patients with thiazide-induced hyponatremia displayed increased free water reabsorption independent of AVP and that this is accompanied by increased renal PGE₂ excretion^{486,486a} (Fig. 49.17). Pharmacogenetic analysis has identified a loss of function variant in *SLCO2A1*, which encodes a prostaglandin transporter in the distal nephron. This prevents access of PGE₂ secreted into tubular fluid by thiazide diuretics from being reabsorbed across the nephron to activate EP1/3 receptors on the basolateral membrane of the collecting duct, where they act as antagonists to AVP. Moreover, the enhanced tubular fluid delivery of PGE₂ enhances its activation of EP4 receptors on the luminal membrane of the collecting ducts. This has the opposite effect of enhancing the response to AVP by enhancing the insertion of aquaporin (AQP) channels into the collecting duct cells, thereby enhancing free water reabsorption at any level of AVP. The outcome is enhanced water reabsorption leading to hyponatremia (Fig. 49.18).

Mild hyponatremia can be treated by withdrawal of diuretics, restriction of the daily intake of free water to 1.0 to 1.5 L, increasing solute intake, restoring of any K⁺ and

THIAZIDE-INDUCED HYPONATREMIA CAUSED BY AN INACTIVATING SINGLE GENE MUTATION OF THE PROSTAGLANDIN TRANSPORTER (PGT) IN THE COLLECTING DUCTS

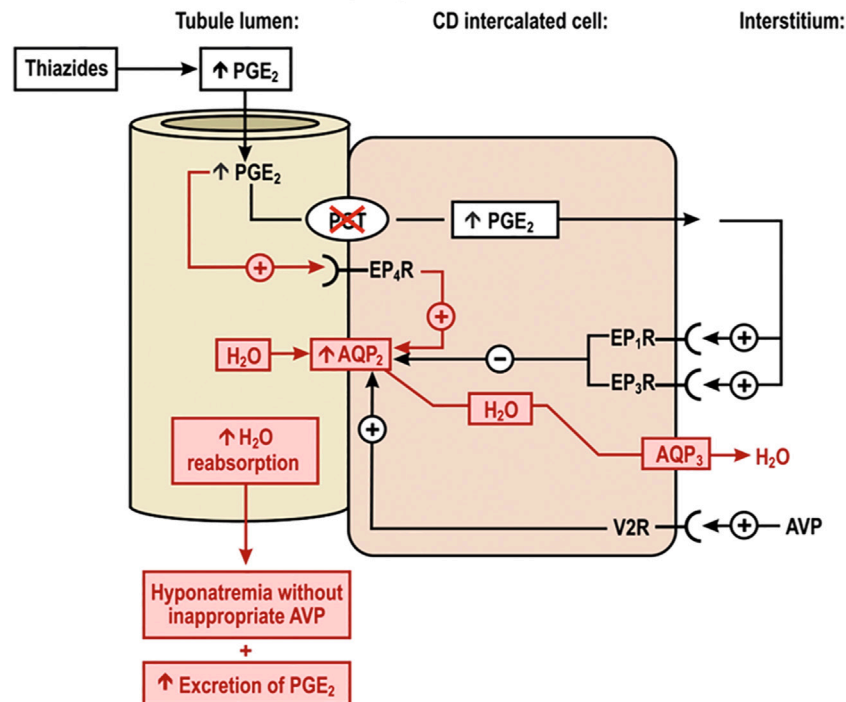


Fig. 49.18 Hypothesis for the role of *SLCO2A1* in contributing to thiazide-induced hyponatremia in individuals carrying the *SLCO2A1* A396T variant. Thiazide diuretics increase renal synthesis and secretion of prostaglandin E₂ into the tubular fluid of the distal nephron. Tubular fluid PGE₂ is normally reabsorbed by the prostaglandin transporter in the distal nephron. PGE₂ activates the PG type 1 and 3 receptors (EP₁R and EP₃R) on the basolateral membrane of the principal cells of the collecting duct (CD) that inhibit the insertion and activation of the aquaporin type 2 channel (AQP₂) into the luminal membrane. Inactivation of the PGT limits PGE₂ from accessing the basolateral EP₁ and EP₃ receptors and increases its luminal concentration and activation of the luminal EP₄ receptor that prompts the insertion and activity of AQP₂. This results in increased reabsorption of free water and the development of hyponatremia coupled with increased excretion of PGE₂. *ADH*, Antidiuretic hormone; *AQP2*, aquaporin-2; *PGE2*, prostaglandin E₂; *PGT*, phosphate glucuronyl transferase; *TIH*, thiazide-induced hyponatremia. (Data from Palmer BF, Clegg DJ. Altered prostaglandin signaling as a cause of thiazide-induced hyponatremia. *Am J Kidney Dis*. 2018;71(6):769–771.)

Mg²⁺ losses, and replenishing of NaCl if the patient is clearly volume depleted.^{478,487}

It is becoming increasingly clear that even mild chronic hyponatremia can produce symptoms.⁴⁸⁸ Therefore the development of diuretic-induced hyponatremia should probably be regarded as an indication to discontinue thiazides.⁴⁸⁹

HYPOKALEMIA

Four mechanisms have been identified that increase renal K⁺ elimination during therapy with thiazides or loop diuretics (Fig. 49.19): increased tubular flow, increased sodium delivery to CNT/CCD, secretion of AVP, aldosterone, and alkalosis. Flow-dependent K⁺ secretion by the distal nephron provides a universal mechanism for increased K⁺ secretion in response to diuretics that act more proximally.¹⁷⁵ The basal and flow-dependent components of K⁺ secretion have been shown to be mediated largely by distinct channels. Whereas basal secretion traverses ROMK channels, flow-dependent K⁺ secretion is mediated by calcium-activated maxi or big K⁺ (BK) channels.^{490–492} Normally, a flow-dependent rise in K⁺

secretion during increased water intake is offset by a drop in AVP concentration that diminishes distal K⁺ secretion. Diuretic therapy, however, is unusual because it combines increased distal tubule flow with maintained or increased AVP release due to nonosmotic stimulation, which is the second mechanism contributing to kaliuresis. Indeed, inhibition of AVP release in normal subjects undergoing a furosemide diuresis inhibits the kaliuresis.²⁸⁰ Nonosmotic AVP release is common in edematous subjects.⁴⁸⁷ Therefore enhanced release of AVP and increased distal tubule fluid delivery combine to promote ongoing K⁺ and H⁺ losses during diuretic therapy for edema. Diuretic-induced aldosterone secretion also promotes distal K⁺ secretion.^{143,175,297} The effects of flow and aldosterone on distal K⁺ secretion are normally counterbalanced during changes in salt intake, just as are the effects of flow and AVP induced by changes in water intake (see Fig. 49.20). Diuretic treatment, however, uncouples the two because it enhances the secretion of aldosterone and AVP but increases distal flow, thereby accounting for the particular importance of aldosterone and AVP in promoting K⁺ loss with diuretics.²⁸⁰

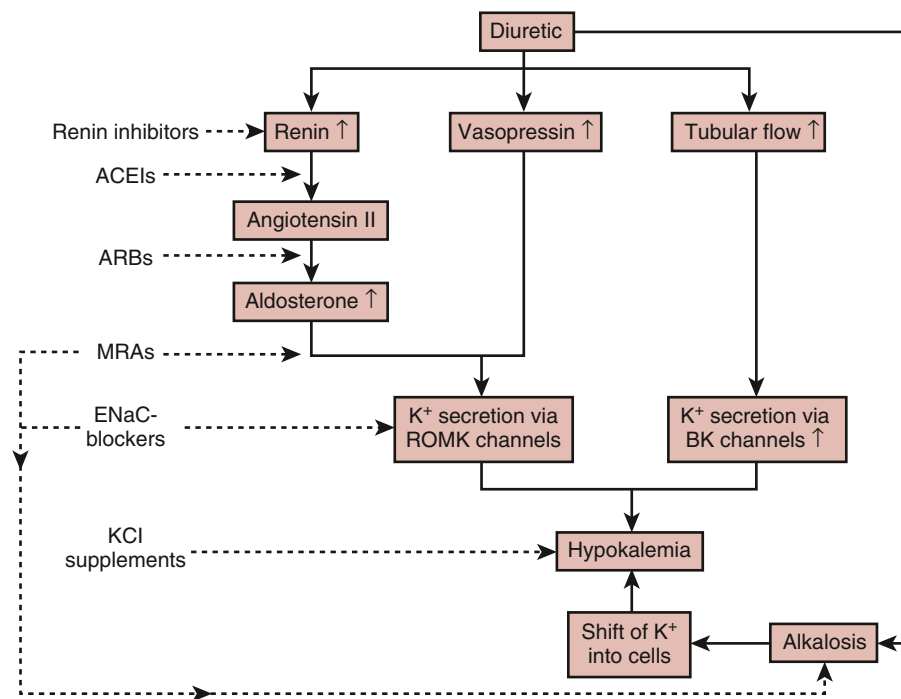


Fig. 49.19 Diagrammatic representation of mechanisms that increase (↑) K⁺ excretion by the collecting ducts or partition K⁺ into cells during therapy with a thiazide or loop diuretic, and strategies for prevention or treatment (*dashed lines*) with direct renin inhibitors, angiotensin-converting enzyme inhibitors (*ACEIs*), angiotensin receptor blockers (*ARBs*), mineralocorticoid receptor antagonists (*MRAs*; e.g., spironolactone), epithelial Na⁺ channel (*ENaC*) blockers (amiloride or triamterene), or KCl supplements. Diuretics stimulate the renin-angiotensin-aldosterone system (RAAS), increase distal tubule flow and release of arginine vasopressin, and generate metabolic alkalosis, all of which enhance K⁺ secretion in the collecting ducts. Two distinct mechanisms are responsible for K⁺ secretion in the collecting ducts, one mediated by the renal outer medulla K⁺ (*ROMK*) channels, and the other by big K⁺ (*BK*) channels.

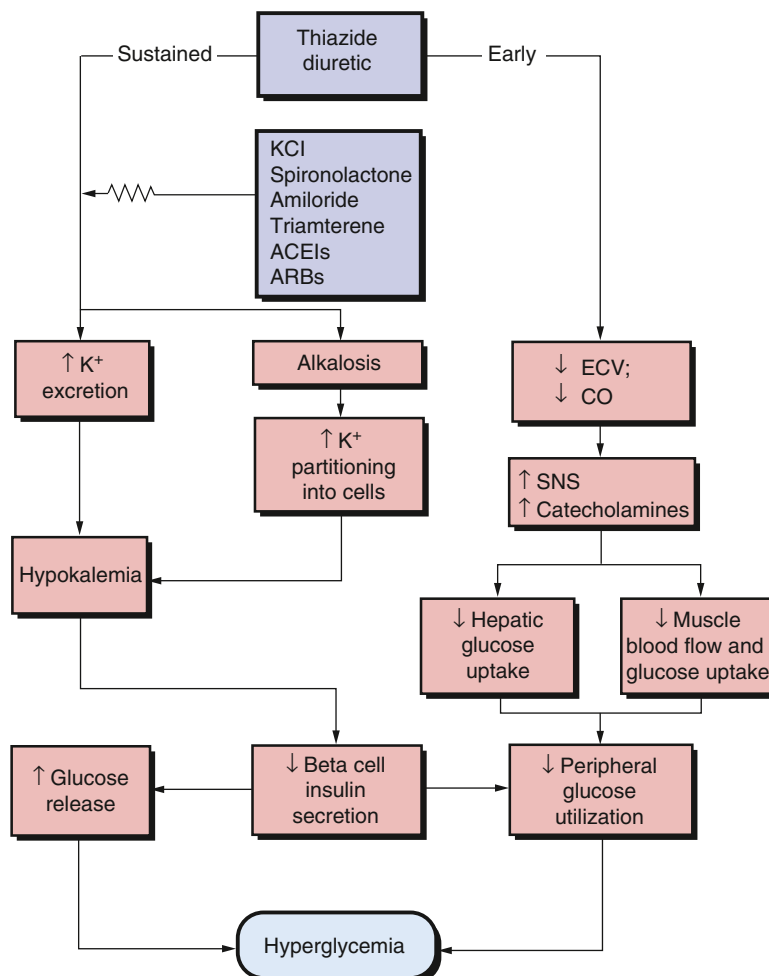


Fig. 49.20 Hypothesis for the hyperglycemic actions of thiazide diuretics. *ACEI*, Angiotensin-converting enzyme inhibitor; *ARB*, angiotensin receptor blocker; *ECV*, extracellular fluid volume; *SNS*, sympathetic nervous system; *CO*, cardiac output; ↑, increase(d); ↓, decrease(d). (From Wilcox CS. Metabolic and adverse effects of diuretics. *Semin Nephrol.* 1999;19:557–568.)
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Finally, diuretic-induced alkalosis enhances the distal secretion of K^+ .¹⁷⁵

The serum potassium concentration (S_K) of hypertensive patients not receiving KCl supplements falls by an average of 0.3 mmol/L with furosemide and 0.6 mmol/L with thiazides.⁴⁹³ In normal subjects receiving a single daily dose of furosemide, there is no detectable change in K^+ balance on a high-salt diet. However, on a low-salt diet, this same medication produced a negative K balance. These findings imply that the primary cause of hypokalemia during loop diuretic administration is dependent on the clinical scenario. With a low-sodium diet or more frequent dosing, renal potassium losses may play a greater role. With a high-sodium diet and less frequent dosing, redistribution of K^+ into cells related to metabolic alkalosis may play a larger role.^{71,126} For thiazide diuretics, which have significantly greater decreases in serum potassium with standard dosing, findings concerning the role of NCC in the physiology of potassium handling may provide some insight. Some studies have demonstrated that acute and chronic deactivation of NCC plays a key role in excreting a potassium load.⁴⁹⁴⁻⁴⁹⁶ This deactivation is critical for delivery of sodium to the CNT and CCD so that ENaC can reabsorb sodium, providing the driving force for potassium secretion via ROMK and Maxi-K channels. Similarly, thiazides can mimic this effect, resulting in significant potassium excretion in the absence of a potassium load, primarily when coupled with aldosterone-induced ENaC stimulation.⁴⁹⁸ This observation helps to explain why higher doses of thiazide diuretics, as used previously, are no longer recommended for patients with normal kidney function. High doses are more likely to lead to hypokalemia, which by, activating NCC, counteracts diuretic effectiveness.⁴⁹⁹

Mild, diuretic-induced hypokalemia (serum potassium, 3.0–3.5 mmol/L) increases the frequency of ventricular ectopy.⁵⁰¹ Some investigators have proposed that this does not pose a risk of clinically significant cardiac dysrhythmia.^{376,502} In contrast, others have reported a dose-dependent risk of cardiac arrest in patients receiving thiazides that is prevented by therapy with a K^+ -sparing diuretic such as amiloride.⁵⁰³ Studies in patients with HF report that mortality increases as S_K falls below 4.0 mmol/L.⁵⁰⁴

Adverse effects of hypokalemia are clearly important in certain cases:

1. Severe hypokalemia ($S_K < 3.0$ mmol/L) requires treatment because it is associated with a doubling of serious ventricular dysrhythmias, muscular weakness, and rhabdomyolysis.⁵⁰⁵
2. Mild hypokalemia can precipitate dangerous dysrhythmias in patients with cardiac dysfunction due to left ventricular hypertrophy, coronary ischemia, HF, prolonged QT interval, anoxia, or ischemia and in patients with known dysrhythmias.
3. Hypokalemia enhances the toxicity of cardiac glycosides by diminishing the renal tubule secretion of digoxin and by enhancing its binding to cardiac Na^+ - K^+ -ATPase, thereby exaggerating its actions on the heart.
4. Hypokalemia stimulates renal ammoniogenesis. This effect is dangerous for patients with cirrhosis and ascites who are prone to hepatic encephalopathy due to hyperammonemia. Moreover, the accompanying

diuretic-induced alkalosis partitions ammonia into the brain.

5. Catecholamines partition K^+ into cells and lower S_K . Myocardial infarction provokes sufficient catecholamine release to lower S_K by approximately 0.5 mmol/L, which is potentiated in patients who have received prior thiazide therapy.⁵⁰⁶
6. Hypokalemia impairs insulin release and predisposes to hyperglycemia.⁵⁰⁷
7. Hypokalemia limits the antihypertensive action of thiazides.⁵⁰⁸ In a placebo-controlled study of hypokalemic subjects receiving thiazide diuretics, coadministration of KCl that restored S_K also reduced BP significantly.⁵⁰⁸

Therefore it is prudent to prevent even mild degrees of hypokalemia by increasing the intake of K^+ with Cl^- (see Fig. 49.19). However, this often requires 40 to 80 mmol daily and, in the presence of alkalosis, hyperaldosteronism, or Mg^{2+} depletion, hypokalemia is quite unresponsive to dietary KCl. A more effective, convenient, and predictable strategy is to prescribe a combined therapy with a distal K^+ -sparing agent such as amiloride or triamterene, which maintains S_K (see Fig. 49.19),^{509,510} prevents diuretic-induced alkalosis, and provides further natriuresis and antihypertensive efficacy. Other strategies include the administration of an ACE inhibitor, ARB, MRA, or SGLTi to counter hyperaldosteronism, which would promote distal K^+ secretion.

HYPERKALEMIA

Diuretics acting in the ASDN decrease K^+ secretion and predispose to hyperkalemia,^{143,511} especially when combined with other drugs that interfere with renal potassium secretion, such as ACE inhibitors or the antibiotic trimethoprim-sulfamethoxazole.^{512,513} Use of a new generation of potassium binders to ameliorate hyperkalemia may enable continued use in some patients.⁵¹⁴

HYPOMAGNESEMIA

Loop diuretics inhibit Mg^{2+} reabsorption in the TAL^{78,179} by reducing the transepithelial voltage that drives Mg^{2+} and Ca^{2+} paracellularly (see Fig. 49.3).⁵¹⁵ Thiazides first enhance Mg^{2+} uptake in the DCT but, during prolonged therapy, there is enhanced renal Mg^{2+} excretion.¹⁵⁴ This is due to downregulation of TRPM6, predominant apical Mg^{2+} channel of the distal nephron.^{151,516} Distal K^+ -sparing agents and spironolactone diminish Mg^{2+} excretion. During prolonged therapy with thiazides and loop diuretics, serum Mg^{2+} concentration falls by 5% to 10%. Diuretic-induced hyponatremia and hypokalemia cannot be reversed fully until any Mg^{2+} deficit is replaced.⁵¹⁷ Mg^{2+} -depleted rats secrete K^+ inappropriately into the distal tubule, independently of aldosterone,⁵¹⁸ possibly because a decrease in intracellular magnesium activates ROMK channels.⁵¹⁹

HYPERCALCEMIA

Thiazides increase the serum concentrations of total and ionized calcium but rarely result in frank hypercalcemia because of a fall in parathyroid hormone concentrations.⁵²⁰ Therefore patients developing hypercalcemia during thiazide diuretic therapy should be screened for hyperparathyroidism. The increased serum calcium can

be ascribed primarily to enhanced Ca^{2+} reabsorption, as discussed previously. Persistent hypercalcemia should prompt a search for a specific cause (e.g., an adenoma of the parathyroid glands).^{520,521} An epidemiologic study reported that of the 221 subjects with thiazide-associated hypercalcemia, 24% had primary hyperparathyroidism.⁵²²

ACID-BASE CHANGES

Metabolic alkalosis induced by thiazides or loop diuretics is an important adverse factor in patients with hepatic cirrhosis and ascites, in whom the alkalosis may provoke hepatic coma by partitioning ammonia into the brain. Diuretic-induced metabolic alkalosis can also cause diminished ventilation in those with underlying pulmonary insufficiency.⁵²³ The generation of metabolic alkalosis with loop diuretics results from contraction of the extracellular HCO_3^- space by the excretion of a relatively HCO_3^- -free urine.^{71,524} The maintenance of metabolic alkalosis involves increased net acid excretion in response to hypokalemia-induced ammoniogenesis and angiotensin II–induced mineralocorticoid excess during continued Na^+ delivery and reabsorption at the distal nephron sites of H^+ secretion.⁵²⁵ Diuretic-induced metabolic alkalosis is best managed with the administration of KCl, but a distal K^+ -sparing diuretic⁵²⁶ or, occasionally, a CA inhibitor, should be considered.²³ Metabolic alkalosis impairs the natriuretic response to loop diuretics¹²⁶ and may thereby contribute to diuretic resistance.

CA inhibitors produce metabolic acidosis. Spironolactone, eplerenone, amiloride, and triamterene can cause hyperkalemic metabolic acidosis, especially in older patients, patients with renal impairment, and those receiving KCl supplements.

METABOLIC ABNORMALITIES

HYPERGLYCEMIA

Diuretic therapy, especially with thiazides, impairs carbohydrate tolerance and occasionally precipitates diabetes mellitus,^{527–529} primarily by decreasing hepatic glucose utilization.⁵³⁰ Hyperglycemia persists⁵³¹ but is reversed rapidly after diuretic discontinuation, even after 14 years of thiazide therapy.⁵³² The increase in blood glucose levels provoked by thiazides is worsened by concurrent β -adrenergic receptor blockade.⁵³⁰ A meta-analysis reported that thiazide diuretics increase fasting blood glucose levels by approximately 30.4 mg/dL (1.69 mmol/L).⁵³³

Thiazides impair glucose uptake into muscle^{534,535} and liver.⁵³⁰ This effect is more pronounced during initiation of therapy. It has been ascribed to a diuretic-induced reduction in cardiac output with reflex activation of the SNS and catecholamine secretion, which reduce hepatic glucose uptake, and muscle blood flow, thereby reducing muscle glucose uptake (Fig. 49.20). During sustained thiazide therapy, there is decreased insulin release, which can be corrected by reversal of hypokalemia with KCl,⁵³⁶ hypomagnesemia with magnesium oxide,⁵³⁷ or administration of amiloride or spironolactone. Studies have provided insights into the mechanism of this effect. Thiazides cause inhibition of the mitochondrial carbonic anhydrase isoform 5b in pancreatic β cells, causing anaplerosis and reduced insulin secretion.⁵³⁸ Experimental K^+ deficiency

causes glucose intolerance and impairs insulin secretion in the absence of diuretics.⁵³⁹ Hydrochlorothiazide and ACE inhibitors have opposite effects on glucose disposal, which are attributed to their opposing effects on S_{K} .⁵³⁵ The increase in serum glucose levels that occurs with thiazide therapy is more pronounced in obese patients and correlates with a fall in intracellular (K^+) or (Mg^{2+}).⁵⁴⁰ It can be prevented by reducing the thiazide dosage⁵⁰⁷ or by KCl replacement.⁵³⁶ Therefore care should be taken to monitor blood glucose levels during thiazide therapy, particularly in obese or diabetic patients, and to prevent hypokalemia.

Thiazide-induced hyperglycemia should be anticipated and prevented. Measures include coadministration of a distal K^+ -sparing diuretic, ACE inhibitor, or ARB, prescribing extra KCl, or reducing the thiazide dosage.^{510,541} Thiazide diuretics are not contraindicated in patients with diabetes mellitus. Indeed, diuretics produce an even greater reduction in the absolute risk for cardiovascular events in hypertensive patients who are diabetic.⁵⁴² Therefore diabetes mellitus is an indication for close surveillance and attention to the coadministration of agents designed to prevent hypokalemia (see Fig. 49.20).

HYPERLIPIDEMIA

Administration of loop diuretics or thiazides initially raises the plasma concentrations of total cholesterol, triglycerides, and low-density lipoprotein cholesterol but reduces the concentration of high-density lipoprotein cholesterol by 5% to 20%.⁵⁴³ Importantly, most studies have shown that serum cholesterol levels return to baseline over 3 to 12 months of thiazide therapy.^{543,544} In fact, when combined with lifestyle management, 4 years of thiazide therapy for hypertension is associated with a modest improvement in the lipid profile.^{545,546}

HYPERURICEMIA

Prolonged thiazide therapy increases the serum urate concentration by approximately 35%. Renal urate clearance falls because of competition for secretion between urate and the diuretic⁵⁴⁷ and because of ECV depletion, both of which reduce proximal tubule reabsorption of urate (see Fig. 49.5).⁸¹ Hyperuricemia is dose related and can lead to gout. A long-term outcome analysis of 3693 patients detected no adverse effects of diuretic-induced hyperuricemia in hypertensive subjects who did not have gout.⁵⁴⁸ However, later studies have correlated raised serum urate levels with a higher cardiovascular death rate.⁵⁴⁹ One possibility is to combine diuretic treatment with the ARB losartan, which has weak uricosuric properties or with an SGLT2 that increases renal urate clearance by 40% in normal subjects even in the presence of a loop diuretic.^{242,550}

OTHER ADVERSE EFFECTS

ERECTILE DYSFUNCTION

The Medical Research Council trial involving 15,000 hypertensive subjects reported that the prevalence of impotence was much higher in those receiving a thiazide than a placebo.⁵²⁷ The Treatment of Mild Hypertension Study reported that problems with penile erection were doubled in subjects receiving a thiazide.⁵⁵¹ A controlled trial has demonstrated the efficacy of sildenafil in reversing

impotence in hypertensive patients receiving multiple antihypertensive agents including diuretics.⁵⁵²

BLADDER DYSFUNCTION

The torrential diuresis induced by loop diuretics, termed the “Niagra effect,” can provoke urinary urgency and incontinence in patients with an overactive bladder. The peak diuresis with an extended-release formulation of torsemide (Soaanz) is reduced by 35%³⁹ that may therefore be a better choice for patients who have bladder dysfunction and require a loop diuretic.

OTOTOXICITY

Loop diuretics can cause deafness that may occasionally be permanent.⁵⁵³ The risk is greater with ethacrynic acid than with other loop diuretics and is also greater when a loop diuretic is combined with another ototoxic drug (e.g., an aminoglycoside).⁵⁵⁴ It is especially common during high-dose bolus intravenous therapy in patients with renal failure and in hypoalbuminemic subjects.⁵⁵⁵ In a crossover trial, no ototoxicity was noted in patients with severe HF when they were given an infusion of 250 to 2000 mg of furosemide over 8 hours, whereas reversible deafness occurred in 25% when the same dose was given as a bolus.³⁵⁶ Furosemide is metabolized in the kidney and, like ethacrynic acid, furosemide and its metabolite accumulate in CKD, whereas bumetanide and torsemide are metabolized in the liver. Therefore ototoxicity in patients with CKD is more likely to be provoked by furosemide.

HAZARDS IN PREGNANCY AND NEWBORNS

Diuretics do not prevent preeclampsia. They have little effect on perinatal mortality.⁵⁵⁷ Thiazide therapy can be continued during pregnancy in patients whose hypertension has been controlled by these agents.⁵⁵⁸

Intensive therapy with loop diuretics for neonates with respiratory distress syndrome increases the prevalence of patent ductus arteriosus because of the increased generation of prostaglandins.^{559,560} Prolonged furosemide therapy in preterm infants can cause renal calcification.⁵⁶¹ Diuretics can be transferred from the mother to the infant in breast milk,⁵⁶¹ in whom they can cause serious fluid and electrolyte abnormalities.⁵⁶²

VITAMIN B DEFICIENCY

Diuretics increase the excretion of water-soluble vitamins.⁵⁶³ Long-term diuretic therapy for HF reduces folate and vitamin B₁ (thiamine) levels⁵⁶⁴ and increases plasma homocysteine levels. Thiamine can improve left ventricular function in some patients with HF being treated with furosemide.⁵⁶⁵

DRUG ALLERGY

A reversible photosensitivity dermatitis occurs rarely during thiazide or furosemide therapy.⁵⁶⁶ High-dose furosemide in renal failure can cause bullous dermatitis.⁵⁶⁷ Diuretics may cause a more generalized dermatitis, sometimes with eosinophilia, purpura, or blood dyscrasia. Occasionally, they cause necrotizing vasculitis or anaphylaxis.⁵⁶⁸ Acute allergic reactions to sulfonamides are mediated via immunoglobulin E, whereas delayed-onset hypersensitivities are mediated by antibodies to specific protein epitopes.⁵⁶⁹ Severe necrotizing pancreatitis is a rare complication.⁵⁷⁰ Acute interstitial

nephritis with fever, rash, and eosinophilia may develop abruptly some months after initiation of therapy with a thiazide or, less often, furosemide.^{571,572} Ethacrynic acid is chemically dissimilar from other loop diuretics and can be a substitute for those that develop allergies.

MALIGNANCY

Thiazide diuretics have been associated with both skin and lip cancers, possibly through their photosensitizing properties.^{573,574} Although previous epidemiologic studies have suggested associations between diuretic use and cancer (renal cell carcinoma and colon carcinoma), a 2011 meta-analysis has refuted the suggestion of a higher risk of cancer or cancer-related death with use of any antihypertensive drug.⁵⁷⁵

ADVERSE DRUG INTERACTIONS

Hyperkalemia in patients receiving distal K⁺-sparing diuretics, spironolactone, or eplerenone can be precipitated by concurrent therapy with KCl, ACE inhibitors, ARBs, heparin, ketoconazole, trimethoprim, or pentamidine. Therefore these drugs should not normally be prescribed in combination, especially in patients with impaired renal function or diabetes. ACE inhibitors increase the risk of severe hyperkalemia in patients with decompensated HF receiving spironolactone.^{576,577} Loop diuretics and aminoglycosides potentiate ototoxicity and nephrotoxicity.^{555,578} Diuretic-induced hypokalemia increases digitalis toxicity fourfold.⁵⁷⁹ Plasma lithium concentrations rise during loop diuretic therapy⁵⁸⁰ because of increased proximal lithium reabsorption.⁵⁸¹ NSAIDs may impair the diuretic, natriuretic, antihypertensive, and venodilating responses to diuretics and predispose to renal vasoconstriction and a drop in the GFR (see earlier discussion, “Adaptation to Diuretic Therapy”). Used together, indomethacin and triamterene may precipitate renal failure.⁵⁸²

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KEY REFERENCES

- Brater DC. Use of diuretics in cirrhosis and nephrotic syndrome. *Semin Nephrol.* 1999;19(6):575–580.
- Brater DC. Pharmacology of diuretics. *Am J Med Sci.* 2000;319(1):38–49.
- Wilcox CS. New insights into diuretic use in patients with chronic renal disease. *J Am Soc Nephrol.* 2002;13(3):798–805.
- Mullens W, Dauw J, Martens P, et al. Acetazolamide in acute decompensated heart failure with volume overload. *N Engl J Med.* 2022;387(13):1185–1195.
- Gamba G, Saltzberg SN, Lombardi M, et al. Primary structure and functional expression of a cDNA encoding the thiazide-sensitive, electroneutral sodium-chloride cotransporter. *Proc Natl Acad Sci USA.* 1993;90(7):2749–2753.
- Ahmad T, Jackson K, Rao VS, et al. Worsening renal function in patients with acute heart failure undergoing aggressive diuresis is not associated with tubular injury. *Circulation.* 2018;137(19):2016–2028.
- Mentz RJ, Anstrom KJ, Eisenstein EL, et al. Effect of torsemide vs furosemide after Discharge on all-cause mortality in patients hospitalized with heart failure: the TRANSFORM-HF randomized clinical trial. *JAMA.* 2023;329(3):214–223.
- Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide I: effects of salt intake and renal compensation. *J Lab Clin Med.* 1983;102:450–458.

126. Loon NR, Wilcox CS. Mild metabolic alkalosis impairs the natriuretic response to bumetanide in normal human subjects. *Clin Sci*. 1998;94(3):287–292.
128. Pichette V, Geadah D, du Souich P. The influence of moderate hypoalbuminaemia on the renal metabolism and dynamics of furosemide in the rabbit. *Br J Pharmacol*. 1996;119:885–890.
142. Fan M, Zhang J, Lee CL, Zhang J, Feng L. Structure and thiazide inhibition mechanism of the human Na-Cl cotransporter. *Nature*. 2023;614(7949):788–793.
151. Nijenhuis T, Vallon V, van der Kemp AW, et al. Enhanced passive Ca²⁺ reabsorption and reduced Mg²⁺ channel abundance explains thiazide-induced hypocalciuria and hypomagnesemia. *J Clin Invest*. 2005;115:1651–1658.
161. Agarwal R, Sinha AD, Cramer AE, et al. Chlorthalidone for hypertension in advanced chronic kidney disease. *N Engl J Med*. 2021;385(27):2507–2519.
173. Bakris GL, Agarwal R, Anker SD, et al. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med*. 2020;383(23):2219–2229.
184. Pitt B, Zannad F, Remme WJ, et al. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med*. 1999;341(10):709–717.
234. Onishi A, Fu Y, Patel R, et al. A role for tubular Na(+)/H(+) exchanger NHE3 in the natriuretic effect of the SGLT2 inhibitor empagliflozin. *Am J Physiol Ren Physiol*. 2020;319(4):F712–F728.
237. Heerspink HJ, Perkins BA, Fitchett DH, et al. Sodium glucose cotransporter 2 inhibitors in the treatment of diabetes mellitus: cardiovascular and kidney effects, potential mechanisms, and clinical applications. *Circulation*. 2016;134(10):752–772.
238. Heerspink HJL, Stefánsson BV, Correa-Rotter R, et al. Dapagliflozin in patients with chronic kidney disease. *N Engl J Med*. 2020;383(15):1436–1446.
259. Ter Maaten JM, Rao VS, Hanberg JS, et al. Renal tubular resistance is the primary driver for loop diuretic resistance in acute heart failure. *Eur J Heart Fail*. 2017;19(8):1014–1022.
260. Kaissling B, Bachmann S, Kriz W. Structural adaptation of the distal convoluted tubule to prolonged furosemide treatment. *Am J Physiol*. 1985;248:F374–F381.
268. Almeshari K, Ahlstrom NG, Capraro FE, et al. A volume-independent component to post-diuretic sodium retention in man. *J Am Soc Nephrol*. 1993;3:1878–1883.
302. Hoorn EJ, Ellison DH. Diuretic resistance. *Am J Kidney Dis*. 2017;69(1):136–142.
318. Anisman SD, Erickson SB, Morden NE. How to prescribe loop diuretics in oedema. *BMJ*. 2019;364:1359.
321. Ellison DH, Felker GM. Diuretic treatment in heart failure. *N Engl J Med*. 2017;377(20):1964–1975.
350. Jentzer JC, DeWald TA, Hernandez AF. Combination of loop diuretics with thiazide-type diuretics in heart failure. *J Am Coll Cardiol*. 2010;56:1527–1534.
351. Rao VS, Planavsky N, Hanberg JS, et al. Compensatory distal reabsorption drives diuretic resistance in human heart failure. *J Am Soc Nephrol*. 2017;28(11):3414–3424.
364. Gines P, Cardenas A, Arroyo V, et al. Management of cirrhosis and ascites. *N Engl J Med*. 2004;350:1646–1654.
369. Biggins SW, Angeli P, Garcia-Tsao G, Isoniemi H, Maiwall R, American Association for the Study of Liver Diseases. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American association for the study of liver diseases. *Hepatology*. 2021;74(2):1014–1048.
420. Stewart RM, Park PK, Hunt JP, et al. Less is more: improved outcomes in surgical patients with conservative fluid administration and central venous catheter monitoring. *J Am Coll Surg*. 2009;208:725–735; discussion 735–727.
423. Voelker JR, Cartwright Brown D, Anderson S, et al. Comparison of loop diuretics in patients with chronic renal insufficiency. *Kidney Int*. 1987;32:572–578.
479. Friedman E, Shadel M, Halkin H, Farfel Z. Thiazide-induced hyponatremia. Reproducibility by single dose rechallenge and an analysis of pathogenesis. *Ann Intern Med*. 1989;110(1):24–30.
486. Ware JS, Wain LV, Channavajjhala SK, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127(9):3367–3374.
499. Ellison DH, Loffing J. Thiazide effects and adverse effects: insights from molecular genetics. *Hypertension*. 2009;54:196–202.
516. Alexander RT, Dimke H. Effect of diuretics on renal tubular transport of calcium and magnesium. *Am J Physiol Ren Physiol*. 2017;312(6):F998–F1015.

REFERENCES

- Bleich M, Greger R. Mechanism of action of diuretics. *Kidney Int Suppl.* 1997;59:S11–S15.
- Brater DC. Use of diuretics in cirrhosis and nephrotic syndrome. *Semin Nephrol.* 1999;19(6):575–580.
- Brater DC. Pharmacology of diuretics. *Am J Med Sci.* 2000;319(1):38–49.
- Wilcox CS. New insights into diuretic use in patients with chronic renal disease. *J Am Soc Nephrol.* 2002;13(3):798–805.
- Wilcox CS, Testani J, Pitt B. The pathophysiology of diuretic resistance and its implications for the management of chronic heart failure. *Hypertension.* 2020;76(4):1045–1054.
- Skott P, Hommel E, Bruun NE, Arnold-Larsen S, Parving HH. The acute effect of acetazolamide on glomerular filtration rate and proximal tubular reabsorption of sodium and water in normal man. *Scand J Clin Lab Invest.* 1989;49(6):583–587.
- Milano S, Carmosino M, Gerbino A, et al. Activation of the thiazide-sensitive sodium-chloride cotransporter by beta3-adrenoreceptor in the distal convoluted tubule. *Front Physiol.* 2021;12:695824.
- Pastorekova S, Parkkila S, Pastorek J, Supuran CT. Carbonic anhydrases: current state of the art, therapeutic applications and future prospects. *J Enzym Inhib Med Chem.* 2004;19(3):199–229.
- Schwartz GJ. Physiology and molecular biology of renal carbonic anhydrase. *J Nephrol.* 2002;15(suppl 5):S61–S74.
- Kaunisto K, Parkkila S, Rajaniemi H, et al. Carbonic anhydrase XIV: luminal expression suggests key role in renal acidification. *Kidney Int.* 2002;61:2111–2118.
- Carta F, Supuran CT. Diuretics with carbonic anhydrase inhibitory action: a patent and literature review (2005–2013). *Expert Opin Ther Pat.* 2013;23(6):681–691.
- Puscas I, Coltau M, Baican M, et al. The inhibitory effect of diuretics on carbonic anhydrases. *Res Commun Mol Pathol Pharmacol.* 1999;105:213–236.
- Ichikawa I, Kon V. Role of peritubular capillary forces in the renal action of carbonic anhydrase inhibitor. *Kidney Int.* 1986;30(6):828–835.
- Unwin RJ, Walter SJ, Giebisch G, Capasso G, Shirley DG. Localization of diuretic effects along the loop of Henle: an in vivo micro-perfusion study in rats. *Clin Sci (Lond).* 2000;98(4):481–488.
- DH E. *Intensive diuretic therapy: high doses, combinations and constant infusions.* San Diego. In: Seldin DW GG, ed. *Diuretic Agents: Clinical Physiology and Pharmacology.* 1997.
- Popovic MM, Schlenker MB, Thiruchelvam D, Redelmeier DA. Serious adverse events of oral and topical carbonic anhydrase inhibitors. *JAMA Ophthalmol.* 2022;140(3):235–242.
- Cogan MG, Maddox DA, Lucci MS, Rector Jr FC. Control of proximal bicarbonate reabsorption in normal and acidotic rats. *J Clin Invest.* 1979;64(5):1168–1180.
- Leyssac PP, Karlén FM, Holstein-Rathlou NH, Skott O. On determinants of glomerular filtration rate after inhibition of proximal tubular reabsorption. *Am J Physiol.* 1994;266(5 Pt 2):R1544–R1549.
- Lorenz JN, Schultheis PJ, Traynor T, Shull GE, Schnermann J. Micropuncture analysis of single-nephron function in NHE3-deficient mice. *Am J Physiol.* 1999;277(3):F447–F453.
- Zahedi K, Barone S, Xu J, Soleimani M. Potentiation of the effect of thiazide derivatives by carbonic anhydrase inhibitors: molecular mechanisms and potential clinical implications. *PLoS One.* 2013;8(11):e79327.
- Grimm PR, Lazo-Fernandez Y, Delpire E, et al. Integrated compensatory network is activated in the absence of NCC phosphorylation. *J Clin Invest.* 2015;125(5):2136–2149.
- Taft DR, Sweeney KR. The influence of protein binding on the elimination of acetazolamide by the isolated perfused rat kidney: evidence of albumin-mediated tubular secretion. *J Pharmacol Exp Therapeut.* 1995;274(2):752–760.
- Miller PD, Berns AS. Acute metabolic alkalosis perpetuating hypercarbia. A role for acetazolamide in chronic obstructive pulmonary disease. *JAMA.* 1977;238(22):2400–2401.
- Teppema LJ, Dahan A. Low-dose acetazolamide reduces the hypoxic ventilatory response in the anesthetized cat. *Respir Physiol Neurobiol.* 2004;140(1):43–51.
- Hohne C, Krebs MO, Seiferheld M, Boemke W, Kaczmarczyk G, Swenson ER. Acetazolamide prevents hypoxic pulmonary vasoconstriction in conscious dogs. *J Appl Physiol.* 2004;97(2):515–521.
- Swenson ER, Robertson HT, Hlastala MP. Effects of carbonic anhydrase inhibition on ventilation-perfusion matching in the dog lung. *J Clin Invest.* 1993;92(2):702–709.
- Wagenaar M, Vos P, Heijdra Y, Teppema L, Folgering H. Comparison of acetazolamide and medroxyprogesterone as respiratory stimulants in hypercapnic patients with COPD. *Chest.* 2003;123(5):1450–1459.
- TH M. Carbonic anhydrase: general perspective and advances in glaucoma research. *Drug Dev Res.* 1987;10:255.
- Vogh BP. The relation of choroid plexus carbonic anhydrase activity to cerebrospinal fluid formation: study of three inhibitors in cat with extrapolation to man. *J Pharmacol Exp Therapeut.* 1980;213(2):321–331.
- Brookes GB, Hodge RA, Booth JB, Morrison AW. The immediate effects of acetazolamide in Meniere's disease. *J Laryngol Otol.* 1982;96(1):57–72.
- Wall M, McDermott MP, Committee NIIHSGW, et al. Effect of acetazolamide on visual function in patients with idiopathic intracranial hypertension and mild visual loss: the idiopathic intracranial hypertension treatment trial. *JAMA.* 2014;311(16):1641–1651.
- Bartsch P, Swenson ER. Clinical practice: acute high-altitude illnesses. *N Engl J Med.* 2013;368(24):2294–2302.
- Low EV, Avery AJ, Gupta V, Schedlbauer A, Grocott MP. Identifying the lowest effective dose of acetazolamide for the prophylaxis of acute mountain sickness: systematic review and meta-analysis. *BMJ.* 2012;345:e6779.
- Nieto Estrada VH, Molano Franco D, Medina RD, Gonzalez Garay AG, Marti-Carvajal AJ, Arevalo-Rodriguez I. Interventions for preventing high altitude illness: Part 1. Commonly-used classes of drugs. *Cochrane Database Syst Rev.* 2017;6(6):CD009761.
- Grissom CK, Roach RC, Surnquist FH, et al. Acetazolamide in the treatment of acute mountain sickness: clinical efficacy and effect on gas exchange. *Ann Intern Med.* 1992;116:461–465.
- White DP, Zwillich CW, Pickett CK, et al. Central sleep apnea: improvement with acetazolamide therapy. *Arch Intern Med.* 1982;142:1816–1819.
- Mullens W, Dauw J, Martens P, et al. Acetazolamide in acute decompensated heart failure with volume overload. *N Engl J Med.* 2022;387(13):1185–1195.
- Griggs RC, Engel WK, Resnick JS. Acetazolamide treatment of hypokalemic periodic paralysis. prevention of attacks and improvement of persistent weakness. *Ann Intern Med.* 1970;73:39–48.
- Suetterlin K, Mannikko R, Hanna MG. Muscle channelopathies: recent advances in genetics, pathophysiology and therapy. *Curr Opin Neurol.* 2014;27:583–590.
- de Groot T, Sinke AP, Kortenoeven ML, et al. Acetazolamide attenuates lithium-induced nephrogenic diabetes insipidus. *J Am Soc Nephrol.* 2016;27(7):2082–2091.
- Supuran CT. Carbonic anhydrases: novel therapeutic applications for inhibitors and activators. *Nat Rev Drug Discov.* 2008;7:168–181.
- Parfitt AM. Acetazolamide and sodium bicarbonate induced nephrocalcinosis and nephrolithiasis; relationship to citrate and calcium excretion. *Arch Intern Med.* 1969;124:736–740.
- Heller I, Halevy J, Cohen S, et al. Significant metabolic acidosis induced by acetazolamide. Not a rare complication. *Arch Intern Med.* 1985;145:1815–1817.
- Kass MA, Kolker AE, Gordon M, et al. Acetazolamide and urolithiasis. *Ophthalmology.* 1981;88:261–265.
- Krivoy N, Ben-Arieh Y, Carter A, et al. Methazolamide-induced hepatitis and pure RBC aplasia. *Arch Intern Med.* 1981;141:1229–1230.
- Mallette LE. Acetazolamide-accelerated anticonvulsant osteomalacia. *Arch Intern Med.* 1977;137:1013–1017.
- Better OS, Rubinstein I, Winaver JM, et al. Mannitol therapy revisited (1940-1997). *Kidney Int.* 1997;51:886–894.
- Warren SE, Blantz RC. Mannitol. *Arch Intern Med.* 1981;141:493–497.
- Cloyd JC, Snyder BD, Cleeremans B, et al. Mannitol pharmacokinetics and serum osmolality in dogs and humans. *J Pharmacol Exp Therapeut.* 1986;236:301–306.
- Solomon R, Werner C, Mann D, et al. Effects of saline, mannitol and furosemide on acute decreases in renal function induced by radiocontrast agents. *N Engl J Med.* 1994;331:1416–1420.
- Conger JD. Interventions in clinical acute renal failure: what are the data? *Am J Kidney Dis.* 1995;26:565–576.

52. Canalese J, Gimson AE, Davis C, et al. Controlled trial of dexamethasone and mannitol for the cerebral oedema of fulminant hepatic failure. *Gut*. 1982;23:625–629.
53. Vukic M, Negovetic L, Kovac D, et al. The effect of implementation of guidelines for the management of severe head injury on patient treatment and outcome. *Acta Neurochir*. 1999;141:1203–1208.
54. Ropper AH. Hyperosmolar therapy for raised intracranial pressure. *N Engl J Med*. 2012;367:746–752.
55. Toung TJ, Hurn PD, Traystman RJ, et al. Global brain water increases after experimental focal cerebral ischemia: effect of hypertonic saline. *Crit Care Med*. 2002;30:644–649.
56. DiFresco V, Landman M, Jaber BL, et al. Dialysis disequilibrium syndrome: an unusual cause of respiratory failure in the medical intensive care unit. *Intensive Care Med*. 2000;26:628–630.
57. Aiyagari V, Deibert E, Diringer MN. Hyponatremia in the neurologic intensive care unit: how high is too high? *J Crit Care*. 2006;21:163–172.
58. Borges HF, Hocks J, Kjellstrand CM. Mannitol intoxication in patients with renal failure. *Arch Intern Med*. 1982;142:63–66.
59. Hannaert P, Alvarez-Guerra M, Pirot D, et al. Rat NKCC2/NKCC1 cotransporter selectivity for loop diuretic drugs. *Naunyn-Schmiedeberg Arch Pharmacol*. 2002;365:193–199.
60. Isenring P, Forbush B. Ion transport and ligand binding by the Na-K-Cl cotransporter, structure-function studies. *Comp Biochem Physiol Mol Integr Physiol*. 2001;130:487–497.
61. Greger R, Schlatter E. Properties of the lumen membrane of the cortical thick ascending limb of Henle's loop of rabbit kidney. *Pflügers Archiv*. 1983;396(4):315–324.
62. Ecelbarger CA, Terris J, Hoyer JR, et al. Localization and regulation of the rat renal Na⁺+K⁺+2Cl[−] cotransporter, BSC-1. *Am J Physiol*. 1996;271:F619–F628.
63. Obermüller N, Kunchaparty S, Ellison DH, et al. Expression of the Na-K-2Cl cotransporter by macula densa and thick ascending limb cells of rat and rabbit nephron. *J Clin Invest*. 1996;98:635–640.
64. Glanville M, Kingscote S, Thwaites DT, et al. Expression and role of sodium, potassium, chloride cotransport (NKCC1) in mouse inner medullary collecting duct (mIMCD-K2) epithelial cells. *Pflügers Archiv*. 2001;443:123–131.
65. Bachmann S, Mutig K. Regulation of renal Na-(K)-Cl cotransporters by vasopressin. *Pflügers Archiv*. 2017;469(7–8):889–897.
66. Riccardi D, Hall AE, Chattopadhyay N, et al. Localization of the extracellular Ca²⁺/ (polyvalent cation)-sensing protein in rat kidney. *Am J Physiol*. 1998;274:F611–F622.
67. Bailly C. Transducing pathways involved in the control of NaCl reabsorption in the thick ascending limb of Henle's loop. *Kidney Int*. 1998;53(suppl 65):S29–S35.
68. Good DW, Knepper MA, Burg MB. Ammonia and bicarbonate transport by thick ascending limb of rat kidney. *Am J Physiol*. 1984;247:F35–F44.
69. Duarte CG, Chomety F, Giebisch G. Effect of amiloride, ouabain, and furosemide on distal tubular function in the rat. *Am J Physiol*. 1971;221:632–640.
70. de Bruijn PI, Larsen CK, Frische S, et al. Furosemide-induced urinary acidification is caused by pronounced H⁺ secretion in the thick ascending limb. *Am J Physiol Ren Physiol*. 2015;309(2):F146–F153.
71. Wilcox CS, Loon NR, Kanthawata S, et al. Generation of alkalosis during furosemide infusion: roles of contraction and acid excretion. *J Nephrol*. 1991;2:81–87.
72. Radtke HW, Rumrich G, Kinne-Saffran E, et al. Dual action of acetazolamide and furosemide on proximal volume absorption in the rat kidney. *Kidney Int*. 1972;1:100–105.
73. Puschett JB, Sylk D, Teredesai PR. Uncoupling of proximal sodium bicarbonate from sodium phosphate transport by bumetanide. *Am J Physiol*. 1978;235:F403–F408.
74. Jung KY, Endou H. Furosemide acts on short loop of descending thin limb, but not on long loop. *J Pharmacol Exp Therapeut*. 1990;253:1184–1188.
75. Velazquez H, Wright FS. Effects of diuretic drugs on Na, Cl, and K transport by rat renal distal tubule. *Am J Physiol*. 1986;250:F1013–F1023.
76. Wilson DR, Honrath U, Sonnenberg H. Furosemide action on collecting ducts: effect of prostaglandin synthesis inhibition. *Am J Physiol*. 1983;244:F666–F673.
77. Earley LE, Friedler RM. Renal tubular effects of ethacrynic acid. *J Clin Invest*. 1964;43:1495.
78. Quamme GA. Effect of furosemide on calcium and magnesium transport in the rat nephron. *Am J Physiol*. 1981;241:F340–F347.
79. Hou J, Renigunta A, Konrad M, et al. Claudin-16 and claudin-19 interact and form a cation-selective tight junction complex. *J Clin Invest*. 2008;118:619–628.
80. Gamba G, Saltzberg SN, Lombardi M, et al. Primary structure and functional expression of a cDNA encoding the thiazide-sensitive, electroneutral sodium-chloride cotransporter. *Proc Natl Acad Sci USA*. 1993;90(7):2749–2753.
81. Weinman EJ, Eknoyan G, Suki WN. The influence of the extracellular fluid volume on the tubular reabsorption of uric acid. *J Clin Invest*. 1975;55:283–291.
82. Araujo M, Solis G, Welch WJ, Wilcox CS. Renal nerve deafferentation attenuates the fall in GFR during intravenous infusion of furosemide in anesthetized rats. *Kidney Blood Press Res*. 2020;45(1):70–83.
83. Shah S, Pitt B, Brater DC, et al. Sodium and fluid excretion with torsemide in healthy subjects is limited by the short duration of diuretic action. *J Am Heart Assoc*. 2017;6(10).
84. Loon NR, Wilcox CS, Unwin RJ. Mechanism of impaired natriuretic response to furosemide during prolonged therapy. *Kidney Int*. 1989;36:682–689.
85. Epstein M, Hollenberg NK, Guttman RD, et al. Effect of ethacrynic acid and chlorothiazide on intrarenal hemodynamics in normal man. *Am J Physiol*. 1971;220:482–487.
86. Spitalowitz S, Chou SY, Faubert PF, et al. Effects of diuretics on inner medullary hemodynamics in the dog. *Circ Res*. 1982;51:703–710.
87. Gerber JG. Role of prostaglandins in the hemodynamic and tubular effects of furosemide. *Fed Proc*. 1983;42:1707–1710.
88. Gerber JG, Nies AS. Furosemide-induced vasodilation: importance of the state of hydration and filtration. *Kidney Int*. 1980;18:454–459.
89. Schnermann J, Homer W. Smith Award lecture: the juxtaglomerular apparatus: from anatomical peculiarity to physiological relevance. *J Am Soc Nephrol*. 2003;14:1681–1694.
90. Wright FS, Schnermann J. Interference with feedback control of glomerular filtration rate by furosemide, trifluoromethoxy, and cyanide. *J Clin Invest*. 1974;53:1695–1708.
91. Wilcox CS. Antihypertensive and renal mechanisms of SGLT2 (Sodium-Glucose linked transporter 2) inhibitors. *Hypertension*. 2020;75(4):894–901.
92. Daskalopoulos G, Laffi G, Morgan T, et al. Immediate effects of furosemide on renal hemodynamics in chronic liver disease with ascites. *Gastroenterology*. 1987;92(6):1859–1863.
93. Dilken O, Ince C, Kapucu A, Heeman PM, Ergin B. Furosemide exacerbated the impairment of renal function, oxygenation and medullary damage in a rat model of renal ischemia/reperfusion induced AKI. *Intensive Care Med Exp*. 2023;11(1):25.
94. Ahmad T, Jackson K, Rao VS, et al. Worsening renal function in patients with acute heart failure undergoing aggressive diuresis is not associated with tubular injury. *Circulation*. 2018;137(19):2016–2028.
95. Epstein FH, Prasad P. Effects of furosemide on medullary oxygenation in younger and older subjects. *Kidney Int*. 2000;57(5):2080–2083.
96. Schnermann J. Juxtaglomerular cell complex in the regulation of renal salt excretion. *Am J Physiol*. 1998;274:R263–R279.
97. Wilcox CS, Peart WS. Release of renin and angiotensin II into plasma and lymph during hyperchloremia. *Am J Physiol Ren Physiol*. 1987;253(4):F734–F741.
98. Murray MD, Haag KM, Black PK, Hall SD, Brater DC. Variable furosemide absorption and poor predictability of response in elderly patients. *Pharmacotherapy*. 1997;17(1):98–106.
99. Kramer WG. Effect of food on the pharmacokinetics and pharmacodynamics of torsemide. *Am J Therapeut*. 1995;2(7):499–503.
100. Ellison DH, Velazquez H, Wright FS. Thiazide-sensitive sodium chloride cotransport in early distal tubule. *Am J Physiol*. 1987;253(3 Pt 2):F546–F554.
101. Brater DC. Update in diuretic therapy: clinical pharmacology. *Semin Nephrol*. 2011;31(6):483–494.
102. Murray MD, Deer MM, Ferguson JA, et al. Open-label randomized trial of torsemide compared with furosemide therapy for patients with heart failure. *Am J Med*. 2001;111:513–520.

103. Cosin J, Díez J, TORIC investigators. Torasemide in chronic heart failure: results of the TORIC study. *Eur J Heart Fail*. 2002;4(4):507–513.
104. Buggey J, Mentz RJ, Pitt B, et al. A reappraisal of loop diuretic choice in heart failure patients. *Am Heart J*. 2015;169(3):323–333.
105. Pitt B, Nicklas J. Loop diuretics in patients with heart failure: time to change to torsemide? *J Cardiovasc Pharmacol*. 2009;53(6):435–437.
106. DiNicolantonio JJ. Should torsemide be the loop diuretic of choice in systolic heart failure? *Future Cardiol*. 2012;8(5):707–728.
107. Mentz RJ, Anstrom KJ, Eisenstein EL, et al. Effect of torsemide vs furosemide after Discharge on all-cause mortality in patients hospitalized with heart failure: the TRANSFORM-HF randomized clinical trial. *JAMA*. 2023;329(3):214–223.
108. Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide I: effects of salt intake and renal compensation. *J Lab Clin Med*. 1983;102:450–458.
109. Inoue M, Okajima K, Itoh K, et al. Mechanism of furosemide resistance in analbuminemic rats and hypoalbuminemic patients. *Kidney Int*. 1987;32:198–203.
110. Chalasani N, Gorski JC, Horlander JCS, et al. Effects of albumin/furosemide mixtures on responses to durosemide in hypoalbuminemic patients. *J Am Soc Nephrol*. 2001;12:1010–1016.
111. Bartel C, Wirtz C, Brandt E, et al. Interaction of thiazide and loop diuretics with the basolateral para-aminohippurate transport system in isolated S2 segments of rabbit kidney proximal tubules. *J Pharmacol Exp Therapeut*. 1993;266:972–977.
112. Uwai Y, Saito H, Hashimoto Y, et al. Interaction and transport of thiazide diuretics, loop diuretics, and acetazolamide via rat renal organic anion transporter rOAT1. *J Pharmacol Exp Therapeut*. 2000;295:261–265.
113. Sweet DH, Bush KT, Nigam SK. The organic anion transporter family: from physiology to ontogeny and the clinic. *Am J Physiol*. 2001;281:F197–F205.
114. Tojo A, Sekine T, Nakajima N, et al. Immunohistochemical localization of multispecific renal organic anion transporter 1 in rat kidney. *J Am Soc Nephrol*. 1999;10:464–471.
115. Eraly SA, Vallon V, Vaughn DA, et al. Decreased renal organic anion secretion and plasma accumulation of endogenous organic anions in OAT1 knock-out mice. *J Biol Chem*. 2006;281:5072–5083.
116. Vallon V, Rieg T, Ahn SY, et al. Overlapping in vitro and in vivo specificities of the organic anion transporters OAT1 and OAT3 for loop and thiazide diuretics. *Am J Physiol Ren Physiol*. 2008;294:F867–F873.
117. Krick W, Wolff NA, Burckhardt G. Voltage-driven p-aminohippurate, chloride, and urate transport in porcine renal brush-border membrane vesicles. *Pflügers Archiv*. 2000;441:125–132.
118. Jutabha P, Anzai N, Kitamura K, et al. Human sodium phosphate transporter 4 (hNPT4/SLC17A3) as a common renal secretory pathway for drugs and urate. *J Biol Chem*. 2010;285:35123–35132.
119. Hasegawa M, Kusuhara H, Adachi M, et al. Multidrug resistance-associated protein 4 is involved in the urinary excretion of hydrochlorothiazide and furosemide. *J Am Soc Nephrol*. 2007;18:37–45.
120. Brater DC. Disposition and response to bumetanide and furosemide. *Am J Cardiol*. 1986;57:20A–25A.
121. Dunn CJ, Fitton A, Brogden R. Torasemide: an update of its pharmacological properties and therapeutic efficacy. *Drugs*. 1995;49:121–142.
122. Blose JS, Adams Jr KF, Patterson JH. Torsemide: a pyridine-sulfonylurea loop diuretic. *Ann Pharmacother*. 1995;29:396–402.
123. Stroupe KT, Forthofer MM, Brater DC, et al. Healthcare costs of patients with heart failure treated with torasemide or furosemide. *Pharmacoeconomics*. 2000;17:429–440.
124. Rose HJ, O'Malley K, Pruitt AW. Depression of renal clearance of furosemide in man by azotemia. *Clin Pharmacol Ther*. 1976;21:141–145.
125. Cemerik D, Wilcox CS, Giebisch G. Intracellular potential and K⁺ activity in rat kidney proximal tubular cells in acidosis and K⁺ depletion. *J Membr Biol*. 1982;69:159–165.
126. Loon NR, Wilcox CS. Mild metabolic alkalosis impairs the natriuretic response to bumetanide in normal human subjects. *Clin Sci*. 1998;94(3):287–292.
127. Besseghir K, Mosig D, Roch-Ramel F. Facilitation by serum albumin of renal tubular secretion of organic ions. *Am J Physiol*. 1989;256:F475–F484.
128. Pichette V, Geadah D, du Souich P. The influence of moderate hypoalbuminaemia on the renal metabolism and dynamics of furosemide in the rabbit. *Br J Pharmacol*. 1996;119:885–890.
129. Deleted in review.
130. Chennavasin P, Seiwel R, Brater DC, et al. Pharmacodynamic analysis of the furosemide-probenecid interaction in man. *Kidney Int*. 1979;16:187–195.
131. Chennavasin P, Seiwel R, Brater DC. Pharmacokinetic-dynamic analysis of the indomethacin-furosemide interaction in man. *J Pharmacol Exp Therapeut*. 1980;215:77–81.
132. Kirchner KA, Martin CJ, Bower JD. Prostaglandin E2 but not I2 restores furosemide response in indomethacin-treated rats. *Am J Physiol*. 1986;250:F980–F985.
133. Viatcheslav N, Bertog M, Canonica J, et al. Critical role of the mineralocorticoid receptor in aldosterone-dependent and aldosterone-independent regulation of ENaC in the distal nephron. *Am J Physiol Renal Physiol*. 2021;321(3):F257–F268.
- 133a. Palmer BF. Regulation of potassium homeostasis. *Clin J Am Soc Nephrol*. 2015;10(6):1050–1060.
134. Obermuller N, Bernstein P, Velazquez H, et al. Expression of the thiazide-sensitive Na-Cl cotransporter in rat and human kidney. *Am J Physiol*. 1995;269:F900–F910.
135. Plotkin MD, Kaplan MR, Verlander JW, et al. Localization of the thiazide sensitive Na-Cl cotransporter, rTSC1, in the rat kidney. *Kidney Int*. 1996;50:174–183.
136. Boer WH, Koomans HA, Dorhout Mees EJ. Acute effects of thiazides, with and without carbonic anhydrase inhibiting activity, on lithium and free water clearance in man. *Clin Sci*. 1989;76:539–545.
137. Colussi G, Bettinelli A, Tedeschi S, et al. A thiazide test for the diagnosis of renal tubular hypokalemic disorders. *Clin J Am Soc Nephrol*. 2007;2:454–460.
138. Walter SJ, Shirley DG. The effect of chronic hydrochlorothiazide administration on renal function in the rat. *Clin Sci*. 1986;70:379–387.
139. Beaumont K, Vaughn DA, Fanestil DD. Thiazide diuretic drug receptors in rat kidney: identification with [³H]metolazone. *Proc Natl Acad Sci USA*. 1988;85:2311–2314.
140. Tran JM, Farrell MA, Fanestil DD. Effect of ions on binding of the thiazide-type diuretic metolazone to kidney membrane. *Am J Physiol*. 1990;258:F908–F915.
141. Chang H, Fujita T. A kinetic model of the thiazide-sensitive Na-Cl cotransporter. *Am J Physiol*. 1999;276:F952–F959.
142. Fan M, Zhang J, Lee CL, Zhang J, Feng L. Structure and thiazide inhibition mechanism of the human Na-Cl cotransporter. *Nature*. 2023;614(7949):788–793.
143. Velazquez H, Wright FS. Control by drugs of renal potassium handling. *Annu Rev Pharmacol Toxicol*. 1986;26:293–309.
144. Okusa MD, Velazquez H, Ellison DH, et al. Luminal calcium regulates potassium transport by the renal distal tubule. *Am J Physiol*. 1990;258:F423–F428.
145. Ellison DH. Divalent cation transport by the distal nephron: insights from Bartter's and Gitelman's syndromes. *Am J Physiol Ren Physiol*. 2000;279:F616–F625.
146. Reilly RF, Huang CL. The mechanism of hypocalciuria with NaCl cotransporter inhibition. *Nat Rev Nephrol*. 2011;7:669–674.
147. Costanzo LS, Windhager EE. Calcium and sodium transport by the distal convoluted tubule of the rat. *Am J Physiol*. 1978;235:F492–F506.
148. den Dekker E, Hoenderop JG, Nilius B, et al. The epithelial calcium channels, TRPV5 & TRPV6: from identification towards regulation. *Cell Calcium*. 2003;33:497–507.
149. Friedman PA. Calcium transport in the kidney. *Curr Opin Nephrol Hypertens*. 1999;8:589–595.
150. Porter RH, Cox BG, Heaney D, et al. Treatment of hypoparathyroid patients with chlorthalidone. *N Engl J Med*. 1978;298:577–581.
151. Nijenhuis T, Vallon V, van der Kemp AW, et al. Enhanced passive Ca²⁺ reabsorption and reduced Mg²⁺ channel abundance explains thiazide-induced hypocalciuria and hypomagnesemia. *J Clin Invest*. 2005;115:1651–1658.

152. Belge H, Gailly P, Schwaller B, et al. Renal expression of parvalbumin is critical for NaCl handling and response to diuretics. *Proc Natl Acad Sci USA*. 2007;104:14849–14854.
153. Cheng CJ, Shiang JC, Hsu YJ, et al. Hypocalciuria in patients with Gitelman syndrome: role of blood volume. *Am J Kidney Dis*. 2007;49:693–700.
154. Dai LJ, Friedman PA, Qumme GA. Cellular mechanism of chlorothiazide and cellular potassium depletion on Mg²⁺ uptake in mouse distal convoluted tubule cells. *Kidney Int*. 1996;51:1008–1017.
155. Schlingmann KP, Weber S, Peters M, et al. Hypomagnesemia with secondary hypocalcemia is caused by mutations in TRPM6, a new member of the TRPM gene family. *Nat Genet*. 2002;31:166–170.
156. Voets T, Nilius B, Hoefs S, et al. TRPM6 forms the Mg²⁺ influx channel involved in intestinal and renal Mg²⁺ absorption. *J Biol Chem*. 2004;279:19–25.
157. Seldin DW, Eknoyan G, Suki WN, et al. Localization of diuretic action from the pattern of water and electrolyte excretion. *Ann N Y Acad Sci*. 1966;139:328–343.
158. Cesar KR, Magaldi AJ. Thiazide induces water absorption in the inner medullary collecting duct of normal and Brattleboro rats. *Am J Physiol*. 1999;277:F756–F760.
159. Kim GH, Lee JW, Oh YK, et al. Antidiuretic effect of hydrochlorothiazide in lithium-induced nephrogenic diabetes insipidus is associated with upregulation of aquaporin-2, Na-Cl co-transporter and epithelial sodium channel. *J Am Soc Nephrol*. 2004;15:2836–2843.
160. Muhlberg W, Mutschler E, Hofner A, et al. The influence of age on the pharmacokinetics and pharmacodynamics of bemetizide and triamterene: a single and multiple dose study. *Arch Gerontol Geriatr*. 2001;32:265–273.
161. Agarwal R, Sinha AD, Cramer AE, et al. Chlorthalidone for hypertension in advanced chronic kidney disease. *N Engl J Med*. 2021;385(27):2507–2519.
162. Welling PG. PHARMACOKINETICS of the thiazide diuretics. *Biopharm Drug Dispos*. 1986;7:501–535.
163. Woodman R, Brown C, Lockette W. Chlorthalidone decreases platelet aggregation and vascular permeability and promotes angiogenesis. *Hypertension*. 2010;56:463–470.
164. Chen P, Chaugai S, Zhao F, et al. Cardioprotective effect of thiazide-like diuretics: a meta-analysis. *Am J Hypertens*. 2015;28(12):1453–1463.
165. Olde Engberink RH, Frenkel WJ, van den Bogaard B, et al. Effects of thiazide-type and thiazide-like diuretics on cardiovascular events and mortality: systematic review and meta-analysis. *Hypertension*. 2015;65(5):1033–1040.
166. Ishani A, Cushman WC, Leatherman SM, et al. Chlorthalidone vs. Hydrochlorothiazide for hypertension-cardiovascular events. *N Engl J Med*. 2022;387(26):2401–2410.
167. Psaty BM, Smith NL, Heckbert SR, et al. Diuretic therapy, the alpha-adducin gene variant, and the risk of myocardial infarction or stroke in persons with treated hypertension. *JAMA*. 2002;287:1680–1689.
168. Turner ST, Schwartz GL, Chapman AB, et al. WNK1 kinase polymorphism and blood pressure response to a thiazide diuretic. *Hypertension*. 2005;46:758–765.
169. Arroyo JP, Lagnaz D, Ronzaud C, et al. Nedd4-2 modulates renal Na⁺-Cl⁻ cotransporter via the aldosterone-SGK1-Nedd4-2 pathway. *J Am Soc Nephrol*. 2011;22:1707–1719.
170. McDonough CW, Burbage SE, Duarte JD, et al. Association of variants in NEDD4L with blood pressure response and adverse cardiovascular outcomes in hypertensive patients treated with thiazide diuretics. *J Hypertens*. 2013;31:698–704.
171. Agarwal R, Filippatos G, Pitt B, et al. Cardiovascular and kidney outcomes with finerenone in patients with type 2 diabetes and chronic kidney disease: the FIDELITY pooled analysis. *Eur Heart J*. 2022;43(6):474–484.
172. Pei H, Wang W, Zhao D, et al. The use of a novel non-steroidal mineralocorticoid receptor antagonist finerenone for the treatment of chronic heart failure: a systematic review and meta-analysis. *Medicine (Baltim)*. 2018;97(16):e0254.
173. Bakris GL, Agarwal R, Anker SD, et al. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med*. 2020;383(23):2219–2229.
174. Filippatos G, Anker SD, Agarwal R, et al. Finerenone reduces risk of incident heart failure in patients with chronic kidney disease and type 2 diabetes: analyses from the FIGARO-DKD trial. *Circulation*. 2022;145(6):437–447.
175. Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol*. 1998;274:F817–F833.
176. McNicholas CM, Canessa CM. Diversity of channels generated by different combinations of epithelial sodium channel subunits. *J Gen Physiol*. 1997;109:681–692.
177. Garty H, Palmer LG. Epithelial sodium channels: function, structure, and regulation. *Physiol Rev*. 1997;77:359–396.
178. Kelly O, Lin C, Ramkumar M, et al. Characterization of an amiloride binding region in the alpha-subunit of ENaC. *Am J Physiol Ren Physiol*. 2003;285:F1279–F1290.
179. Ryan MP, Devane J, Ryan MF, et al. Effects of diuretics on the renal handling of magnesium. *Drugs*. 1984;28(suppl 1):167–181.
180. Bärack L, Kuhl A, Hillisch A, et al. Discovery of BAY 94-8862: a nonsteroidal antagonist of the mineralocorticoid receptor for the treatment of cardiorenal diseases. *ChemMedChem*. 2012;7(8):1385–1403.
181. Perez-Ayuso RM, Arroyo V, Planas R, et al. Randomized comparative study of efficacy of furosemide versus spironolactone in nonazotemic cirrhosis with ascites. Relationship between the diuretic response and the activity of the renin-aldosterone system. *Gastroenterology*. 1983;84:961–968.
182. Williams B, MacDonald TM, Morant S, et al. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet*. 2015;386(10008):2059–2068.
183. Carey RM, Calhoun DA, Bakris GL, et al. Resistant hypertension: detection, evaluation, and management: a scientific statement from the American heart association. *Hypertension*. 2018;72(5):e53–e90.
184. Pitt B, Zannad F, Remme WJ, et al. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med*. 1999;341(10):709–717.
185. Beldhuis IE, Myhre PL, Bristow M, et al. Spironolactone in patients with heart failure, preserved ejection fraction, and worsening renal function. *J Am Coll Cardiol*. 2021;77(9):1211–1221.
186. Zannad F, McMurray JJ, Krum H, et al. Eplerenone in patients with systolic heart failure and mild symptoms. *N Engl J Med*. 2011;364(1):11–21.
187. Mutschler E, Gilfrich HJ, Knauf H, et al. PHARMACOKINETICS of triamterene. *Clin Exp Hypertens*. 1983;5:249–269.
188. Villeneuve JP, Rocheleau F, Raymond G. Triamterene kinetics and dynamics in cirrhosis. *Clin Pharmacol Ther*. 1984;35:831–837.
189. Somogyi AA, Hovens CM, Muirhead MR, et al. Renal tubular secretion of amiloride and its inhibition by cimetidine in humans and in an animal model. *Drug Metab Dispos*. 1989;17:190–196.
190. Knauf H, Mohrke W, Mutschler E. Delayed elimination of triamterene and its active metabolite in chronic renal failure. *Eur J Clin Pharmacol*. 1983;24:453–456.
191. Lynn KL, Bailey RR, Swainson CP, et al. Renal failure with potassium-sparing diuretics. *N Z Med J*. 1985;98:629–633.
192. Gardiner P, Schrode K, Quinlan D, et al. Spironolactone metabolism: steady-state serum levels of the sulfur-containing metabolites. *J Clin Pharmacol*. 1989;29:342–347.
193. Cook CS, Hauswald C, Oppermann JA, et al. Involvement of cytochrome P-450III_A in metabolism of potassium canrenoate to an epoxide: mechanism of inhibition of the epoxide formation by spironolactone and its sulfur-containing metabolite. *J Pharmacol Exp Therapeut*. 1993;266:1–7.
194. McInnes GT. Relative potency of amiloride and spironolactone in healthy man. *Clin Pharmacol Ther*. 1982;31:472–477.
195. Rahn KH. Clinical pharmacology of diuretics. *Clin Exp Hypertens*. 1983;5:157–166.
196. Stier Jr CT. Eplerenone: a selective aldosterone blocker. *Cardiovasc Drug Rev*. 2003;21:169–184.
197. Keating GM, Plosker GL. Eplerenone: a review of its use in left ventricular systolic dysfunction and heart failure after acute myocardial infarction. *Drugs*. 2004;64:2689–2707.

198. Cook CS, Berry LM, Bible RH, et al. PHARMACOKINETICS and metabolism of [14C]epplerone after oral administration to humans. *Drug Metab Dispos.* 2003;31:1448–1455.
199. Schombelan M, Sebastian A, Biglieri EG, et al. Amelioration of hypokalemia by amiloride in diverse syndromes associated with renal potassium wasting. *Kidney Int.* 1982;21:157.
200. Rosner MH, Gupta R, Ellison D, et al. Management of cirrhotic ascites: physiological basis of diuretic action. *Eur J Intern Med.* 2006;17:8–19.
201. Santos J, Planas R, Pardo A, et al. Spironolactone alone or in combination with furosemide in the treatment of moderate ascites in nonazotemic cirrhosis. a randomized comparative study of efficacy and safety. *J Hepatol.* 2003;39:187–192.
202. Bruce A, Runyon MD. Management of adult patients with ascites due to cirrhosis: update 2012. *Hepatology.* 2013;57(4):1651–1653.
203. Nishizaka MK, Zaman MA, Calhoun DA. Efficacy of low-dose spironolactone in subjects with resistant hypertension. *Am J Hypertens.* 2003;16:925–930.
204. Calhoun DA, Nishizaka MK, Zaman MA, et al. Hyperaldosteronism among black and white subjects with resistant hypertension. *Hypertension.* 2002;40:892–896.
205. Ouzan J, Perault C, Lincoff AM, et al. The role of spironolactone in the treatment of patients with refractory hypertension. *Am J Hypertens.* 2002;15:333–339.
206. Sato A, Hayashi K, Saruta T. Antiproteinuric effects of mineralocorticoid receptor blockade in patients with chronic renal disease. *Am J Hypertens.* 2005;18:44–49.
207. Shavit L, Lifschitz MD, Epstein M. Aldosterone blockade and the mineralocorticoid receptor in the management of chronic kidney disease: current concepts and emerging treatment paradigms. *Kidney Int.* 2012;81:955–968.
208. Shen W, Alshehri M, Desai S, Wilcox C. The effect of amiloride on proteinuria in patients with proteinuric kidney disease. *Am J Nephrol.* 2021;52(5):368–377.
209. Pitt B, Remme W, Zannad F, et al. Eplerenone, a selective aldosterone blocker, in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med.* 2003;348:1309–1321.
210. Iraqi W, Rossignol P, Angioi M, et al. Extracellular cardiac matrix biomarkers in patients with acute myocardial infarction complicated by left ventricular dysfunction and heart failure: insights from the Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study (EPHESUS) study. *Circulation.* 2009;119:2471–2479.
211. Juurlink DN, Mamdani MM, Lee DS, et al. Rates of hyperkalemia after publication of the randomized aldactone evaluation study. *N Engl J Med.* 2004;351:543–551.
212. Rose LI, Underwood RH, Newmark SR, et al. Pathophysiology of spironolactone-induced gynecomastia. *Ann Intern Med.* 1977;87:398–403.
213. Effectiveness of spironolactone added to an angiotensin-converting enzyme inhibitor and a loop diuretic for severe chronic congestive heart failure (the Randomized Aldactone Evaluation Study [RALES]). *Am J Cardiol.* 1996;78:902–907.
214. Hulter HN, Licht JH, Glynn RD, et al. Pathophysiology of chronic renal tubular acidosis induced by administration of amiloride. *J Lab Clin Med.* 1980;95:637–653.
215. Ettinger B. Excretion of triamterene and its metabolite in triamterene stone patients. *J Clin Pharmacol.* 1985;25:365–368.
216. Favre L, Glasson P, Vallotton MB. Reversible acute renal failure from combined triamterene and indomethacin: a study in healthy subjects. *Ann Intern Med.* 1982;96:317–320.
217. Jose PA, Felder RA. What can we learn from the selective manipulation of dopaminergic receptors about the pathogenesis and treatment of hypertension? *Curr Opin Nephrol Hypertens.* 1996;5:447–451.
218. Beale RJ, Hollenberg SM, Vincent JL, et al. Vasopressor and inotropic support in septic shock: an evidence-based review. *Crit Care Med.* 2004;32(suppl):S455–S465.
219. Joannidis M, Druml W, Forni LG, et al. Prevention of acute kidney injury and protection of renal function in the intensive care unit: update 2017: expert opinion of the Working Group on Prevention, AKI section, European Society of Intensive Care Medicine. *Intensive Care Med.* 2017;43(6):730–749.
220. Decaux G, Soupart A, Vassart G. Non-peptide arginine-vasopressin antagonists: the vaptans. *Lancet.* 2008;371:1624–1632.
221. Ohnishi A, Orita Y, Okahara R, et al. Potent aquaretic agent. a novel nonpeptide selective vasopressin 2 antagonist (OPC-31260) in men. *J Clin Invest.* 1993;92:2653–2659.
222. Goldsmith SR. Is there a cardiovascular rationale for the use of combined vasopressin V1a/V2 receptor antagonists? *Am J Med.* 2006;119(suppl 1):S93–S96.
223. Hoorn EJ, Zietse R. Vasopressin-receptor antagonists. *Future Cardiol.* 2010;6:523–534.
224. Gines P, Wong F, Watson H, et al. Effects of satavaptan, a selective vasopressin V receptor antagonist, on ascites and serum sodium in cirrhosis with hyponatremia: a randomized trial. *Hepatology.* 2008;48:204–213.
225. Schrier RW, Gross P, Gheorghiade M, et al. Tolvaptan, a selective oral vasopressin V2-receptor antagonist, for hyponatremia. *N Engl J Med.* 2006;355:2099–2112.
226. Konstam MA, Gheorghiade M, Burnett Jr JC, et al. Effects of oral tolvaptan in patients hospitalized for worsening heart failure: the EVEREST Outcome Trial. *JAMA.* 2007;297(12):1319–1331.
227. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med.* 2014;371:993–1004.
228. Deleted in review.
229. Packer M. Critical reanalysis of the mechanisms underlying the cardiorenal benefits of SGLT2 inhibitors and Reaffirmation of the nutrient deprivation signaling/autophagy hypothesis. *Circulation.* 2022;146(18):1383–1405.
230. Deleted in review.
231. Thomson SC, Rieg T, Miracle C, et al. Acute and chronic effects of SGLT2 blockade on glomerular and tubular function in the early diabetic rat. *Am J Physiol Regul Integr Comp Physiol.* 2012;302(1):R75–R83.
232. Griffin M, Rao VS, Ivey-Miranda J, et al. Empagliflozin in heart failure: diuretic and cardiorenal effects. *Circulation.* 2020;142(11):1028–1039.
233. Packer M, Wilcox CS, Testani JM. Critical analysis of the effects of SGLT2 inhibitors on renal tubular sodium, water and chloride homeostasis and their role in influencing heart failure outcomes. *Circulation.* 2023;148(4):354–372.
234. Onishi A, Fu Y, Patel R, et al. A role for tubular Na⁽⁺⁾/H⁽⁺⁾ exchanger NHE3 in the natriuretic effect of the SGLT2 inhibitor empagliflozin. *Am J Physiol Ren Physiol.* 2020;319(4):F712–F728.
235. Niu Y, Liu R, Guan C, et al. Structural basis of inhibition of the human SGLT2-MAP17 glucose transporter. *Nature.* 2022;601(7892):280–284.
236. Coady MJ, El Tarazi A, Santer R, et al. MAP17 is a necessary activator of renal Na⁺/Glucose cotransporter SGLT2. *J Am Soc Nephrol.* 2017;28(1):85–93.
237. Heerspink HJ, Perkins BA, Fitchett DH, et al. Sodium glucose cotransporter 2 inhibitors in the treatment of diabetes mellitus: cardiovascular and kidney effects, potential mechanisms, and clinical applications. *Circulation.* 2016;134(10):752–772.
238. Heerspink HJL, Stefánsson BV, Correa-Rotter R, et al. Dapagliflozin in patients with chronic kidney disease. *N Engl J Med.* 2020;383(15):1436–1446.
239. Chen S, Wang Q, Christodoulou A, et al. Sodium glucose cotransporter-2 inhibitor empagliflozin reduces infarct size independently of sodium glucose cotransporter-2. *Circulation.* 2023;147(3):276–279.
240. Fukushima K, Kitamura S, Tsuji K, Wada J. Sodium-glucose cotransporter 2 inhibitors work as a “regulator” of autophagic activity in overnutrition diseases. *Front Pharmacol.* 2021;12:761842. Published 2021 Oct 21.
241. Yang X, Shen Q, Xu X. Effects of SGLT2 inhibitors on potassium homeostasis: a narrative review. *Diabetes Res Clin Pract.* 2023;196:110239.
242. Wilcox CS, Shen W, Boulton DW, Leslie BR, Griffen SC. Interaction between the sodium-glucose-linked transporter 2 inhibitor dapagliflozin and the loop diuretic bumetanide in normal human subjects. *J Am Heart Assoc.* 2018;7(4):e007046.
243. Wanner C, Inzucchi SE, Lachin JM, EMPA-REG OUTCOME Investigators, et al. Empagliflozin and progression of kidney disease in type 2 diabetes. *N Engl J Med.* 2016;375(4):323–334.
244. Anders HJ, Davis JM, Thurau K. Nephron protection in diabetic kidney disease. *N Engl J Med.* 2016;375(21):2096–2098.

245. Esteva-Font C, Anderson MO, Verkman AS. Urea transporter proteins as targets for small-molecule diuretics. *Nat Rev Nephrol*. 2015;11(2):113–123.
246. Li F, Lei T, Zhu J, et al. A novel small-molecule thienouquinolin urea transporter inhibitor acts as a potential diuretic. *Kidney Int*. 2013;83:1076–1086.
247. Cil O, Esteva-Font C, Tas ST, et al. Salt-sparing diuretic action of a water-soluble urea analog inhibitor of urea transporters UT-A and UT-B in rats. *Kidney Int*. 2015;88(2):311–320.
248. Kharade SV, Flores D, Lindsley CW, et al. ROMK inhibitor actions in the nephron probed with diuretics. *Am J Physiol Ren Physiol*. 2016;310(8):F732–F737.
249. Araujo M, Welch WJ, Zhou X, et al. Inhibition of ROMK blocks macula densa tubuloglomerular feedback yet causes renal vasoconstriction in anesthetized rats. *Am J Physiol Ren Physiol*. 2017;312(6):F1120–F1127.
250. Wall SM, Lazo-Fernandez Y. The role of pendrin in renal physiology. *Annu Rev Physiol*. 2015;77:363–378.
251. Cil O, Haggie PM, Phuan PW, et al. Small-molecule inhibitors of pendrin potentiate the diuretic action of furosemide. *J Am Soc Nephrol*. 2016;27(12):3706–3714.
252. Patel-Chamberlin M, Varasteh Kia M, Xu J, et al. The role of epithelial sodium channel ENAC and the apical Cl⁻/HCO₃⁻ exchanger pendrin in compensatory salt reabsorption in the setting of Na-Cl cotransporter (NCC) inactivation. *PLoS One*. 2016;11(3):e0150918.
253. Kelly RA, Wilcox CS, Mitch WE, et al. Response of the kidney to furosemide II: effect of captopril on sodium balance. *Kidney Int*. 1983;24:233–239.
254. Wilcox CS, Guzman NJ, Mitch WE, et al. Na⁺, K⁺, and BP homeostasis in man during furosemide: effects of prazosin and captopril. *Kidney Int*. 1987;31:135–141.
255. Wilcox CS, Loon NR, Ameer B, et al. Renal and hemodynamic responses to bumetanide in hypertension: effects of nitrendipine. *Kidney Int*. 1989;36:719–725.
256. Bak M, Shalmi M, Petersen JS, et al. Effects of angiotensin-converting enzyme inhibition on renal adaptations to acute furosemide administration in conscious rats. *J Pharmacol Exp Therapeut*. 1993;266:33–40.
257. Nowack R, Fliser D, Richter J, et al. Effects of angiotensin-converting enzyme inhibition on renal sodium handling after furosemide injection. *Clin Invest*. 1993;71:622–627.
258. Shirley DG, Walter SJ, Unwin RJ. Mechanism of the impaired natriuretic response to frusemide during sodium depletion: a micro-puncture study in rats. *Clin Sci*. 1996;91:299–305.
259. Ter Maaten JM, Rao VS, Hanberg JS, et al. Renal tubular resistance is the primary driver for loop diuretic resistance in acute heart failure. *Eur J Heart Fail*. 2017;19(8):1014–1022.
260. Kaissling B, Bachmann S, Kriz W. Structural adaptation of the distal convoluted tubule to prolonged furosemide treatment. *Am J Physiol*. 1985;248:F374–F381.
261. Beck FX, Ohno A, Muller E, et al. Inhibition of angiotensin-converting enzyme modulates structural and functional adaptation to loop diuretic-induced diuresis. *Kidney Int*. 1997;51:36–43.
262. Kobayashi S, Clemmons DR, Nogami H, et al. Tubular hypertrophy due to work load induced by furosemide associated with increases of IGF-I and IGFBP-1. *Kidney Int*. 1995;47:818–828.
263. Scherzer P, Wald H, Popovtzer MM. Enhanced glomerular filtration and Na⁺-K⁺-ATPase with furosemide administration. *Am J Physiol*. 1987;252:F910–F915.
264. Kim J, Welch WJ, Cannon JK, et al. Immunocytochemical response of type A and type B intercalated cells to increased sodium chloride delivery. *Am J Physiol*. 1992;262:F288–F302.
265. Barlet-Bas C, Khadouri C, Marsy S, et al. Enhanced intracellular sodium concentration in kidney cell recruits a latent pool of Na-K-ATPase whose size is modulated by corticosteroids. *J Biol Chem*. 1990;265:7799–7803.
266. Stanton BA, Kaissling B. Adaptation of distal tubule and collecting duct to increased Na delivery II: Na⁺ and K⁺ transport. *Am J Physiol*. 1988;255:F1269–F1275.
267. DeWardener HE. Idiopathic edema: role of diuretic abuse. *Kidney Int*. 1981;19:881.
268. Almeshari K, Ahlstrom NG, Capraro FE, et al. A volume-independent component to post-diuretic sodium retention in man. *J Am Soc Nephrol*. 1993;3:1878–1883.
269. Ecelbarger CA, Terris J, Hoyer JR, Nielsen S, Wade JB, Knepper MA. Localization and regulation of the rat renal Na⁺(+)-K⁺(+)-2Cl⁻ cotransporter, BSC-1. *Am J Physiol*. 1996;271(3 Pt 2):F619–F628.
270. Fanestil DD, Vaughn DA, Blakely P. Metabolic acid-base influences on renal thiazide receptor density. *Am J Physiol*. 1997;272:R2004–R2008.
271. van Meyel JJ, Smits P, Russel FG, et al. Diuretic efficiency of furosemide during continuous administration versus bolus injection in healthy volunteers. *Clin Pharmacol Ther*. 1992;51:440–444.
272. Rudy DW, Voelker JR, Greene PK, et al. Loop diuretics for chronic renal insufficiency: a continuous infusion is more efficacious than bolus therapy. *Ann Intern Med*. 1991;115:360–366.
273. Haris A, Rado JP. Patterns of potassium wasting in response to stepwise combinations of diuretics in nephrotic syndrome. *Int J Clin Pharm Ther*. 1999;37:332–340.
274. Kamel KS, Halperin ML. Use of urine electrolytes and urine osmolality in the clinical diagnosis of fluid, electrolytes, and acid-base disorders. *Kidney Int Rep*. 2021;6(5):1211–1224.
275. Imbs JL, Schmidt M, Velly J, et al. Comparison of the effect of two groups of diuretics on renin secretion in the anaesthetized dog. *Clin Sci Mol Med*. 1977;52:171–182.
276. Wilson TW, Loadholt CB, Privitera PJ, et al. Furosemide increases urine 6-keto-prostaglandin F1 alpha. relation to natriuresis, vasodilation, and renin release. *Hypertension*. 1982;4:634–641.
277. Dzau VJ, Colucci WS, Williams GH, et al. Sustained effectiveness of converting-enzyme inhibition in patients with severe congestive heart failure. *N Engl J Med*. 1980;302:1373–1379.
278. Good JM, Brady AJB, Noormohamed FH, et al. Effect of intense angiotensin II suppression on the diuretic response to furosemide during chronic ACE inhibition. *Circulation*. 1994;90:220–224.
279. Wilcox CS, Williams CM, Smith TB, et al. Diagnostic uses of angiotensin-converting enzyme inhibitors in renovascular hypertension. *Am J Hypertens*. 1988;1:344S–349S.
280. Wilcox CS, Mitch WE, Kelly RA, et al. Factors affecting potassium balance during furosemide administration. *Clin Sci*. 1984;67:195–203.
281. Johnson CI, McGrath BP, Matthews PG. Interaction between captopril and hydrochlorothiazide in hypertension. *Med J Aust*. 1979;2:18.
282. Kaji DK, Chase Jr HS, Eng JP, et al. Prostaglandin E2 inhibits Na-K-2Cl cotransport in medullary thick ascending limb cells. *Am J Physiol*. 1996;271:C354–C361.
283. Brater DC, Anderson S, Baird B, Campbell WB. Effects of ibuprofen, naproxen, and sulindac on prostaglandins in men. *Kidney Int*. 1985;27(1):66–73.
284. Sakairi Y, Jacobson HR, Noland TD, et al. Luminal prostiglandin E receptors regulate salt and water transport in rabbit cortical collecting duct. *Am J Physiol*. 1995;269:F257–F265.
285. Kramer HJ, Dusing R, Stinnesbeck B, et al. Interaction of conventional and antidiuretic diuretics with the renal prostaglandin system. *Clin Sci*. 1980;59:67–70.
286. Tiggeler RG, Koene RA, Wijdeveld PG. Inhibition of furosemide-induced natriuresis by indomethacin in patients with the nephrotic syndrome. *Clin Sci Mol Med*. 1977;52:149–151.
287. Kirchner KA, Brandon S, Mueller RA, et al. Mechanism of attenuated hydrochlorothiazide response during indomethacin administration. *Kidney Int*. 1987;31:1097–1103.
288. Favre L, Glasson P, Riondel A, et al. Interaction of diuretics and non-steroidal anti-inflammatory drugs in man. *Clin Sci*. 1983;64:407–415.
289. Data JL, Rane A, Gerkens J, et al. The influence of indomethacin on the pharmacokinetics, diuretic response and hemodynamics of furosemide in the dog. *J Pharmacol Exp Therapeut*. 1978;206:431–438.
290. Johnston GD, Hiatt WR, Nies AS, et al. Factors modifying the early nondiuretic vascular effects of furosemide in man. The possible role of renal prostaglandins. *Circ Res*. 1983;53:630–635.
291. Frolich JC, Hollifield JW, Dormois JC, et al. Suppression of plasma renin activity by indomethacin in man. *Circ Res*. 1976;39:447–452.
292. Kammerl MC, Nusing RM, Richthammer W, et al. Inhibition of COX-2 counteracts the effects of diuretics in rats. *Kidney Int*. 2001;60:1684–1691.
293. Kawada N, Solis G, Ivey N, et al. Cyclooxygenase-1-deficient mice have high sleep-to-wake blood pressure ratios and renal vasoconstriction. *Hypertension*. 2005;45:1131–1138.

294. Danielsen H, Pedersen EB, Madsen M, et al. Abnormal renal sodium excretion in the nephrotic syndrome after furosemide: relation to glomerular filtration rate. *Acta Med Scand*. 1985;217:513–518.
295. Goldsmith SR, Francis GS, Cowley AW, et al. Increased plasma arginine vasopressin levels in patients with congestive heart failure. *J Am Coll Cardiol*. 1983;1:1385–1390.
296. Ghose RR. Plasma arginine vasopressin in hyponatraemic patients receiving diuretics. *Postgrad Med J*. 1985;61:1043–1046.
297. Field MJ, Stanton BA, Giebisch GH. Influence of ADH on renal potassium handling: a micropuncture and micropfusion study. *Kidney Int*. 1984;25:502–511.
298. Bugaj V, Pochynuk O, Stockand JD. Activation of the epithelial Na⁺ channel in the collecting duct by vasopressin contributes to water reabsorption. *Am J Physiol Ren Physiol*. 2009;297:F1411–F1418.
299. Mutig K, Saritas T, Uchida S, et al. Short-term stimulation of the thiazide-sensitive Na⁺-Cl⁻ cotransporter by vasopressin involves phosphorylation and membrane translocation. *Am J Physiol Ren Physiol*. 2009;298:F502–F509.
300. Petersen JS, DiBona GF. Reflex control of renal sympathetic nerve activity during furosemide diuresis in rats. *Am J Physiol*. 1994;266:R537–R545.
301. Fett DL, Caverio PG, Burnett JC. Low-dose atrial natriuretic factor and furosemide in experimental acute congestive heart failure. *J Am Soc Nephrol*. 1993;4:162–167.
302. Hoorn EJ, Ellison DH. Diuretic resistance. *Am J Kidney Dis*. 2017;69(1):136–142.
303. van der Heijden AG, Huysmans FT, van Hamersvelt HW. Foot volume increase on nifedipine is not prevented by pretreatment with diuretics. *J Hypertens*. 2004;22:425–430.
304. Testani JM, Hanberg JS, Cheng S, et al. Rapid and highly accurate prediction of poor loop diuretic natriuretic response in patients with heart failure. *Circulation Heart fail*. 2016;9(1):e002370.
305. Rao VS, Ivey-Miranda JB, Cox ZL, et al. Natriuretic equation to predict loop diuretic response in patients with heart failure. *J Am Coll Cardiol*. 2021;77(6):695–708.
306. Caravaca Pérez P, Nuche J, Morán Fernández L, et al. Potential role of natriuretic response to furosemide stress test during acute heart failure. *Circulation Heart fail*. 2021;14(6):e008166.
307. Davenport A. Ultrafiltration in diuretic-resistant volume overload in nephrotic syndrome and patients with ascites due to chronic liver disease. *Cardiology*. 2001;96:190–195.
308. Ellison DH. The physiologic basis of diuretic synergism: its role in treating diuretic resistance [see comments]. *Ann Intern Med*. 1991;114:886–894.
309. Brater DC, Pressley RH, Anderson SA. Mechanisms of the synergistic combination of metolazone and bumetanide. *J Pharmacol Exp Therapeut*. 1985;233:70–74.
310. Knauf H, Mutschler E. Diuretic effectiveness of hydrochlorothiazide and furosemide alone and in combination in chronic renal failure. *J Cardiovasc Pharmacol*. 1995;26:394–400.
311. Fliser D, Schroter M, Neubeck M, et al. Coadministration of thiazides increases the efficacy of loop diuretics even in patients with advanced renal failure. *Kidney Int*. 1994;46:482–488.
312. Nakahama H, Orita Y, Yamazaki M, et al. Pharmacokinetic and pharmacodynamic interactions between furosemide and hydrochlorothiazide in nephrotic patients. *Nephron*. 1988;49:223–227.
313. Channer KS, McLean KA, Lawson-Matthew P, et al. Combination diuretic treatment in severe heart failure: a randomised controlled trial. *Br Heart J*. 1994;71:146–150.49.
314. Jonassen TEN, Gronbeck L, Shalmi M, et al. Supra-additive natriuretic synergism between bendroflumethiazide and furosemide in rats. *J Pharmacol Exp Therapeut*. 1995;275:558–565.
315. Wollam GL, Tarazi RC, Bravo EL, et al. Diuretic potency of combined hydrochlorothiazide and furosemide therapy in patients with azotemia. *Am J Med*. 1982;72:929–938.
316. Jeunemaitre X, Charru A, Chatellier G, et al. Long-term metabolic effects of spironolactone and thiazides combined with potassium-sparing agents for treatment of essential hypertension. *Am J Cardiol*. 1988;62:1072–1077.
317. Francis GS, Siegel RM, Goldsmith SR, et al. Acute vasoconstrictor response to intravenous furosemide in patients with chronic congestive heart failure. *Ann Intern Med*. 1985;103:1–6.
318. Anisman SD, Erickson SB, Morden NE. How to prescribe loop diuretics in oedema. *BMJ*. 2019;364:1359.
319. Andreoli TE. Edematous states: an overview. *Kidney Int*. 1997;51(suppl 59):S2–S10.
320. Rasool A, Palevsky PM. Treatment of edematous disorders with diuretics. *Am J Med Sci*. 2000;319:25–37.
321. Ellison DH, Felker GM. Diuretic treatment in heart failure. *N Engl J Med*. 2017;377(20):1964–1975.
322. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American heart association joint committee on clinical practice guidelines. *J Am Coll Cardiol*. 2022;79(17):e263–e421.
323. Martin PY, Schrier RW. Sodium and water retention in heart failure: pathogenesis and treatment. *Kidney Int*. 1997;59(suppl):S57–S61.
324. O'Connor CM, Starling RC, Hernandez AF, et al. Effect of nesiritide in patients with acute decompensated heart failure. *N Engl J Med*. 2011;365(1):32–43.
325. Kittleson MM. Nesiritide and me. *Circulation Heart fail*. 2018;11(8):e005440.
326. Johnson W, Omland T, Hall C, et al. Neurohormonal activation rapidly decreases after intravenous therapy with diuretics and vasodilators for class IV heart failure. *J Am Coll Cardiol*. 2002;39:1623–1629.
327. Testani JM, Chen J, McCauley BD, et al. Potential effects of aggressive decongestion during the treatment of decompensated heart failure on renal function and survival. *Circulation*. 2010;122:265–272.
328. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364:797–805.
329. Noe LL, Vreeland MG, Pezzella SM, et al. A pharmacoeconomic assessment of torsemide and furosemide in the treatment of patients with congestive heart failure. *Clin Ther*. 1999;21:854–866.
330. Tomiyama H, Nakayama T, Watanabe G, et al. Effects of short-acting and long-acting loop diuretics on heart rate variability in patients with chronic compensated congestive heart failure. *Am Heart J*. 1999;137:543–548.
331. Mentz RJ, Hasselblad V, DeVore AD, et al. Torsemide versus furosemide in patients with acute heart failure (from the ASCEND-HF trial). *Am J Cardiol*. 2016;117(3):404–411.
332. Brater DC, Seiwel R, Anderson S, Chennavasani P, et al. Absorption and disposition of furosemide in congestive heart failure. *Kidney Int*. 1982;22:171–176.
333. Brater DC, Day B, Burdette A, et al. Bumetanide and furosemide in heart failure. *Kidney Int*. 1984;26:183–189.
334. Vargo DL, Kramer WG, Black PK, et al. Bioavailability, pharmacokinetics, and pharmacodynamics of torsemide and furosemide in patients with congestive heart failure. *Clin Pharmacol Ther*. 1995;57:601–609.
335. Costanzo MR, Ronco C, Abraham WT, et al. Extracorporeal ultrafiltration for fluid overload in heart failure: current status and prospects for further research. *J Am Coll Cardiol*. 2017;69(19):2428–2445.
336. Bart BA, Goldsmith SR, Lee KL, et al. Ultrafiltration in decompensated heart failure with cardiorenal syndrome. *N Engl J Med*. 2012;367:2296–2304.
337. Kabach M, Alkhwam H, Shah S, et al. Ultrafiltration versus intravenous loop diuretics in patients with acute decompensated heart failure: a meta-analysis of clinical trials. *Acta Cardiol*. 2017;72(2):132–141.
338. Costanzo MR, Negoianu D, Jaski BE, et al. Aquapheresis versus intravenous diuretics and hospitalizations for heart failure. *JACC Heart Fail*. 2016;4(2):95–105.
339. Trullàs JC, Morales-Rull JL, Casado J, et al. CLOROTIC trial investigators. Combining loop with thiazide diuretics for decompensated heart failure: the CLOROTIC trial. *Eur Heart J*. 2023;44(5):411–421.
340. Chen HH, Anstrom KJ, Givertz MM, et al. Low-dose dopamine or low-dose nesiritide in acute heart failure with renal dysfunction: the ROSE acute heart failure randomized trial. *JAMA*. 2013;310:2533–2543.
341. Dikshit K, Vyden JK, Forrester JS, et al. Renal and extrarenal hemodynamic effects of furosemide in congestive heart failure after acute myocardial infarction. *N Engl J Med*. 1973;288:1087–1090.

342. Johnston GD, Nicholls DP, Leahey WJ, et al. The effects of captopril on the acute vascular responses to frusemide in man. *Clin Sci*. 1983;65:359–363.
343. Verma SP, Silke B, Hussain M, et al. First-line treatment of left ventricular failure complicating acute myocardial infarction: a randomised evaluation of immediate effects of diuretic, venodilator, arteriodilator, and positive inotropic drugs on left ventricular function. *J Cardiovasc Pharmacol*. 1987;10:38–46.
344. Cotter G, Metzker E, Kaluski E, et al. Randomised trial of high-dose isosorbide dinitrate plus low-dose furosemide versus high-dose furosemide plus low-dose isosorbide dinitrate in severe pulmonary oedema. *Lancet*. 1998;351:389–393.
345. Duvernoy CS, Bates ER. Management of cardiogenic shock attributable to acute myocardial infarction in the reperfusion era. *J Intensive Care Med*. 2005;20:188–198.
346. McBride BF, White CM. Acute decompensated heart failure: a contemporary approach to pharmacotherapeutic management. *Pharmacotherapy*. 2003;23:997–1020.
347. Faris R, Flather M, Purcell H, et al. Current evidence supporting the role of diuretics in heart failure: a meta analysis of randomised controlled trials. *Int J Cardiol*. 2002;82:149–158.
348. Janardhanan R, Desai AS, Solomon SD. Therapeutic approaches to diastolic dysfunction. *Curr Hypertens Rep*. 2009;11(4):283–291.
349. Knauf H. Functional state of the nephron and diuretic dose-response rationale for low dose combination therapy. *Cardiology*. 1994;84(suppl 2):18–26.
350. Jentzer JC, DeWald TA, Hernandez AF. Combination of loop diuretics with thiazide-type diuretics in heart failure. *J Am Coll Cardiol*. 2010;56:1527–1534.
351. Rao VS, Planavsky N, Hanberg JS, et al. Compensatory distal reabsorption drives diuretic resistance in human heart failure. *J Am Soc Nephrol*. 2017;28(11):3414–3424.
352. McMurray JJ. Clinical practice: systolic heart failure. *N Engl J Med*. 2010;362:228–238.
353. Butler J, Anker SD, Lund LH, et al. Patisomer for the management of hyperkalemia in heart failure with reduced ejection fraction: the DIAMOND trial. *Eur Heart J*. 2022;43(41):4362–4373.
354. Kasama S, Toyama T, Kumakura H, et al. Effect of spironolactone on cardiac sympathetic nerve activity and left ventricular remodeling in patients with dilated cardiomyopathy. *J Am Coll Cardiol*. 2003;41:574–581.
355. Pitt B. Do diuretics and aldosterone receptor antagonists improve ventricular remodeling? *J Card Fail*. 2002;8:S491–S493.
356. Edelmann F, Wachter R, Schmidt AG, et al. Effect of spironolactone on diastolic function and exercise capacity in patients with heart failure with preserved ejection fraction: the Aldo-DHF randomized controlled trial. *JAMA*. 2013;309:781–791.
357. Pitt B, Pfeffer MA, Assmann SF, et al. Spironolactone for heart failure with preserved ejection fraction. *N Engl J Med*. 2014;370:1383–1392.
358. Butler J, Forman DE, Abraham WT, et al. Relationship between heart failure treatment and development of worsening renal function among hospitalized patients. *Am Heart J*. 2004;147:331–338.
359. Solomon SD, McMurray JJV, Claggett B, et al. Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med*. 2022;387(12):1089–1098.
360. Solomon SD, Rizkala AR, Gong J, et al. Angiotensin receptor neprilysin inhibition in heart failure with preserved ejection fraction: rationale and design of the PARAGON-HF trial. *JACC Heart Fail*. 2017;5(7):471–482.
361. Sarnak MJ. A patient with heart failure and worsening kidney function. *Clin J Am Soc Nephrol*. 2014;9:1790–1798.
362. Steinness E, Olesen KH. Cardiac arrhythmias induced by hypokalaemia and potassium loss during maintenance digoxin therapy. *Br Heart J*. 1976;38:167–172.
363. Schrier RW, Gurevich AK, Cadnapaphornchai MA. Pathogenesis and management of sodium and water retention in cardiac failure and cirrhosis. *Semin Nephrol*. 2001;21:157–172.
364. Gines P, Cardenas A, Arroyo V, et al. Management of cirrhosis and ascites. *N Engl J Med*. 2004;350:1646–1654.
365. Lieberman FL, Denison EK, Reynolds TB. The relationship of plasma volume, portal hypertension, ascites and renal sodium retention in cirrhosis: the overflow theory of ascites formation. *Ann N Y Acad Sci*. 1970;170:202–212.
366. Diez J, Simon MA, Prieto J. Analysis of segmental tubular Na⁺ handling: a rational approach to the use of spironolactone in cirrhotic patients with ascites. In: Puschett J, Greenberg A, eds. *Diuretics III: Chemistry, Pharmacology and Clinical Applications: Proceedings of the Third International Conference on Diuretics, Held April 2–7, 1989*. Mexico City, Mexico. New York: Elsevier Science; 1990:394–395.
367. Earley LE, Martino JA. Influence of sodium balance on the ability of diuretics to inhibit tubular reabsorption. a study of factors that influence renal tubular sodium reabsorption in man. *Circulation*. 1970;42:323–334.
368. Runyon BA, AASLD. Introduction to the revised American Association for the Study of Liver Diseases Practice Guideline management of adult patients with ascites due to cirrhosis 2012. *Hepatology*. 2013;57:1651–1653.
369. Biggins SW, Angeli P, Garcia-Tsao G, Isoniemi H, Maiwall R, American Association for the Study of Liver Diseases. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American association for the study of liver diseases. *Hepatology*. 2021;74(2):1014–1048.
370. Chen JL, Wang L, Yao XM, et al. Association of urinary plasminogen-plasmin with edema and epithelial sodium channel activation in patients with nephrotic syndrome. *Am J Nephrol*. 2019;50(2):92–104.
371. Pariente EA, Bataille C, Bercoff E, et al. Acute effects of captopril on systemic and renal hemodynamics and on renal function in cirrhotic patients with ascites. *Gastroenterology*. 1985;88:1255–1259.
372. Kim WR, Biggins SW, Kremers WK, et al. Hyponatremia and mortality among patients on the liver-transplant waiting list. *N Engl J Med*. 2008;359:1018–1026.
373. Henriksen JH, Schlichting P. Increased extravasation and lymphatic return rate of albumin during diuretic treatment of ascites in patients with liver cirrhosis. *Scand J Clin Lab Invest*. 1981;41:589–599.
374. Shelton EL, Yang H-C, Zhong J, Salzman MM, Kon V. Renal lymphatic vessel dynamics. *Am J Physiol Ren Physiol*. 2020;319(6):F1027–F1036.
375. Rector Jr WG. Ascites kinetics in cirrhosis: effects of rapid volume expansion and diuretic administration. *J Lab Clin Med*. 1988;111:166–172.
376. Levy M. Physiological factors constraining the mobilization of ascites. In: Puschett J, Greenberg A, eds. *Diuretics III: Chemistry, Pharmacology, and Clinical Applications, Proceedings of the Third International Conference on Diuretics, Held April 2–7, 1989*. Mexico City, Mexico. New York: Elsevier Science; 1990:376–382.
377. Shear L, Ching S, Gabuzda GJ. Compartmentalization of ascites and edema in patients with hepatic cirrhosis. *N Engl J Med*. 1970;282:1391–1396.
378. Pockros PJ, Reynolds TB. Rapid diuresis in patients with ascites from chronic liver disease: the importance of peripheral edema. *Gastroenterology*. 1986;90:1827–1833.
379. Gines P, Arroyo V, Quintero E, et al. Comparison of paracentesis and diuretics in the treatment of cirrhotics with tense ascites. Result of a randomized study. *Gastroenterology*. 1987;93:234–241.
380. Runyon BA. Management of adult patients with ascites due to cirrhosis. *Hepatology*. 2004;39:841–856.
381. Fredrick MJ, Pound DC, Hall SD, et al. Furosemide absorption in patients with cirrhosis. *Clin Pharmacol Ther*. 1991;49:241–247.
382. Pinzani M, Daskalopoulos G, Laffi G, et al. Altered furosemide pharmacokinetics in chronic alcoholic liver disease with ascites contributes to diuretic resistance. *Gastroenterology*. 1987;92:294–298.
383. Uemura M, Matsumoto M, Tsujii T, et al. Effects of “body compression” on parameters related to ascites formation: therapeutic trial in cirrhotic patients. *J Gastroenterol*. 1999;34:75–82.
384. Wong W, Liu P, Blendis L, et al. Long-term renal sodium handling in patients with cirrhosis treated with transjugular intrahepatic portosystemic shunts for refractory ascites. *Am J Med*. 1999;106:315–322.
385. Orth SR, Ritz E. The nephrotic syndrome. *N Engl J Med*. 1998;338:1202–1211.
386. Schrier RW, Fassett RG. A critique of the overfill hypothesis of sodium and water retention in the nephrotic syndrome. *Kidney Int*. 1998;53:1111–1117.
387. Meltzer JI, Keim HJ, Laragh JH, et al. Nephrotic syndrome: vasoconstriction and hypervolemic types indicated by renin-sodium profiling. *Ann Intern Med*. 1979;91:688–696.

388. Bernard DB, Alexander EA, Couser WG, et al. Renal sodium retention during volume expansion in experimental nephrotic syndrome. *Kidney Int.* 1978;14:478–485.
389. Kirchner KA, Voelker JR, Brater DC. Tubular resistance to furosemide contributes to the attenuated diuretic response in nephrotic rats. *J Am Soc Nephrol.* 1992;2:1201–1207.
390. Ichikawa I, Rennke HG, Hoyer JR, et al. Role for intrarenal mechanisms in the impaired salt excretion of experimental nephrotic syndrome. *J Clin Invest.* 1983;71:91–103.
391. Humphreys MH. Mechanisms and management of nephrotic edema. *Kidney Int.* 1994;45:266–281.
392. Brown EA, Markandu ND, Sagnella GA, et al. Evidence that some mechanism other than the renin system causes sodium retention in nephrotic syndrome. *Lancet.* 1982;2(8310):1237–1240.
393. Kapur G, Valentini RP, Imam AA, et al. Treatment of severe edema in children with nephrotic syndrome with diuretics alone—a prospective study. *Clin J Am Soc Nephrol.* 2009;4:907–913.
394. Keller E, Hoppe-Seyler G, Schollmeyer P. Disposition and diuretic effect of furosemide in the nephrotic syndrome. *Clin Pharmacol Ther.* 1982;32:442–449.
395. Akcicek F, Yalniz T, Basci A, et al. Diuretic effect of frusemide in patients with nephrotic syndrome: is it potentiated by intravenous albumin? [see comments]. *BMJ.* 1995;310:162–163.
396. Fliser D, Zurbruggen I, Mutschler E, et al. Coadministration of albumin and furosemide in patients with the nephrotic syndrome. *Kidney Int.* 1999;55:629–634.
397. Rane A, Villeneuve JP, Stone WJ, et al. Plasma binding and disposition of furosemide in the nephrotic syndrome and in uremia. *Clin Pharmacol Ther.* 1978;24:199–207.
398. Ribelink TJ, Bijlsma JA, Koomans HA. Iso-oncotic volume expansion in the nephrotic syndrome. *Clin Sci.* 1993;84:627–632.
399. Mees EJD. Does it make sense to administer albumin to the patient with nephrotic oedema? *Nephrol Dial Transplant.* 1996;11:1224–1226.
400. Kalay Z, Sahin OE, Copur S, et al. SGLT-2 inhibitors in nephrotic-range proteinuria: emerging clinical evidence. *Clin Kidney J.* 2023;16(1):52–60.
401. Esnault VL, Ekhlās A, Delcroix C, et al. Diuretic and enhanced sodium restriction results in improved antiproteinuric response to RAS blocking agents. *J Am Soc Nephrol.* 2005;16:474–481.
402. Pichette V, du Souich P. Role of the kidneys in the metabolism of furosemide: its inhibition by probenecid. *J Am Soc Nephrol.* 1996;7:345–349.
403. Sjöström PA, Odland BG, Beermann BA, et al. PHARMACOKINETICS and effects of frusemide in patients with the nephrotic syndrome. *Eur J Pharmacol.* 1989;37:173–180.
404. Green TP, Mirkin BL. Furosemide disposition in normal and proteinuric rats: urinary drug-protein binding as a determinant of drug excretion. *J Pharmacol Exp Therapeut.* 1981;218:122–127.
405. Kirchner KA, Voelker JR, Brater DC. Binding inhibitors restore furosemide potency in tubule fluid containing albumin. *Kidney Int.* 1991;40:418–424.
406. Agarwal R, Gorski JC, Sundblad K, et al. Urinary protein binding does not affect response to furosemide in patients with nephrotic syndrome. *J Am Soc Nephrol.* 2000;11:1100–1105.
407. Garin EH. A comparison of combinations of diuretics in nephrotic edema. *Am J Dis Child.* 1987;141:769–771.
408. Sønningsen P, Bistrup C, Friis UG, et al. Plasmin in nephrotic urine activates the epithelial sodium channel. *J Am Soc Nephrol.* 2009;20:299–310.
409. Sønningsen P, Andersen H, Nielsen LH, Jensen BL. Urinary serine proteases and activation of ENaC in kidney—implications for physiological renal salt handling and hypertensive disorders with albuminuria. *Pflügers Archiv.* 2015;467(3):531–542.
410. Schork A, Woern M, Kalbacher H, et al. Association of plasminuria with overhydration in patients with CKD. *Clin J Am Soc Nephrol.* 2016;11(5):761–769.
411. Streeten DH. Idiopathic edema. pathogenesis, clinical features, and treatment. *Endocrinol Metab Clin N Am.* 1995;24:531–547.
412. MacGregor GA, Markandu ND, Roulston JE, et al. Is “idiopathic” edema idiopathic? *Lancet.* 1979;1(8113):397–400.
413. Young JB, Brownjohn AM, Lee MR. Diuretics and idiopathic oedema. *Nephron.* 1986;43:311–312.
414. Kim YG, Kim B, Kim MK, et al. Medullary nephrocalcinosis associated with long-term furosemide abuse in adults. *Nephrol Dial Transplant.* 2001;16:2303–2309.
415. Smirnakis KV, Yu AS. Diuretics in acute renal failure. In: Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. 2nd ed. Philadelphia: Saunders Inc; 2003:33–37.
416. Cantarovich F, Rangoonwala B, Lorenz H, et al. High-dose furosemide for established ARF: a prospective, randomized, double-blind, placebo-controlled, multicenter trial. *Am J Kidney Dis.* 2004;44:402–409.
417. Mehta RL, Pascual MT, Sorokos S, et al. Diuretics, mortality and nonrecovery of renal function in acute renal failure. *J Am Med Assoc.* 2002;288:2547–2553.
418. Uchino S, Doig GS, Bellomo R, et al. Diuretics and mortality in acute renal failure. *Crit Care Med.* 2004;32:1669–1677.
419. Mehta RL, Pascual MT, Soroko S, et al. Diuretics, mortality, and nonrecovery of renal function in acute renal failure. *JAMA.* 2002;288:2547–2553.
420. Stewart RM, Park PK, Hunt JP, et al. Less is more: improved outcomes in surgical patients with conservative fluid administration and central venous catheter monitoring. *J Am Coll Surg.* 2009;208:725–735; discussion 735–727.
421. Rose HJ, O'Malley K, Pruitt AW. Depression of renal clearance of furosemide in man by azotemia. *Clin Pharmacol Ther.* 1977;21:141–146.
422. Brater DC, Anderson SA, Brown Cartwright D. Response to furosemide in chronic renal insufficiency: rationale for limited doses. *Clin Pharmacol Ther.* 1986;40:134–139.
423. Voelker JR, Cartwright Brown D, Anderson S, et al. Comparison of loop diuretics in patients with chronic renal insufficiency. *Kidney Int.* 1987;32:572–578.
424. Buerkert J, Martin D, Prasad J, et al. Response of deep nephrons and the terminal collecting duct to a reduction in renal mass. *Am J Physiol.* 1979;236:F454–F464.
425. Kwon TH, Frokiaer J, Fernandez-Llama P, et al. Altered expression of Na transporters NHE-3, NaPi-II, Na-K-ATPase, BSC-1, and TSC in CRF rat kidneys. *Am J Physiol.* 1999;277:F257–F270.
426. Copeland JG, Campbell DW, Plachetka JR, et al. Diuresis with continuous infusion of furosemide after cardiac surgery. *Am J Surg.* 1983;146:796–799.
427. Agarwal R, Sinha AD, Pappas MK, Ammous F. Chlorthalidone for poorly controlled hypertension in chronic kidney disease: an interventional pilot study. *Am J Nephrol.* 2014;39(2):171–182.
428. Humes HD. Insights into ototoxicity. analogies to nephrotoxicity. *Ann NY Acad Sci.* 1999;884:15–18.
429. Hawkins RG, Houston MC. Is population-wide diuretic use directly associated with the incidence of end-stage renal disease in the United States? A hypothesis. *Am J Hypertens.* 2005;18:744–749.
430. Akram M, Reddan D. Diuretic use and ESRD, precipitating factor or epiphenomenon. *Am J Hypertens.* 2005;18:739–740.
431. Bragg-Gresham JL, Fissell RB, Mason NA, et al. Diuretic use, residual renal function, and mortality among hemodialysis patients in the Dialysis Outcomes and Practice Pattern Study (DOPPS). *Am J Kidney Dis.* 2007;49:426–431.
432. Lim LM, et al. Untargeted metabolomics analysis reveals key pathways responsible for the synergistic killing of colistin and doripenem combination against *Acinetobacter baumannii*. *Sci Rep.* 2017.
433. Sica DA, Gehr TW. Diuretic use in stage 5 chronic kidney disease and end-stage renal disease. *Curr Opin Nephrol Hypertens.* 2003;12:483–490.
434. Wilcox CS, Granges F, Kirk G, et al. Effects of saline infusion on titratable acid generation and ammonia secretion. *Am J Physiol.* 1984;247:F506–F519.
435. Na KY, Kim GH, Joo KW, et al. Chronic furosemide or hydrochlorothiazide administration increases H⁺-ATPase B1 subunit abundance in rat kidney. *Am J Physiol Ren Physiol.* 2007;292:F1701–F1709.
436. Rastogi SP, Crawford C, Wheeler R, et al. Effect of furosemide on urinary acidification in distal renal tubular acidosis. *J Lab Clin Med.* 1984;104:271–282.
437. Weinstein AM. A mathematical model of distal nephron acidification: diuretic effects. *Am J Physiol Ren Physiol.* 2008;295:F1353–F1364.
438. Schambelan M, Sebastian A, Hulter HN. Mineral corticoid excess and deficiency syndromes. In: Brenner BM, Stein JH, eds. *Acid-Base and Potassium Homeostasis (Contemporary Issues in Nephrology)*. Vol II. New York: Churchill Livingstone; 1978:232.

439. Sands JM, Naruse M, Baum M, et al. Apical extracellular calcium/polyvalent cation-sensing receptor regulates vasopressin-elicited water permeability in rat kidney inner medullary collecting duct. *J Clin Invest*. 1997;99:1399–1405.
440. LeGrand SB, Leskusi D, Zama I. Narrative review: furosemide for hypercalcemia: an unproven yet common practice. *Ann Intern Med*. 2008;149:259–263.
441. Borghi L, Meschi T, Guerra A, et al. Randomized prospective study of a nonthiazide diuretic indapamide, in preventing calcium stone recurrences. *J Cardiovasc Pharmacol*. 1993;22:S78–S86.
442. Dhayat NA, Bonny O, Roth B, et al. Hydrochlorothiazide and prevention of kidney-stone recurrence. *N Engl J Med*. 2023;388(9):781–791.
443. Pak CY, Peterson R, Sakhaee K, et al. Correction of hypocitraturia and prevention of stone formation by combined thiazide and potassium citrate therapy in thiazide-unresponsive hypercalciuric nephrolithiasis. *Am J Med*. 1985;79:284–288.
444. Leppla D, Browne R, Hill K, et al. Effect of amiloride with or without hydrochlorothiazide on urinary calcium and saturation of calcium salts. *J Clin Endocrinol Metab*. 1983;57:920–924.
445. Frassetto LA, Nash E, Morris Jr RC, et al. Comparative effects of potassium chloride and bicarbonate on thiazide-induced reduction in urinary calcium excretion. *Kidney Int*. 2000;58:748–752.
446. Barry EL, Gesek FA, Kaplan MR, et al. Expression of the sodium-chloride cotransporter in osteoblast-like cells: effects of thiazide diuretics. *Am J Physiol*. 1997;272:C109–C116.
447. Aubin R, Menard P, Lajeunesse D. Selective effect of thiazides on the human osteoblast-like cell line MG-63. *Kidney Int*. 1996;50:1476–1482.
448. Lajeunesse D, Delalandre A, Guggino SE. Thiazide diuretics affect osteocalcin production in human osteoblasts at the transcription level without affecting vitamin D3 receptors. *J Bone Miner Res*. 2000;15:894–901.
449. Dvorak MM, De Joussineau C, Carter DH, et al. Thiazide diuretics directly induce osteoblast differentiation and mineralized nodule formation by interacting with a sodium chloride co-transporter in bone. *J Am Soc Nephrol*. 2007;18:2509–2516.
450. Hsu YJ, Yang SS, Cheng CJ, et al. Thiazide-sensitive Na⁺-Cl⁻ cotransporter (NCC) gene inactivation results in increased duodenal Ca²⁺ absorption, enhanced osteoblast differentiation and elevated bone mineral density. *J Bone Miner Res*. 2015;30(1):116–127.
451. Lalande A, Roux S, Denne MA, et al. Indapamide, a thiazide-like diuretic, decreases bone resorption in vitro. *J Bone Miner Res*. 2001;16:361–370.
452. Sigurdsson G, Franzson L. Increased bone mineral density in a population-based group of 70-year-old women on thiazide diuretics, independent of parathyroid hormone levels. *J Intern Med*. 2001;250:51–56.
453. Cauley JA, Cummings SR. Thiazide diuretics preserve bone mass and reduce the risk of fractures in elderly women: a prospective study. *J Bone Miner Res*. 1989;4(suppl 1):216–217.
454. Schoofs MW, van der Klift M, Hofman A, et al. Thiazide diuretics and the risk for hip fracture. *Ann Intern Med*. 2003;139:476–482.
455. Puttnam R, Davis BR, Pressel SL, et al. Association of 3 different antihypertensive medications with hip and pelvic fracture risk in older adults: secondary analysis of a randomized clinical trial. *JAMA Intern Med*. 2017;177(1):67–76.
456. Reid IR, Ames RW, Orr-Walker BJ, et al. Hydrochlorothiazide reduces loss of cortical bone in normal postmenopausal women: a randomized controlled trial. *Am J Med*. 2000;109:362–370.
457. Rejnmark L, Vestergaard P, Heickendorff L, et al. Effects of thiazide- and loop-diuretics, alone or in combination, on calcitropic hormones and biochemical bone markers: a randomized controlled study. *J Intern Med*. 2001;250:144–153.
458. Carbone LD, Johnson KC, Bush AJ, et al. Loop diuretic use and fracture in postmenopausal women: findings from the Women's Health Initiative. *Arch Intern Med*. 2009;169:132–140.
459. Lim LS, Fink HA, Kuskowski MA, et al. Loop diuretic use and increased rates of hip bone loss in older men: the Osteoporotic Fractures in Men Study. *Arch Intern Med*. 2008;168:735–740.
460. Thillemann TM, Pedersen AB, Mehnert F, et al. Use of diuretics and risk of implant failure after primary total hip arthroplasty: a nationwide population-based study. *Bone*. 2009;45:499–504.
461. Colussi G, Rombola G, De Ferrari ME, et al. Correction of hypokalemia with antialdosterone therapy in Gitelman's syndrome. *Am J Nephrol*. 1994;14:127–135.
462. Blanchard A, Vargas-Poussou R, Vallet M, et al. Indomethacin, amiloride, or eplerenone for treating hypokalemia in Gitelman syndrome. *J Am Soc Nephrol*. 2015;26:468–475.
463. Crawford JD, Kennedy GC. Chlorothiazide in diabetes insipidus. *Nature*. 1959;183:891.
464. Al Nofal A, Lteif A. Thiazide diuretics in the management of young children with central diabetes insipidus. *J Pediatr*. 2015;167(3):658–661.
465. Spannow J, Thomsen K, Petersen JS, et al. Influence of renal nerves and sodium balance on the acute antidiuretic effect of bendroflumethiazide in rats with diabetes insipidus. *J Pharmacol Exp Therapeut*. 1997;282:1155–1162.
466. Gronbeck L, Marples D, Nielsen S, et al. Mechanism of antidiuresis caused by bendroflumethiazide in conscious rats with diabetes insipidus. *Br J Pharmacol*. 1998;123:737–745.
467. Bedford JJ, Wegger S, Ellis G, et al. Lithium-induced nephrogenic diabetes insipidus: renal effects of amiloride. *Clin J Am Soc Nephrol*. 2008;3:1324–1331.
468. Kortenoeven ML, Li Y, Shaw S, et al. Amiloride blocks lithium entry through the sodium channel thereby attenuating the resultant nephrogenic diabetes insipidus. *Kidney Int*. 2009;76:44–53.
469. Rao R, Zhang MZ, Zhao M, et al. Lithium treatment inhibits renal GSK-3 activity and promotes cyclooxygenase 2-dependent polyuria. *Am J Physiol Ren Physiol*. 2005;288:F642–F649.
470. Blantz RC. Pathophysiology of pre-renal azotemia. *Kidney Int*. 1998;53:512–523.
471. Canton AD, Fuiano G, Conte G, et al. Mechanism of increased plasma urea after diuretic therapy in uraemic patients. *Clin Sci*. 1985;68:255–261.
472. Kamm DE, Genin M, Kuchmy B, et al. Diuretic-induced azotemia: unmasking the role of increased peripheral catabolism through evisceration-hepatectomy. *Kidney Int Suppl*. 1983;16:S58–S60.
473. Thomas RD, Newill A, Morgan DB. The cause of the raised plasma urea of acute heart failure. *Postgrad Med J*. 1979;55:10–14.
474. Kitiyakara C, Chabrashvili T, Jose P, et al. Effects of dietary salt intake on plasma arginine. *Am J Physiol*. 2001;280:R1069–R1075.
475. Lapi F, Azoulay L, Yin H, et al. Concurrent use of diuretics, angiotensin converting enzyme inhibitors, and angiotensin receptor blockers with non-steroidal anti-inflammatory drugs and risk of acute kidney injury: nested case-control study. *BMJ*. 2013;346:e8525.
476. Dreischulte T, Morales DR, Bell S, Guthrie B. Combined use of nonsteroidal anti-inflammatory drugs with diuretics and/or renin-angiotensin system inhibitors in the community increases the risk of acute kidney injury. *Kidney Int*. 2015;88(2):396–403.
477. Szatalowicz VL, Miller PD, Lacher JW, et al. Comparative effect of diuretics on renal water excretion in hyponatraemic oedematous disorders. *Clin Sci*. 1982;62:235–238.
478. Spital A. Diuretic-induced hyponatremia. *Am J Nephrol*. 1999;19(4):447–452.
479. Friedman E, Shadel M, Halkin H, Farfel Z. Thiazide-induced hyponatremia. Reproducibility by single dose rechallenge and an analysis of pathogenesis. *Ann Intern Med*. 1989;110(1):24–30.
480. Clark BA, Shannon RP, Rosa RM, et al. Increased susceptibility to thiazide-induced hyponatremia in the elderly. *J Am Soc Nephrol*. 1994;5:1106–1111.
481. Sonnenblick M, Friedlander Y, Rosin AJ. Diuretic-induced severe hyponatremia. Review and analysis of 129 reported patients. *Chest*. 1993;103:601–606.
482. Booker JA. Severe symptomatic hyponatremia in elderly outpatients: the role of thiazide therapy and stress. *J Am Geriatr Soc*. 1984;32:108–113.
483. Chow KM, Szeto CC, Wong TY, Leung CB, Li PK. Risk factors for thiazide-induced hyponatraemia. *QJM*. 2003;96(12):911–917.
484. Ashraf N, Locksley R, Arief AL. Thiazide-induced hyponatremia associated with death or neurologic damage in outpatients. *Am J Med*. 1981;70:1163–1168.
485. Frenkel NJ, Vogt L, De Rooij SE, et al. Thiazide-induced hyponatraemia is associated with increased water intake and impaired urea-mediated water excretion at low plasma antidiuretic hormone and urine aquaporin-2. *J Hypertens*. 2015;33(3):627–633.
486. Ware JS, Wain LV, Channavajjhala SK, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127(9):3367–3374.

- 486a. Palmer BF, Clegg DJ. Altered prostaglandin signaling as a cause of thiazide-induced hyponatremia. *Am J Kidney Dis*. 2018;71(6):769–771.
487. Janicic N, Verbalis JG. Evaluation and management of hyposmolality in hospitalized patients. *Endocrinol Metab Clin N Am*. 2003;32:459–481.
488. Renneboog B, Musch W, Vandemergel X, et al. Mild chronic hyponatremia is associated with falls, unsteadiness, and attention deficits. *Am J Med*. 2006;119(71):e1–e8.
489. Jiang JY, Wong MC, Ali MK, et al. Association of antihypertensive monotherapy with serum sodium and potassium levels in Chinese patients. *Am J Hypertens*. 2009;22:243–249.
490. Carrisoza-Gaytan R, Ray EC, Flores D, et al. Intercalated cell BKA subunit is required for flow-induced K⁺ secretion. *JCI Insight*. 2020;5(8).
491. Pluznick JL, Wei P, Carmines PK, et al. Renal fluid and electrolyte handling in BKCa-beta1-/- mice. *Am J Physiol Ren Physiol*. 2003;284:F1274–F1279.
492. Taniguchi J, Imai M. Flow-dependent activation of maxi K⁺ channels in apical membrane of rabbit connecting tubule. *J Membr Biol*. 1998;164:35–45.
493. Morgan DB, Davidson C. Hypokalaemia and diuretics: an analysis of publications. *BMJ*. 1980;280:905–908.
494. Penton D, Czogalla J, Wengi A, et al. Extracellular K⁺ rapidly controls NaCl cotransporter phosphorylation in the native distal convoluted tubule by Cl⁻ dependent and independent mechanisms. *J Physiol*. 2016;594(21):6319–6331.
495. Rengarajan S, Lee DH, Oh YT, et al. Increasing plasma [K⁺] by intravenous potassium infusion reduces NCC phosphorylation and drives kaliuresis and natriuresis. *Am J Physiol Renal Physiol*. 2014;306(9):F1059–F1068.
496. Terker AS, Zhang C, McCormick JA, et al. Potassium modulates electrolyte balance and blood pressure through effects on distal cell voltage and chloride. *Cell Metab*. 2015;21(1):39–50.
497. Deleted in review.
498. Hunter RW, Craigie E, Homer NZ, Mullins JJ, Bailey MA. Acute inhibition of NCC does not activate distal electrogenic Na⁺ reabsorption or kaliuresis. *Am J Physiol Ren Physiol*. 2014;306(4):F457–F467.
499. Ellison DH, Loffing J. Thiazide effects and adverse effects: insights from molecular genetics. *Hypertension*. 2009;54:196–202.
500. Deleted in review.
501. Freis ED, Papademetriou V. How dangerous are diuretics? *Drugs*. 1985;30:469–474.
502. Robertson JW, Isles CG, Brown I, et al. Mild hypokalaemia is not a risk factor in treated hypertensives. *J Hypertens*. 1986;4:603–608.
503. Siscovick DS, Raghunathan TE, Psaty BM, et al. Diuretic therapy for hypertension and the risk of primary cardiac arrest. *N Engl J Med*. 1994;330:1852–1857.
504. Collins AJ, Pitt B, Reaven N, et al. Association of serum potassium with all-cause mortality in patients with and without heart failure, chronic kidney disease, and/or diabetes. *Am J Nephrol*. 2017;46(3):213–221.
505. Siegel D, Hulley SB, Black DM, et al. Diuretics, serum and intracellular electrolyte levels, and ventricular arrhythmias in hypertensive men. *J Am Med Assoc*. 1992;267:1083–1089.
506. Rosa RM, Silva P, Young JB, et al. Adrenergic modulation of extrarenal potassium disposal. *N Engl J Med*. 1980;302:431–434.
507. Weinberger MH. Mechanisms of diuretic effects on carbohydrate tolerance, insulin sensitivity and lipid levels. *Eur Heart J*. 1992;13:5–9.
508. Krishna GG, Miller E, Kapoor S. Increased blood pressure during potassium depletion in normotensive men. *N Engl J Med*. 1989;320:1177–1182.
509. Maronde RF, Milgrom M, Vlachakis N, et al. Response to thiazide-induced hypokalemia to amiloride. *J Am Med Assoc*. 1983;249:237–241.
510. Brown MJ, Williams B, Morant SV, British Hypertension Society's Prevention and Treatment of Hypertension with Algorithm-based Therapy (PATHWAY) Studies Group, et al. Effect of amiloride, or amiloride plus hydrochlorothiazide, versus hydrochlorothiazide on glucose tolerance and blood pressure (PATHWAY-3): a parallel-group, double-blind randomised phase 4 trial. *Lancet Diabetes Endocrinol*. 2016;4(2):136–147.
511. Hropot M, Fowler N, Karlmark B, et al. Tubular action of diuretics: distal effects on electrolyte transport and acidification. *Kidney Int*. 1985;28:477–489.
512. Antoniou T, Gomes T, Mamdani MM, et al. Trimethoprim-sulfamethoxazole induced hyperkalaemia in elderly patients receiving spironolactone: nested case-control study. *BMJ*. 2011;343:d5228.
513. Schepkens H, Vanholder R, Billioud JM, et al. Life-threatening hyperkalaemia during combined therapy with angiotensin-converting enzyme inhibitors and spironolactone: an analysis of 25 cases. *Am J Med*. 2001;110(6):438–441.
514. Agarwal R, Rossignol P, Romero A, et al. Patiromer versus placebo to enable spironolactone use in patients with resistant hypertension and chronic kidney disease (AMBER): a phase 2, randomised, double-blind, placebo-controlled trial. *Lancet*. 2019;394(10208):1540–1550.
515. Dai LJ, Ritchie G, Kerstan D, et al. Magnesium transport in the renal distal convoluted tubule. *Physiol Rev*. 2001;81:51–84.
516. Alexander RT, Dimke H. Effect of diuretics on renal tubular transport of calcium and magnesium. *Am J Physiol Ren Physiol*. 2017;312(6):F998–F1015.
517. Dyckner T, Wester PO. Effects of magnesium infusions in diuretic induced hyponatraemia. *Lancet*. 1981;1(8220):585–586.
518. Quamme GA. Renal magnesium handling: new insights in understanding old problems. *Kidney Int*. 1997;52:1180–1195.
519. Huang CL, Kuo E. Mechanism of hypokalemia in magnesium deficiency. *J Am Soc Nephrol*. 2007;18:2649–2652.
520. Suki WN. Effects of diuretics on calcium metabolism. *Adv Exp Med Biol*. 1982;151:493–500.
521. Wermers RA, Kearns AE, Jenkins GD, et al. Incidence and clinical spectrum of thiazide-associated hypercalcemia. *Am J Med*. 2007;120(10):911.e9–911.e15.
522. Griebeler ML, Kearns AE, Ryu E, et al. Thiazide-associated hypercalcemia: incidence and association with primary hyperparathyroidism over two decades. *J Clin Endocrinol Metab*. 2016;101(3):1166–1173.
523. Brijker F, Heijdra YF, van den Elshout FJ, et al. Discontinuation of furosemide decreases PaCO₂ in patients with COPD. *Chest*. 2002;121:377–382.
524. Cannon PJ, Heineman HO, Albert MS, et al. “Contraction” alkalosis after diuresis of edematous patients with ethacrynic acid. *Ann Intern Med*. 1965;62:979–990.
525. Wagner CA, Geibel JP. Acid-base transport in the collecting duct. *J Nephrol*. 2002;15(suppl 5):S112–S127.
526. Levine DZ, Iacovitti M, Buckman S, et al. Distal tubule unidirectional HCO₃ reabsorption in vivo during acute and chronic metabolic alkalosis in the rat. *Am J Physiol*. 1994;266:F919–F925.
527. Adverse reactions to bendrofluazide and propranolol for the treatment of mild hypertension: report of medical research Council working party on mild to moderate hypertension. *Lancet*. 1981;2(8246):539–543.
528. Fruman BL. Impairment of glucose intolerance produced by diuretics and other drugs. *Pharmacol Ther*. 1981;12:613.
529. Shen L, Shah BR, Reyes EM, et al. Role of diuretics, beta blockers, and statins in increasing the risk of diabetes in patients with impaired glucose tolerance: reanalysis of data from the NAVIGATOR study. *BMJ*. 2013;347:f6745.
530. Dornhorst A, Powell SH, Pensky J. Aggravation by propranolol of hyperglycaemic effect of hydrochlorothiazide in type II diabetics without alteration of insulin secretion. *Lancet*. 1985;1(8421):123–126.
531. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs diuretic: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA*. 2002;288:2981–2997.
532. Murphy MB, Lewis PJ, Kohner E, et al. Glucose intolerance in hypertensive patients treated with diuretics; a fourteen-year follow-up. *Lancet*. 1982;2(8311):1293–1295.
533. Hirst JA, Farmer AJ, Feakins BG, Aronson JK, Stevens RJ. Quantifying the effects of diuretics and β -adrenoceptor blockers on glycaemic control in diabetes mellitus—a systematic review and meta-analysis. *Br J Clin Pharmacol*. 2015;79(5):733–743.
534. Bergstrom J, Hultman E. The effect of thiazides, chlorthalidone and furosemide on muscle electrolytes and muscle glycogen in normal subjects. *Acta Med Scand*. 1966;180:363–376.
535. Pollare T, Lithell H, Berne C. A comparison of the effects of hydrochlorothiazide and captopril on glucose and lipid metabolism in patients with hypertension. *N Engl J Med*. 1989;321:868–873.

536. Helderma JH, Elahi D, Andersen DK, et al. Prevention of the glucose intolerance of thiazide diuretics by maintenance of body potassium. *Diabetes*. 1983;32:106–111.
537. Paolisso G, Gambardella A, Balbi V, et al. Effects of magnesium and nifedipine infusions on insulin action, substrate oxidation, and blood pressure in aged hypertensive patients. *Am J Hypertens*. 1993;6:920–926.
538. Sharma A, Singh AK, Bakris G, Bushinsky DA, Pitt B, Agarwal R. Effect of finerenone on serum potassium and occurrence of hyperkalemia in patients with chronic kidney disease and type 2 diabetes: a FIDELIO-DKD subgroup analysis. *J Am Soc Nephrol*. 2023;34(7):1179–1190.
539. Rowe JW, Tobin JD, Rosa RM, et al. Effect of experimental potassium deficiency on glucose and insulin metabolism. *Metabolism*. 1980;29:498–502.
540. Siegel D, Saliba P, Haffner S. Glucose and insulin levels during diuretic therapy in hypertensive men. *Hypertension*. 1994;23:688–694.
541. Stears AJ, Woods SH, Watts MM, et al. A double-blind, placebo-controlled, crossover trial comparing the effects of amiloride and hydrochlorothiazide on glucose tolerance in patients with essential hypertension. *Hypertension*. 2012;59:934–942.
542. Curb JD, Pressel SL, Cutler JA. Effects of diuretic-based antihypertensive treatment on cardiovascular disease risk in older diabetic patients with isolated systolic hypertension. *J Am Med Assoc*. 1996;276:1886–1892.
543. Ames RP. The effects of antihypertensive drugs on serum lipids and lipoproteins II: non-diuretic drugs. *Drugs*. 1986;32:335–357.
544. Ott SM, LaCroix AZ, Ichikawa LE, et al. Effect of low-dose thiazide diuretics on plasma lipids: results from a double-blind, randomized clinical trial in older men and women. *J Am Geriatr Soc*. 2005;51:1003–1013.
545. Grimm RHJ, Flack JM, Grandits GA. Antihypertensive drugs had either a neutral or beneficial effect on lipids. *J Am Med Assoc*. 1996;275:1549–1556.
546. Manttari M, Tenkanen L, Manninen V, et al. Antihypertensive therapy in dyslipidemic men: effects on coronary heart disease incidence and total mortality. *Hypertension*. 1995;25:47–52.
547. Roch-Ramel F, Guisan B, Diezi J. Effects of uricosuric and antiuricosuric agents on urate transport in human brush-border membrane vesicles. *J Pharmacol Exp Therapeut*. 1997;280:839–845.
548. Langford HG, Blaufox MD, Borhani NO, et al. Is thiazide-induced uric acid elevation harmful? Analysis of data from the Hypertension Detection and Follow-up Program. *Arch Intern Med*. 1987;147:645–649.
549. Fang J, Alderman MH. Serum uric acid and cardiovascular mortality: the NHANES I epidemiologic follow-up study, 1971–1992. National Health and Nutrition Examination Survey. *J Am Med Assoc*. 2000;283:2404–2410.
550. Rayner BL, Trinder YA, Baines D, et al. Effect of losartan versus candesartan on uric acid, renal function, and fibrinogen in patients with hypertension and hyperuricemia associated with diuretics. *Am J Hypertens*. 2006;19:208–213.
551. Grimm RHJ, Grandits GA, Prineas RJ, et al. Long-term effects on sexual function of five antihypertensive drugs and nutritional hygienic treatment in hypertensive men and women. Treatment of Mild Hypertension Study (TOMHS). *Hypertension*. 1997;29:8–14.
552. Pickering TG, Shepherd AMM, Puddy I, et al. Sildenafil citrate for erectile dysfunction in men receiving multiple antihypertensive agents: a randomized controlled trial. *Am J Hypertens*. 2004;17:1135–1142.
553. Rybak LP. Ototoxicity of loop diuretics. *Otolaryngol Clin*. 1993;26:829–844.
554. Molnar J, Somberg JC. The clinical pharmacology of ethacrynic acid. *Am J Therapeut*. 2009;16:86–92.
555. Rybak LP, Whitworth C, Scott V. Furosemide ototoxicity is enhanced in anabunimic rats. *Arch Otolaryngol Head Neck Surg*. 1993;119:758–761.
556. Dormans TP, Gerlag PG. Combination of high-dose furosemide and hydrochlorothiazide in the treatment of refractory congestive heart failure. *Eur Heart J*. 1996;17:1867–1874.
557. Collins R, Yusuf S, Peto R. Overview of randomised trials of diuretics in pregnancy. *BMJ*. 1985;290:19–23.
558. Davison JM. Edema in pregnancy. *Kidney Int*. 1997;51(suppl 59):S90–S96.
559. Green TP, Thompson TR, Johnson DE, et al. Furosemide promotes patent ductus arteriosus in premature infants with the respiratory-distress syndrome. *N Engl J Med*. 1983;308:743–748.
560. Prandota J. Clinical pharmacology of furosemide in children: a supplement. *Am J Therapeut*. 2001;8:275–289.
561. Hufnagle KG, Khan SN, Penn D, et al. Renal calcifications: a complication of long-term furosemide therapy in preterm infants. *Pediatrics*. 1982;70:360–363.
562. Bailie MD, Linshaw MA, Stygles VG. Diuretic pharmacology in infants and children. *Pediatr Clin*. 1981;28(217):231–240.
563. Lubetsky A, Winaver J, Seligmann H, et al. Urinary thiamine excretion in the rat: effects of furosemide, other diuretics and volume load. *J Lab Clin Med*. 1999;134:192–193.
564. Suter PM, Haller J, Hany A, et al. Diuretic use: a risk for subclinical thiamine deficiency in elderly patients. *J Nutr Health Aging*. 2000;4:69–71.
565. Shimon I, Almog S, Vered Z, et al. Improved left ventricular function after thiamine supplementation in patients with congestive heart failure receiving long-term furosemide therapy. *Am J Med*. 1995;98:485–490.
566. Addo HA, Ferguson J, Frain Bell W. Thiazide-induced photosensitivity: a study of 33 subjects. *Br J Dermatol*. 1987;116:749–760.
567. Heydenreich G, Pindborg T, Schmidt H. Bullous dermatosis among patients with chronic renal failure on high dose frusemide. *Acta Med Scand*. 1977;202:61–64.
568. Dominguez-Ortega J, Martinez-Alonso JC, Dominguez-Ortega C, et al. Anaphylaxis to oral furosemide. *Allergol Immunopathol*. 2003;31:345–347.
569. Cribb AE, Pohl LR, Spielberg SP, et al. Patients with delayed-onset sulfonamide hypersensitivity reactions have antibodies recognizing endoplasmic reticulum luminal proteins. *J Pharmacol Exp Therapeut*. 1997;282:1064–1072.
570. Eckhauser ML, Dokler M, Imbembo AL. Diuretic-associated pancreatitis: a collective review and illustrative cases. *Am J Gastroenterol*. 1987;82:865–870.
571. Lyons H, Pinn VW, Cortell S, et al. Allergic interstitial nephritis causing reversible renal failure in four patients with idiopathic nephrotic syndrome. *N Engl J Med*. 1973;288:124–128.
572. Magil AB, Ballon HS, Cameron EC, et al. Acute interstitial nephritis associated with thiazide diuretics. clinical and pathologic observations in three cases. *Am J Med*. 1980;69:939–943.
573. Pottegård A, Hallas J, Olesen M, et al. Hydrochlorothiazide use is strongly associated with risk of lip cancer. *J Intern Med*. 2017;282(4):322–331.
574. Jensen AØ, Thomsen HF, Engebjerg MC, Olesen AB, Sørensen HT, Karagas MR. Use of photosensitising diuretics and risk of skin cancer: a population-based case-control study. *Br J Cancer*. 2008;99(9):1522–1528.
575. Bangalore S, Kumar S, Kjeldsen SE, et al. Antihypertensive drugs and risk of cancer: network meta-analyses and trial sequential analyses of 324,168 participants from randomised trials. *Lancet Oncol*. 2011;12:65–82.
576. Anton C, Cox AR, Watson RD, et al. The safety of spironolactone treatment in patients with heart failure. *J Clin Pharm Therapeut*. 2003;28:285–287.
577. Cruz CS, Cruz AA, Marcilio de Sousa CA. Hyperkalaemia in congestive heart failure patients using ACE inhibitors and spironolactone. *Nephrol Dial Transplant*. 2003;18:1814–1819.
578. Lawson DH, Macadam RF, Singh MH, et al. Effect of furosemide on antibiotic-induced renal damage in rats. *J Infect Dis*. 1972;126:593–600.
579. Shapiro S, Slone D, Lewis GP, et al. The epidemiology of digoxin toxicity. A study in three Boston Hospitals. *J Chron Dis*. 1969;22:361–371.
580. Juurlink DN, Mamdani MM, Kopp A, et al. Drug-induced lithium toxicity in the elderly: a population-based study. *J Am Geriatr Soc*. 2004;52:794–798.
581. Petersen V, Hvidt S, Thomsen K, et al. Effect of prolonged thiazide treatment on renal lithium clearance. *BMJ*. 1974;3:143–145.
582. Weinberg MS, Quigg RJ, Salant DJ, et al. Anuric renal failure precipitated by indomethacin and triamterene. *Nephron*. 1983;40:216–218.